ESE 3123 Complete Solutions

Thermo-Fluid Devices · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' material (Fahim sir, Shameem sir) and the textbooks they follow (Cengel & Cimbala, Bansal, Incropera, Holman)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 3 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 ...) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). Part A is mostly pumps and flow meters: the venturi/Pitot numericals (A4), affinity laws (A7) and the reciprocating pump numerical (A8) are short and come every year, and each has a calculator. In Part B the heat exchanger numericals (B5, B6) are the safest marks; the normal-shock calculator (B3) gives every downstream property in one go. Leave dimensional analysis (A3) and nozzle derivations (B2) for last.

Section A: Fluid flow, measurement and pumps

Block 1: very likely and easy · 2.1 h

Topic	Chance	Level	Covered
A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter	Very likely 100%	Easy	22%
A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height	Very likely 100%	Easy	37%
A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals	Very likely 100%	Easy	50%
A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws	Very likely 100%	Easy	56%
A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller	Very likely 87%	Easy	67%

Block 2: very likely, medium effort · 0.8 h

Topic	Chance	Level	Covered

A1 Pipe flow theory: laminar and turbulent flow, Reynolds number, entrance region, losses, Darcy-Weisbach, laminar velocity profile	Very likely 100%	Medium	81%

✔ Pass line: 81% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem	Very likely 87%	Hard	88%

Block 4: likely · 1.2 h

Topic	Chance	Level	Covered
A9 Compressors, fans and blowers	Likely 67%	Easy	93%
A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules	Likely 60%	Medium	100%

Section B: Compressible flow and heat exchangers

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B5 Heat exchanger LMTD numericals: area, length, outlet temperature, fouling	Very likely 100%	Easy	17%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B4 Heat exchanger fundamentals: classification, temperature profiles, overall U, fouling factor, LMTD derivations	Very likely 100%	Medium	38%
B1 Compressible flow basics: definitions, Mach number classes, speed of sound, stagnation properties, adiabatic Bernoulli	Very likely 100%	Medium	53%
B3 Shock waves: normal and oblique shocks, Rayleigh line, normal-shock numericals	Very likely 93%	Medium	69%
B6 Effectiveness-NTU method	Very likely 93%	Medium	83%

✔ Pass line: 83% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.4 h)

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered

B2 Isentropic flow in nozzles and diffusers: area-velocity relation, choking, back pressure, nozzle numericals	Very likely 100%	Hard	100%

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1. One-shot notes
2. **Section A: Fluid flow, measurement and pumps**: A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter, A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height, A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals, A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws, A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller, A1 Pipe flow theory: laminar and turbulent flow, Reynolds number, entrance region, losses, Darcy-Weisbach, laminar velocity profile, A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem, A9 Compressors, fans and blowers, A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules
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One-shot notes: ESE 3123 Thermo-Fluid Devices

Pipe flow (A1)

Re = ρVD/μ = VD/ν: laminar < 2300, transition 2300-4000, turbulent > 4000. Entrance length $L_h \approx 0.05ReD$ (laminar), 10D (turbulent); beyond it the flow is fully developed.

Darcy-Weisbach $h_L = f(L/D)(V^2/2g)$; laminar $f = 64/Re$; turbulent from Moody/Colebrook (f depends on Re and ε/D). Minor loss $h = K\,V^2/2g$.

Laminar profile: $u = u_{max}(1 - r^2/R^2)$, $V_{avg} = u_{max}/2$, $\Delta P = 32\mu LV/D^2$ (Hagen-Poiseuille). Turbulent profile is flatter.

Pipe numericals (A2)

Steps: Re → f → h_L → $\Delta P = \rho g h_L$ → $W = Q\Delta P$.

2019 2b: Re 1777, f 0.036, h_L 4.46 m, ΔP 43.7 kPa, W 0.28 W. 2023 6d: Re 1836, ΔP 188.2 kPa, h_L 19.2 m, W 0.71 W.

Series: same Q, h_L add. Parallel: same h_L, Q add.

2021 1c ratio 45.7. 3-inch pipe: 10S ID 3.260 in; sch 40 ID 3.068 in; XXS ID 2.300 in.

Dimensional analysis (A3)

Fundamental dimensions M, L, T (θ). Homogeneity: every term has the same dimensions. Rayleigh: write $Q = kA^aB^b\ldots$, equate powers of M, L, T.

Buckingham π: n variables, m dimensions ⇒ (n − m) π groups; choose m repeating variables (geometric, kinematic, dynamic: usually D, V, ρ), each $\pi = D^aV^b\rho^cX$. Drag gives $F/(\rho V^2D^2) = f(Re)$.

Flow meters (A4)

Venturi: $Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$, $h = x(S_m/S - 1)$; C_d 0.95-0.99. Orifice same with C_d 0.6-0.65 (vena contracta). Pitot $V = C_v\sqrt{(2gh)}$.

Rotameter: float rises in tapered tube until weight = drag + buoyancy (variable area). Turbine meter: rotor speed ∝ Q.

Venturi Q 125.8 L/s. Pitot 2020: h 2.451 m, V 6.93 m/s; 2022: 2.328 m, 6.76 m/s. Air venturi 2019 6d: 3.12 kg/s (3.25 if incompressible).

Centrifugal pump (A5)

Parts: impeller, casing (volute, vortex, diffuser), suction pipe with foot valve and strainer, delivery pipe with valve. Converts mechanical energy into pressure energy by centrifugal action.

Work per unit weight = $V_{w2}u_2/g$ (radial entry, $V_{w1} = 0$). Manometric head $H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/2g$.

$\eta_{mano} = gH_m/(V_{w2}u_2)$; η_{mech} = impeller power/shaft power; $\eta_o = \rho gQH_m/P$.

Pump performance (A6)

Curves: H, P, η vs Q at constant N; BEP at peak η. Affinity: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.

$N_s = N\sqrt{Q}/H^{3/4}$ (per stage, per eye). Cavitation: local $P < P_{vap}$ ⇒ bubbles collapse, pitting, noise, drop of H and η. $NPSH_A = (P_{atm} - P_v)/\rho g - h_s - h_{fs} \geq NPSH_R$.

Priming: fill suction pipe and casing with liquid before starting (air gives too little head).

Pump numericals (A7)

Series: H adds. 2020 3d: 31.9 m per pump ⇒ 3.14 ⇒ 4 pumps. Multistage: $n = H/H_{stage}$; 2021 3d: N_s 322, 18.8 m/stage, 10 stages, D 58.7 cm. 2022 4d: N_s 672.7, 19 stages, D 36.7 cm (alt. 10 stages, 35.8 cm). Affinity: 2023 3d: 866 rpm, 92.4 gpm, 9.24 hp. 2019 7d: 1072 rpm, 122.5 gpm, 14.7 hp.

Reciprocating pump (A8)

$Q_{th} = ALN/60$ (single acting), $2ALN/60$ (double). Slip = $Q_{th} - Q_{act}$; % slip = $(1 - C_d)100$; negative slip when long suction, high speed. $P = \rho gQ_{th}(h_s + h_d)$.

2020 2b: C_d 0.960, slip 3.96%, P 4.21 kW. 2021 4d: slip −1.86%, C_d 1.019, P 7.38 kW. 2023 4d: slip 8.33%, C_d 0.917. 2019 8c: N 83.2 rpm, H 117.1 m, P 130.2 kW.

Plunger: piston-type, high pressure. Diaphragm: flexible membrane, leak-free, slurries/chemicals.

Compressors and fans (A9)

Fan (pressure ratio < 1.1), blower (1.1-1.2), compressor (> 1.2). Reciprocating (high P, low Q), rotary screw/vane, centrifugal and axial (large Q). Isothermal efficiency = isothermal work/actual work; FD fan pushes air into the furnace, ID fan pulls gas out through the stack.

Compressible basics (B1)

Ma = V/c; incompressible < 0.3, subsonic < 1, sonic = 1, transonic 0.8-1.2, supersonic 1-5, hypersonic > 5.

$c = \sqrt{(K/\rho)} = \sqrt{(kRT)}$; air 300 K: 347 m/s. $T_0/T = 1 + (k - 1)Ma^2/2$; $P_0/P = (T_0/T)^{k/(k-1)}$. Adiabatic Bernoulli: $[k/(k-1)]p/\rho g + V^2/2g + z = C$.

Air critical: T^*/T_0 0.8333, P^*/P_0 0.5283, ρ^*/ρ_0 0.6339. 2021 5d: 324.8 m/s vs 291.4 m/s incompressible.

Nozzles (B2)

$dA/A = (Ma^2 - 1)dV/V$: subsonic nozzle converges, supersonic nozzle diverges, Ma = 1 only at the throat.

$\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}[2/(k + 1)]^{(k+1)/2(k-1)} = 0.6847A^*P_0/\sqrt{(RT_0)}$ air. Back pressure below P* ⇒ choked, no further rise in $\dot{m}$.

2022 5c: T* 301 K, P* 118 kPa, A^*/A_1 0.629. 2020 4c CO_2: V 164.9, ρ 13.9, A 13.1 cm^2, Ma 0.494. 2023 1d: $\dot{m}$ 2.86 kg/s, V_e 845 m/s, A_e 33.75 cm^2. 2019 5d: 6.78 and 7.11 kg/s.

Shocks (B3)

Normal shock: supersonic → subsonic, P, T, ρ, s up, P_0 down, T_0 same. Oblique: inclined, flow turned, weaker.

$Ma_2^2 = (Ma_1^2 + 5)/(7Ma_1^2 - 1)$ (air); $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$ (Rayleigh line). Shock states = Rayleigh ∩ Fanno.

Ma 2: Ma_2 0.577, P ratio 4.5, T ratio 1.6875. Ma 2.5: 0.513, 7.125, 2.1375. Ma 2.2: 0.547, 5.48, 1.857.

HX fundamentals (B4)

Recuperator, regenerator, direct contact; parallel, counter, cross flow; double pipe, shell-and-tube, plate, compact.

$1/UA = 1/h_iA_i + R_{f,i}/A_i + \ln(D_o/D_i)/2\pi kL + R_{f,o}/A_o + 1/h_oA_o$; thin clean: $U = 1/(1/h_i + 1/h_o)$. $R_f = 1/U_{dirty} - 1/U_{clean}$.

$Q = UA\Delta T_{lm}$, $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$. Counter flow: bigger LMTD, $T_{c,o}$ can exceed $T_{h,o}$.

LMTD numericals (B5)

Energy balance first, then ΔT_{lm}, $A = Q/U\Delta T_{lm}$, $L = A/\pi D$.

2020 5c: 301 kW, LMTD 92.0, L 108.5 m. 2020 6b: 28.4 m^2. 2022 7c: 15.8 m^2. 2021 7c: U 43.7, L 92.5 m. 2023 7c: L 0.968 m; R_f 9.5×10^{-4}. 2019 4c: U 37.7, L 66.6 m. 2023 8c: $T_{w,o}$ 18.5 °C, UA 785 W/K, L 5.0 m.

Effectiveness-NTU (B6)

$\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})]$; $NTU = UA/C_{min}$; $C_r = C_{min}/C_{max}$.

Parallel: $\varepsilon = [1 - e^{-NTU(1+C_r)}]/(1 + C_r)$. Counter: $\varepsilon = [1 - e^{-NTU(1-C_r)}]/[1 - C_re^{-NTU(1-C_r)}]$; $NTU = [1/(C_r - 1)]\ln[(\varepsilon - 1)/(\varepsilon C_r - 1)]$.

2021 8c: $\dot{m}_w$ 0.403 kg/s, ε 0.75, NTU 2.35, A 2.18 m^2. 2022 8d: NTU 2.50, ε 0.8325, Q 133.3 kW, $T_{c,o}$ 86.6 °C.

Definitions to know by heart

Cavitation, **priming**, **NPSH**, **specific speed** (speed of a geometrically similar pump giving 1 m^3/s against 1 m), **slip**, **manometric efficiency**, **vena contracta**, **C_d**, **Mach number**, **stagnation property**, **choking**, **shock wave**, **fouling factor**, **effectiveness**, **NTU**, **heat capacity rate**.

Sketches to practise

1 Reynolds regimes and Moody chart; 2 entrance region; 3 laminar vs turbulent profile; 4 venturimeter; 5 orifice with vena contracta; 6 Pitot tube; 7 rotameter; 8 centrifugal pump and casings; 9 velocity triangles; 10 pump curves; 11 NPSH/suction; 12 reciprocating pump and indicator diagram; 13 C-D nozzle pressure plot; 14 back-pressure effect; 15 normal shock and Rayleigh-Fanno; 16 parallel vs counter temperature profiles; 17 ε-NTU chart.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AODD	air-operated double diaphragm (pump)
ASME	American Society of Mechanical Engineers
BEP	best efficiency point
BHP	brake horsepower
C-D	converging-diverging
FD	forced draught
gpm	gallons per minute
hp	horsepower
HX	heat exchanger
ID	induced draught (or inside diameter)
kPa	kilopascal
LMTD	log mean temperature difference
MPa	megapascal
NPSH	net positive suction head
NTU	number of transfer units
rpm	revolutions per minute
TEMA	Tubular Exchanger Manufacturers Association
WHP	water horsepower
XXS	double extra strong (pipe schedule)

Section A: Fluid flow, measurement and pumps

A4 Flow measurement: venturimeter, orifice meter, nozzle meter, Pitot tube, rotameter, turbine meter

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(b)** [15] What is an orifice meter? Derive an expression for the flow rate.
- **2023 Q5(c)** [6] Relative accuracy of venturi meter, nozzle meter and orifice meter.
- **2023 Q5(d)** [10] Same venturimeter as 2022 2(d): flow rate.
- **2022 Q2(b)** [6] Commonly used flow-rate measuring instruments; differences between venturimeter and orifice meter.
- **2022 Q2(c)** [12] Derive an expression for the volumetric flow rate through a nozzle meter.
- **2022 Q2(d)** [11] Horizontal venturimeter, inlet 30 cm, throat 15 cm, differential manometer 20 cm of mercury, C_d = 0.98: rate of flow.
- **2022 Q1(c)** [10] Submarine "Titan", same set-up with 190 mm of mercury: speed.
- **2021 Q2(c)** [4] Briefly discuss vena contracta.
- **2021 Q2(b)** [10] What are hydraulic coefficients? Deduce an expression for the actual velocity at a point using a Pitot tube.
- **2020 Q1(b)** [12] Define hydraulic diameter. Deduce an expression for the actual flow rate through a venturimeter.
- **2020 Q1(c)** [6] Submarine 15 m below the sea surface; Pitot tube to a mercury U-tube shows 200 mm; S_{Hg} = 13.6, S_{sea} = 1.026: speed of the submarine.
- **2019 Q6(c)** [12] Define venturimeter. Describe it and find an expression for the discharge through a pipe.
- **2019 Q3(b)** [12] Derive an expression for the volumetric flow rate through an orifice meter.
- **2019 Q3(a)** [12] What is a Pitot tube? How is it used to measure velocity at a point in a pipe or channel?
- **2019 Q3(c)** [11] Working principle of a rotameter.
- **2019 Q6(d)** [10] Air through a venturimeter, inlet 400 mm, throat 150 mm; inlet 137 kN/m^2 abs, 15 °C; throat 127 kN/m^2 abs; R = 287, γ = 1.4: mass flow rate.

LEARN IT FROM ZERO

Flow meters, from zero

All obstruction meters (venturi, orifice, nozzle) use one idea: **squeeze the flow, it speeds up, its pressure drops** (Bernoulli). Measure the pressure drop with a manometer and you know the speed.

A differential manometer reading x of a heavy liquid (mercury, S_h) under a lighter one (S_o) is a head of flowing liquid $\mathbf{h = x(S_h/S_o - 1)}$. Then Bernoulli plus continuity give $\mathbf{Q = C_d a_1 a_2 \sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)}}$. C_d corrects for friction and the vena contracta: venturi 0.95–0.99 (smooth, little loss), nozzle about 0.95, orifice about 0.6 (cheap but big loss).

A **Pitot tube** faces the flow; the fluid stops at its tip, so the extra (stagnation) head is $V^2/2g$, giving $\mathbf{V = C_v\sqrt{(2gh)}}$.

CALCULATOR Venturimeter with a mercury manometer (2022 2(d), 2023 5(d))

Change any number; every step is recalculated and shown with the numbers put in.

1. Head difference: $h = x(S_h/S_o - 1)$ = **0.2** × (**13.6** / **1** - 1) = **2.520** m
2. a_1: π × **0.3** × **0.3** / 4 = **0.07069** m^2
3. a_2: π × **0.15** × **0.15** / 4 = **0.01767** m^2
4. Actual flow: $Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$ = **0.98** × **0.07069** × **0.01767** × √(2 × 9.81 × **2.52**) / √(**0.07069** × **0.07069** - **0.01767** × **0.01767**) = **0.1258** m^3/s
5. In litres per second: **0.1258** × 1000 = **125.8** L/s

CALCULATOR Pitot tube speed (2020 1(c), 2022 1(c))

Change any number; every step is recalculated and shown with the numbers put in.

1. Head: $h = x(S_h/S_o - 1)$ = **0.2** × (**13.6** / **1.026** - 1) = **2.451** m
2. Speed: $V = C_v\sqrt{(2gh)}$ = **1** × √(2 × 9.81 × **2.451**) = **6.93** m/s
3. Speed: **6.935** × 3.6 = **24.96** km/h

The depth (15 m) does not matter: both manometer limbs feel the same sea pressure.

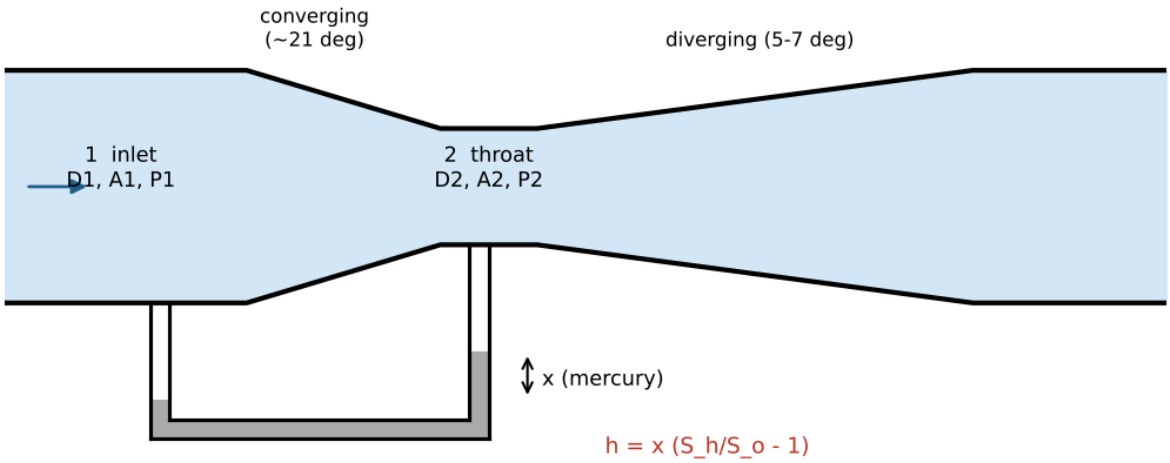
Before you start: how flow meters work

Put an obstruction in the pipe: the area falls, so by continuity (Q = AV) the velocity rises, and by Bernoulli the pressure falls. A manometer reads the pressure drop; from it we get the velocity and the flow rate. Real fluids lose some energy, so the ideal flow is multiplied by a coefficient of discharge $C_d < 1$.

Differential U-tube manometer with a heavier liquid (mercury, $S_h = 13.6$) under a flowing liquid S_o: pressure head difference $\mathbf{h = x(S_h/S_o - 1)}$ metres of the flowing liquid, x = difference in mercury levels.

Common flow-rate measuring instruments (2022 2b)

- Obstruction (differential-pressure) meters: venturimeter, orifice meter, flow nozzle
- Velocity probes: Pitot tube, Pitot-static tube (point velocity), hot-wire anemometer
- Variable area meter: rotameter
- Turbine and paddle-wheel meters; positive-displacement (nutating disc) meters
- Electromagnetic, ultrasonic (Doppler/transit-time), vortex-shedding and Coriolis (mass) meters



Venturimeter with a differential manometer

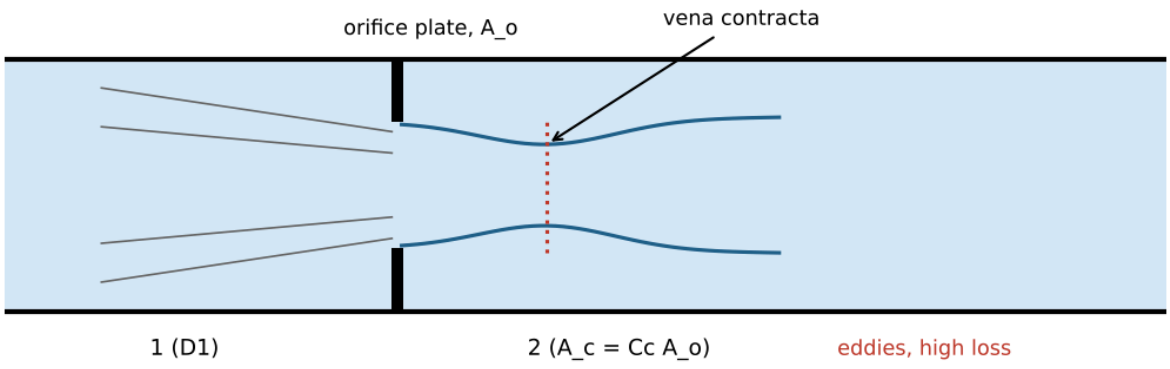
Venturimeter: description and discharge (2020 1b, 2019 6c)

A **venturimeter** measures the discharge through a pipe. It has a short **converging** cone (about 20°), a cylindrical **throat** (d/D = 0.4-0.7) and a long **diverging** cone (5-7°, long so that the flow does not separate and most of the pressure is recovered). Pressure taps at the inlet (1) and throat (2) go to a differential manometer. (Hydraulic diameter $D_h = 4A/p$: see A1.)

Step (write this)	Why / what it means
Bernoulli 1 → 2 (horizontal, $z_1 = z_2$): $p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g$	Ideal flow.
$h = (p_1 - p_2)/\rho g = (V_2^2 - V_1^2)/2g$	h = venturi head, from the manometer.
Continuity: $A_1V_1 = A_2V_2 \Rightarrow V_1 = V_2A_2/A_1$	
$2gh = V_2^2(1 - A_2^2/A_1^2) \Rightarrow V_2 = A_1\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$	
$Q_{th} = A_2V_2 = \mathbf{A_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}}$	Theoretical discharge.
$\mathbf{Q_{act} = C_dA_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}}$, $C_d = 0.96\text{-}0.99$	Friction makes the actual flow a little smaller.

Nozzle meter (2022 2c)

A flow nozzle is a smooth converging nozzle with a short cylindrical throat placed in the pipe; it has no long diverging cone (cheaper and shorter than a venturi, used for high-pressure steam). Bernoulli between the pipe (1) and the nozzle throat (2) and continuity give exactly the venturi result: $\mathbf{Q = C_dA_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}}$ with C_d = 0.95-0.98 (write the same six steps as above with A_2 = nozzle throat area).



Orifice meter and the vena contracta

Orifice meter (2019 3b, 2023 5b)

An **orifice meter** is a thin plate with a sharp-edged circular hole (diameter about 0.5D) clamped between pipe flanges. Taps: one about 1.5-2D upstream (section 1), one at the vena contracta (section 2, about D/2 downstream). Cheap and compact, but the sudden expansion after the jet wastes much energy.

Step (write this)	Why / what it means
Bernoulli 1 → 2: $(p_1 - p_2)/\rho g + (z_1 - z_2) = h = (V_2^2 - V_1^2)/2g \Rightarrow V_2 = \sqrt{(2gh + V_1^2)}$	Section 2 is at the vena contracta.
Jet area $A_2 = C_c A_o$; continuity $A_1V_1 = A_2V_2 \Rightarrow V_1 = (A_oC_c/A_1)V_2$	C_c = coefficient of contraction.
$V_2^2[1 - (A_o/A_1)^2C_c^2] = 2gh \Rightarrow V_2 = \sqrt{(2gh)}/\sqrt{(1 - (A_o/A_1)^2C_c^2)}$	
$Q = A_2V_2 = A_oC_c\sqrt{(2gh)}/\sqrt{(1 - (A_o/A_1)^2C_c^2)}$	Contains the unknown C_c.
Define $C_d = C_c\sqrt{(1 - (A_o/A_1)^2)}/\sqrt{(1 - (A_o/A_1)^2C_c^2)}$	Lumps contraction and friction (include C_v for losses).
$Q = C_dA_oA_1\sqrt{(2gh)}/\sqrt{(A_1^2 - A_o^2)}$, $C_d \approx 0.61$-0.65	Same form as the venturi with A_o in place of the throat area.

Vena contracta (2021 2c)

The **vena contracta** is the section of a jet just downstream of an orifice (about $0.5D_o$ from the plate) where the jet area is the smallest, the velocity the largest and the pressure the lowest. It forms because the streamlines converge towards the hole and, by inertia, keep converging for a short distance after it instead of turning sharply. $C_c = A_c/A_o \approx 0.61$-$0.69$.

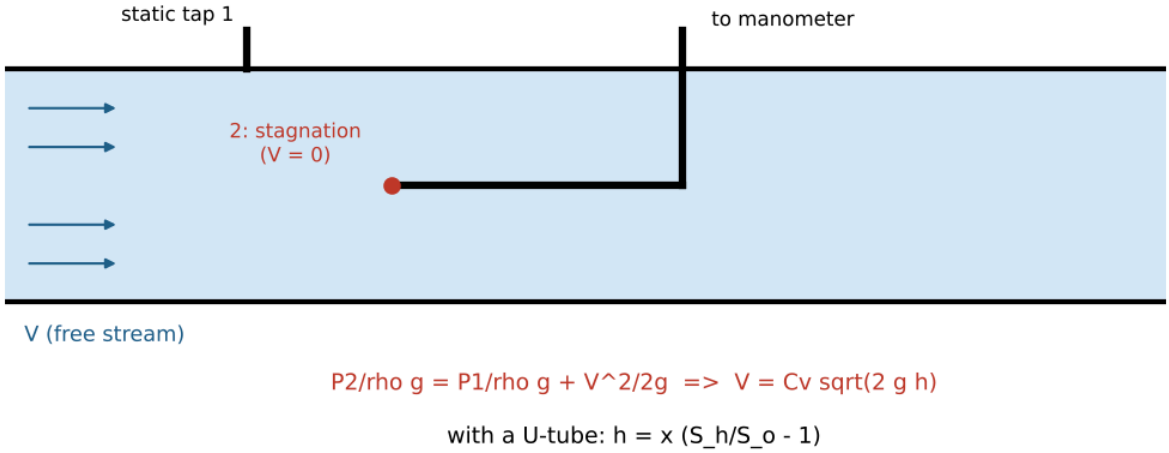
Venturimeter vs orifice meter (2022 2b) and relative accuracy (2023 5c)

Feature	Venturimeter	Nozzle meter	Orifice meter
Construction	converging cone, throat, long diverging cone	short smooth converging nozzle	thin plate with a sharp-edged hole
C_d	0.96-0.99	0.95-0.98	0.61-0.65
Permanent pressure loss	low (10-15% of Δp)	moderate (40-60%)	high (60-80%)
Accuracy	highest (±0.5-1%), C_d well known and steady	good (±1-2%)	lowest (±1-3%); edge wear and deposits change C_d
Space and cost	long, costly	medium	compact, cheapest, easy to change
Use	large water and process lines where pumping cost matters	high-pressure steam lines	general/lab use, gas lines, where loss is acceptable

Relative accuracy: venturi > nozzle > orifice. The smooth venturi has almost no contraction or separation, so its C_d is close to 1 and changes little with Re; the orifice has a strong vena contracta and eddies, so its C_d is low and depends on Re, edge sharpness and tap position.

Hydraulic coefficients (2021 2b)

Coefficient	Definition	Typical value
Coefficient of contraction C_c	area of jet at vena contracta / area of orifice	0.61-0.69
Coefficient of velocity C_v	actual velocity at vena contracta / theoretical velocity √(2gH)	0.95-0.99
Coefficient of discharge C_d	actual discharge / theoretical discharge = $C_c \times C_v$	0.61-0.65
Coefficient of resistance C_r	head lost in the orifice / head remaining in the jet = $1/C_v^2 - 1$	small



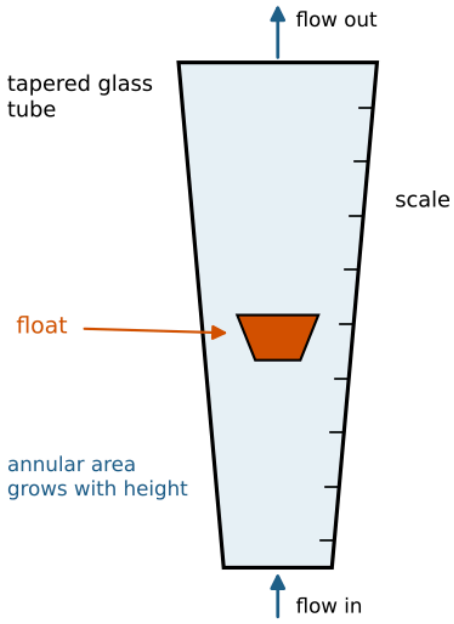
Pitot tube

Pitot tube: velocity at a point (2021 2b, 2019 3a)

A **Pitot tube** is a small bent tube with its open end facing the flow. At the tip the fluid is brought to rest (stagnation point), so the kinetic energy changes into pressure energy and the liquid in the tube rises above the free surface by h. Measuring h gives the velocity at that point. Arrangements: Pitot tube with a vertical piezometer; Pitot-static tube (two concentric tubes: stagnation hole at the nose

and static holes on the side) connected to a differential U-tube manometer, used in pipes and on aircraft.

Step (write this)	Why / what it means
Point 1 upstream (velocity V_1, pressure head H below the surface); point 2 at the tip ($V_2 = 0$, pressure head H + h)	Same streamline, same level.
$p_1/\rho g + V_1^2/2g = p_2/\rho g \Rightarrow H + V_1^2/2g = H + h$	Bernoulli, $z_1 = z_2$.
$V_1 = \sqrt{(2gh)}$ (theoretical)	Dynamic head $h = V^2/2g$.
$\mathbf{V_{act} = C_v\sqrt{(2gh)}}$, $C_v \approx 0.98$	Coefficient of the Pitot tube; with a U-tube $h = x(S_h/S_o - 1)$.



Rotameter

Rotameter (2019 3c)

A **rotameter** is a variable-area flow meter: a vertical tapered transparent tube (wider at the top) containing a float. The fluid enters at the bottom and flows up through the annular gap between float and tube. The float rises until the upward forces (drag of the fluid + buoyancy) balance its weight. As it rises the annular area grows, so the velocity and drag in the gap fall; for every flow rate there is one balance height. The flow rate is read directly from the scale on the tube at the top edge of the float. Simple, direct reading, low pressure loss, good for small gas and liquid flows (laboratories, oxygen and anaesthesia lines); must be vertical, glass is fragile, accuracy only ±2-5%.

Numerical 1: venturimeter 30 cm × 15 cm, 20 cm of mercury (2022 2d, 2023 5d)

Step (write this)	Why / what it means
$A_1 = (\pi/4)(0.30)^2 = 0.07069\ m^2$; $A_2 = (\pi/4)(0.15)^2 = 0.01767\ m^2$	
$h = x(S_h/S_o - 1) = 0.20(13.6/1 - 1) = 2.52$ m of water	Mercury under water.

$Q = C_d A_1 A_2 \sqrt{(2gh)} / \sqrt{(A_1^2 - A_2^2)} = 0.98 \times 0.07069 \times 0.01767 \times \sqrt{(2 \times 9.81 \times 2.52)} / \sqrt{(0.07069^2 - 0.01767^2)}$	
$Q = 0.1258\ m^3/s = 125.8\ L/s$	Bansal Problem 6.10 (125.76 L/s).

Numerical 2: speed of a submarine by Pitot tube (2020 1c: 200 mm; 2022 1c "Titan": 190 mm)

Step (write this)	Why / what it means
$h = x(S_{Hg}/S_{sea} - 1) = 0.200(13.6/1.026 - 1) = 2.451$ m of sea water	The 15 m depth cancels: it adds equally to both limbs.
$V = \sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 2.451)} =$ **6.93 m/s = 24.96 km/h**	2020 answer (take $C_v = 1$).
2022: $h = 0.190(13.6/1.026 - 1) = 2.328$ m; $V = \sqrt{(2 \times 9.81 \times 2.328)} =$ **6.76 m/s = 24.3 km/h**	Relative velocity of water past the submarine = speed of the submarine.

Numerical 3: mass flow of air through a venturimeter (2019 6d)

Given: $D_1 = 0.40$ m, $D_2 = 0.15$ m, $p_1 = 137$ kPa, $T_1 = 288.15$ K, $p_2 = 127$ kPa, $R = 287$, $\gamma = 1.4$. Air is compressible, so use the adiabatic (isentropic) energy equation instead of Bernoulli.

Step (write this)	Why / what it means
$\rho_1 = p_1/RT_1 = 137{,}000/(287 \times 288.15) = 1.657\ kg/m^3$	Ideal gas.
$\rho_2 = \rho_1(p_2/p_1)^{1/\gamma} = 1.657(127/137)^{0.714} = 1.569\ kg/m^3$; $T_2 = 282.0$ K	p/ρ^γ = constant.
Energy: $V_2^2/2 - V_1^2/2 = [\gamma/(\gamma - 1)](p_1/\rho_1 - p_2/\rho_2) = 3.5(82{,}699 - 80{,}927) = 6{,}201$ J/kg	Adiabatic Bernoulli (see B1).
Continuity: $\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \Rightarrow V_1 = (1.569 \times 0.01767)/(1.657 \times 0.1257)\ V_2 = 0.1332 V_2$	$A_1 = 0.1257$, $A_2 = 0.01767\ m^2$.
$V_2^2(1 - 0.1332^2) = 2 \times 6{,}201 \Rightarrow V_2 = 112.4$ m/s	
$\dot{m} = \rho_2 A_2 V_2 = 1.569 \times 0.01767 \times 112.4 =$ **3.12 kg/s**	Treating the air as incompressible would give 3.25 kg/s, 4% high.

Memory trick: "V-N-O": Venturi, Nozzle, Orifice in falling order of C_d (0.98, 0.96, 0.62), accuracy and cost, and rising order of pressure loss.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 2(d) Given: $d_1 = 30$ cm, $d_2 = 15$ cm, $x = 20$ cm of mercury, $C_d = 0.98$

Head: $h = x(S_h/S_o - 1)$

$= 0.20(13.6/1 - 1)$

$=$ **2.52 m of water**

Areas: $a = \pi d^2/4$

$= a_1 = 0.070686$, $a_2 = 0.017671$

$=$ **m^2**

Flow: $Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$

$= 0.98 \times 0.070686 \times 0.017671 \times \sqrt{(2 \times 9.81 \times 2.52)}/\sqrt{(0.070686^2 - 0.017671^2)}$

$=$ **$0.1258\ m^3/s = 125.8\ L/s$**

A6 Pump performance: characteristic curves, affinity laws, specific speed, cavitation and NPSH, priming, suction height Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q3(a)** [8] What are the affinity laws? Illustrate and explain the characteristic curve of a centrifugal pump.
- **2023 Q4(a)** [8] Explain cavitation in pump operation and how NPSH (net positive suction head) is related to cavitation.
- **2022 Q4(b)** [8] Significance of the characteristic curves of a centrifugal pump; main and operating characteristic curves.
- **2022 Q4(a)** [12] What is specific speed? Show that it depends on the manometric head and the discharge.
- **2022 Q4(c)** [7] Define cavitation. Why and where does it occur? Significance of NPSH in the pump industry.
- **2021 Q4(a)** [7] What is priming? Why is it necessary?
- **2021 Q4(b)** [10] Derive an expression for the maximum suction height of a centrifugal pump.
- **2020 Q3(a)** [7] Differences between fans, compressors and blowers. Draw the main and operating characteristic curves of a centrifugal pump.
- **2020 Q3(b)** [10] Define specific speed of a centrifugal pump. Deduce the expression for it.
- **2020 Q3(c)** [5] What is NPSH? Its significance.
- **2020 Q2(c)** [12] What will happen if the impeller of a centrifugal pump is made to rotate in the presence of air?
- **2019 Q7(b)** [10] Discuss pump affinity laws. When does an engineer use them?
- **2019 Q8(a)** [8] Define cavitation. Effects of cavitation and precautions against it.

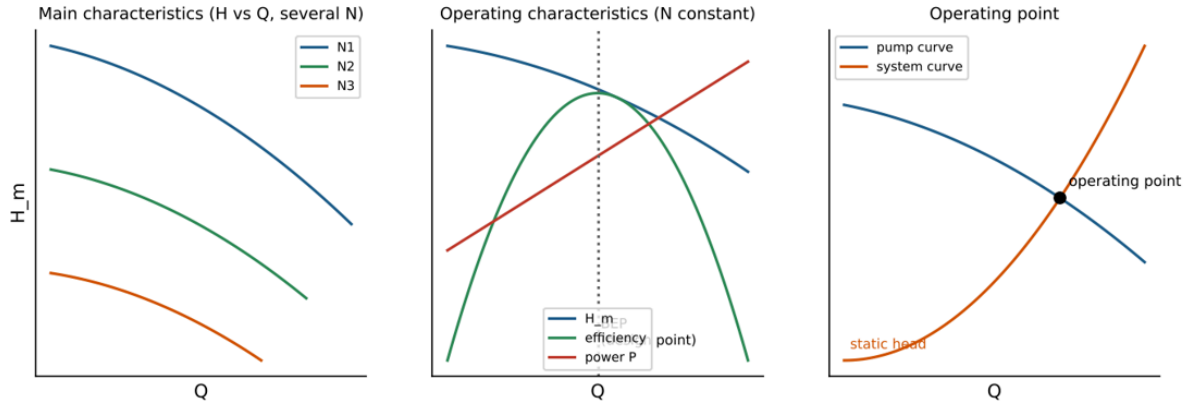
★ REMEMBER IT

- **Affinity laws** (same pump, speed changes): **$Q \propto N$, $H \propto N^2$, $P \propto N^3$** (with diameter: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$). Remember "1, 2, 3".
- Specific speed $N_s = N\sqrt{Q}/H^{3/4}$: speed of a geometrically similar pump that delivers 1 m^3/s against 1 m. Low N_s = radial (high head), high N_s = axial (high flow).
- **Cavitation**: suction pressure falls below vapour pressure → bubbles form → they collapse in the high-pressure zone → pitting, noise, vibration, lower head. Avoid it by keeping $NPSH_{available} > NPSH_{required}$.
- **Priming**: fill the suction pipe and casing with liquid first. Air is ~800 times lighter, so the impeller would make only a tiny head in air and could not lift water.

Before you start: reading a pump's behaviour

A pump never runs at just one point. When the valve opens wider, Q rises and the head the pump can give falls. The manufacturer's test results, plotted as curves, show how head, power and efficiency change with flow and speed. Two tools let us predict new conditions without new tests: the **affinity laws** (same pump, new speed or diameter) and the **specific speed** (which type of pump suits a duty).

Pumps can only push liquid, never vapour. If the pressure at the impeller eye falls below the vapour pressure, the liquid boils locally: that is **cavitation**, and NPSH is the margin that prevents it.

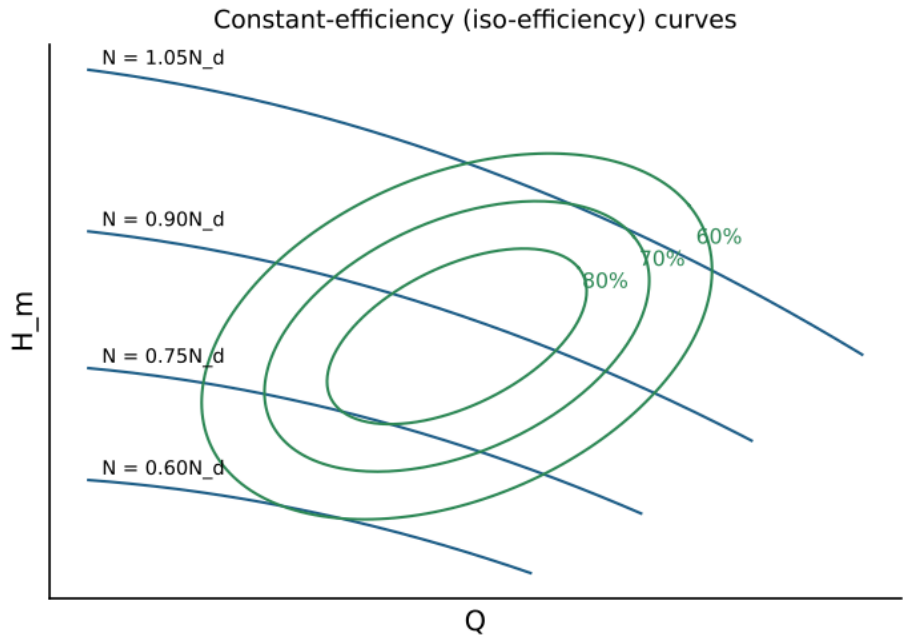


Main characteristic curves, operating characteristic curves and the operating point

Characteristic curves and their significance (2020 3a, 2022 4b, 2023 3a)

Significance: characteristic curves show the actual behaviour of a pump found by tests: the head, power and efficiency it gives at each discharge and speed. They are used to choose a pump for a duty, to find its **best efficiency point** (BEP), to predict how it will behave at other speeds, and to find the operating point with a given pipe system.

- **Main characteristic curves:** the speed N is kept constant at several values (N_1, N_2, N_3), and H_m, shaft power P and efficiency are plotted against Q for each speed (by throttling the delivery valve). H_m falls as Q rises; higher N gives higher curves.
- **Operating characteristic curves:** at the design speed only, H_m, P and η are plotted against Q on one graph. η rises to a maximum at the design point (BEP) and then falls; P rises with Q (it is not zero at Q = 0 because of losses); H_m is highest at shut-off (Q = 0).
- **Constant-efficiency (iso-efficiency, Muschel) curves:** from the main curves, points of equal η are joined on the H-Q plane; they show the range of speed and flow for good efficiency.
- **Operating point:** the system curve $H = H_{static} + KQ^2$ (pipe losses) crosses the pump H-Q curve at the point where the pump will run.



Iso-efficiency curves

Fans, compressors and blowers (2020 3a): see the table in A9: fan specific ratio up to 1.11, blower 1.11-1.20, compressor above 1.20.

Affinity laws (2023 3a, 2019 7b)

For geometrically similar pumps (or the same pump) running at similar operating points (same efficiency), the flow coefficient Q/ND^3, head coefficient gH/N^2D^2 and power coefficient $P/\rho N^3D^5$ stay constant. Hence:

Change	Flow	Head	Power
speed N (same D)	$Q_2/Q_1 = N_2/N_1$	$H_2/H_1 = (N_2/N_1)^2$	$P_2/P_1 = (N_2/N_1)^3$
impeller diameter D (same N)	$\propto D$ (trimmed impeller) or D^3 (similar pumps)	$\propto D^2$	$\propto D^3$ (trimmed) or D^5

When engineers use them: to predict the new flow, head and power when the motor speed is changed (variable-speed drives); to find the speed needed when the system head changes (new heat exchanger, longer pipe); to trim an impeller to a new duty; to scale a model test to a prototype; and to draw new H-Q curves at other speeds. Doubling the speed doubles Q, quadruples H and needs 8 times the power.

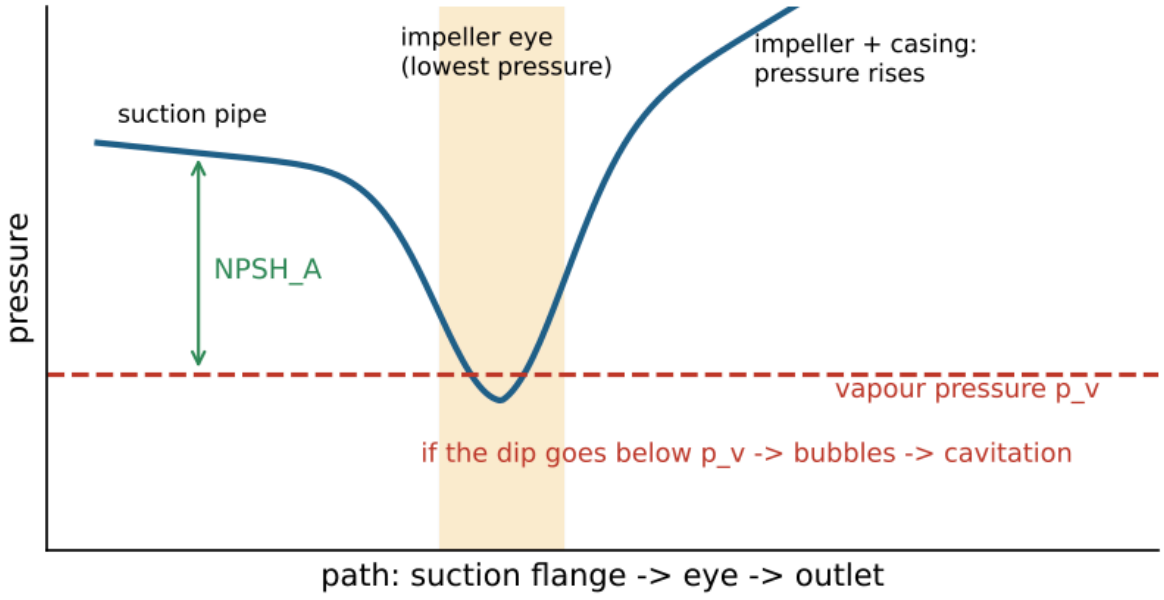
Specific speed: definition and derivation (2020 3b, 2022 4a)

Specific speed N_s of a centrifugal pump is the speed of a geometrically similar pump that would deliver **1 m^3/s against a head of 1 m**. It has no direct physical meaning but is the same for all similar pumps, so it tells the type of impeller: radial (low N_s, 10-80), mixed flow (80-160), axial (above 160, SI units with Q in m^3/s).

Step (write this)	Why / what it means
`Q = (area) × (velocity of flow) =` $\pi DB\ V_f$	D = impeller diameter, B = width.

$B \propto D$ (similar pumps) $\Rightarrow Q \propto D^2 V_f$	
Tangential velocity $u = \pi DN/60 \propto DN$, and $u \propto V_f \propto \sqrt{H}$ ($u = K_u\sqrt{(2gH)}$)	Velocities scale with √(head).
$\Rightarrow D \propto \sqrt{H}/N$ and $V_f \propto \sqrt{H}$	
$Q \propto (\sqrt{H}/N)^2\sqrt{H} = H^{3/2}/N^2 \Rightarrow Q = K\, H^{3/2}/N^2$	K = constant for similar pumps.
Definition: $H = 1$ m and $Q = 1$ m^3/s when $N = N_s$ $\Rightarrow 1 = K/N_s^2$, $K = N_s^2$	
$Q = N_s^2 H^{3/2}/N^2 \Rightarrow \mathbf{N_s = N\sqrt{Q}/H_m^{3/4}}$	Q at best efficiency; H_m per stage.

Dependence on H_m and Q (2022 4a): the result shows that, for a given speed, N_s rises as √Q and falls as $H_m^{3/4}$: high-head, low-flow duties need low-N_s radial pumps; low-head, high-flow duties need high-N_s axial (propeller) pumps.



Pressure along the pump and NPSH

Cavitation and NPSH (2020 3c, 2022 4c, 2023 4a, 2019 8a)

Cavitation is the formation of vapour bubbles in a flowing liquid where the local absolute pressure falls below the vapour pressure at that temperature, followed by the sudden collapse of these bubbles when they are carried into a region of higher pressure. **Where:** at the eye and inlet of the vanes of a centrifugal pump (lowest pressure), at the outlet of reaction turbines (draft tube), on ship propellers and in valves. **Why:** high suction lift, high liquid temperature (higher p_v), high speed or flow beyond the BEP (best efficiency point), long or narrow suction pipe (friction), low atmospheric pressure at altitude, clogged strainer.

Effects: collapsing bubbles produce very high local pressures (up to thousands of bar) that hammer the metal: **pitting and erosion** of impeller vanes, noise and vibration, and a sudden fall in head, discharge and efficiency.

Precautions: keep the pressure everywhere above p_v, i.e. $NPSH_A > NPSH_R$: low suction lift (place the pump near or below the liquid level), short, large-diameter suction pipe with few bends, avoid

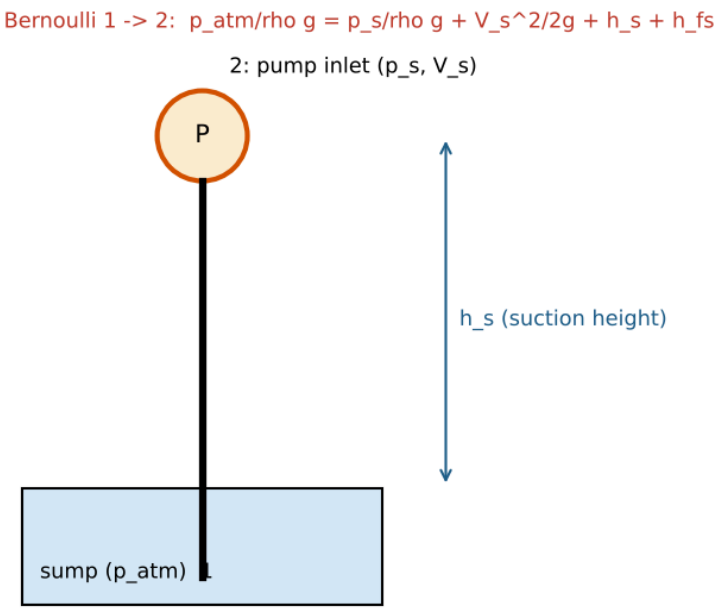
hot liquid, do not exceed the design speed/flow, keep strainers clean, use cavitation-resistant materials (stainless steel, aluminium bronze).

NPSH (net positive suction head) is the total head at the pump suction above the vapour pressure head: **$NPSH = p_s/\rho g + V_s^2/2g - p_v/\rho g = p_{atm}/\rho g - p_v/\rho g - h_s - h_{fs}$**. **$NPSH_A$** (available) is fixed by the system; **$NPSH_R$** (required) is given by the manufacturer for the pump at each flow. **Significance:** it is the criterion for cavitation-free operation, used to fix the pump elevation and the suction pipe; the pump industry specifies $NPSH_R$ on every pump curve, and a margin $NPSH_A \geq NPSH_R$ + 0.5-1 m is kept. Thoma's cavitation factor $\sigma = NPSH/H_m$; cavitation starts when $\sigma < \sigma_c$.

Priming (2021 4a), and running the impeller in air (2020 2c)

Priming is the filling of the suction pipe, the casing and the part of the delivery pipe up to the delivery valve with the liquid to be pumped, so that all air (or vapour) is removed before starting the pump. Methods: filling from an overhead tank or by hand through a funnel (foot valve holds it), using a vacuum pump, or self-priming designs.

Why it is necessary (and what happens in air): the head developed by the impeller, $H = V_{w2}u_2/g$, depends only on the velocities, not on the fluid. The **pressure** it produces is ρgH, which is proportional to the density of the fluid in the impeller. Air is about 800 times lighter than water (1.2 vs 1000 kg/m^3), so an impeller running in air produces only about 1/800 of the pressure: e.g. a pump designed for 40 m of water makes only 40/800 = 0.05 m (5 cm) of water pressure difference. That cannot lift water up the suction pipe (needs several metres), so no liquid is delivered: the pump simply churns the air. Running dry also overheats and damages the seals, wearing rings and bearings because there is no liquid to cool and lubricate them. Hence the pump must be primed.



Suction side of a centrifugal pump

Maximum suction height (2021 4b)

Step (write this)	Why / what it means
`Bernoulli from the sump surface (1) to the pump inlet (2), sump as datum:` $p_{atm}/\rho g + 0 + 0 = p_s/$	h_s = suction height, h_{fs} = friction loss in the suction pipe.

$\rho g + V_s^2/2g + h_s + h_{fs}$	
$p_s/\rho g = p_{atm}/\rho g - (h_s + h_{fs} + V_s^2/2g)$	Suction pressure falls as h_s rises.
No cavitation: $p_s/\rho g \geq p_v/\rho g$ (strictly + $NPSH_R$)	Limit.
$h_{s,max} = (p_{atm} - p_v)/\rho g - h_{fs} - V_s^2/2g$ (- $NPSH_R$)	Maximum suction height.
For cold water at sea level: 10.3 - 0.24 = about 10 m in theory; in practice 6-7 m	Never above 10.3 m.

Memory trick: Affinity laws "1-2-3": $Q \propto N^1$, $H \propto N^2$, $P \propto N^3$. Specific speed: "N root Q over H to the three-quarters".

A8 Reciprocating and other positive-displacement pumps: working, slip, classification, plunger vs diaphragm, numericals Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(c)** [10] Define slip and percentage slip. Using diagrams describe the working of a reciprocating pump.
- **2023 Q4(b)** [8] Advantages and disadvantages of plunger pumps and diaphragm pumps: applications and performance.
- **2023 Q4(d)** [9] Single-acting pump, 50 rpm, stroke 40 cm, piston 25 cm, discharges 0.015 m^3/s: theoretical discharge, slip and % slip, C_d, power.
- **2022 Q3(a)** [10] Define hydraulic machines. Explain the working principle of a reciprocating pump.
- **2021 Q4(c)** [8] Classification of reciprocating pumps. Differences between centrifugal and reciprocating pumps.
- **2021 Q4(d)** [10] Double-acting pump at 40 rpm discharges 2 m^3/min; stroke 500 mm, piston 250 mm; delivery 18 m, suction 5 m: slip and power.
- **2020 Q2(a)** [6] What is the reason for negative slip in a reciprocating pump?
- **2020 Q2(b)** [12] Single-acting pump, plunger 250 mm, stroke 350 mm, 60 rpm, delivers 16.5 L/s against suction 5 m and delivery 20 m: theoretical discharge, C_d, slip, % slip, power.
- **2019 Q8(b)** [12] Define slip and percentage slip of a reciprocating pump. Describe its working with neat sketches.
- **2019 Q8(c)** [10] Three-throw pump delivers 0.1 m^3/s against 100 m; cylinder 25 cm × 50 cm stroke; friction 1 m suction, 16 m delivery; delivery velocity 1.4 m/s; η 90%, slip 2%: speed and power.

UNDERSTAND **Read this to understand; you do not need to write it**

A reciprocating pump is a syringe on a crank: the piston pulls liquid in through the suction valve, then pushes it out through the delivery valve. Theoretical flow is the swept volume times strokes: $\mathbf{Q_{th} = ALN/60}$ (double-acting × 2, three-throw × 3). Real flow is a little less because valves leak and close late: $\mathbf{slip = Q_{th} - Q_{act}}$. If a long delivery pipe's momentum keeps the delivery valve open, $Q_{act} > Q_{th}$: **negative slip**.

CALCULATOR **Reciprocating pump: discharge, slip, C_d, power (2020 2(b), 2021 4(d), 2023 4(d))**

Change any number; every step is recalculated and shown with the numbers put in.

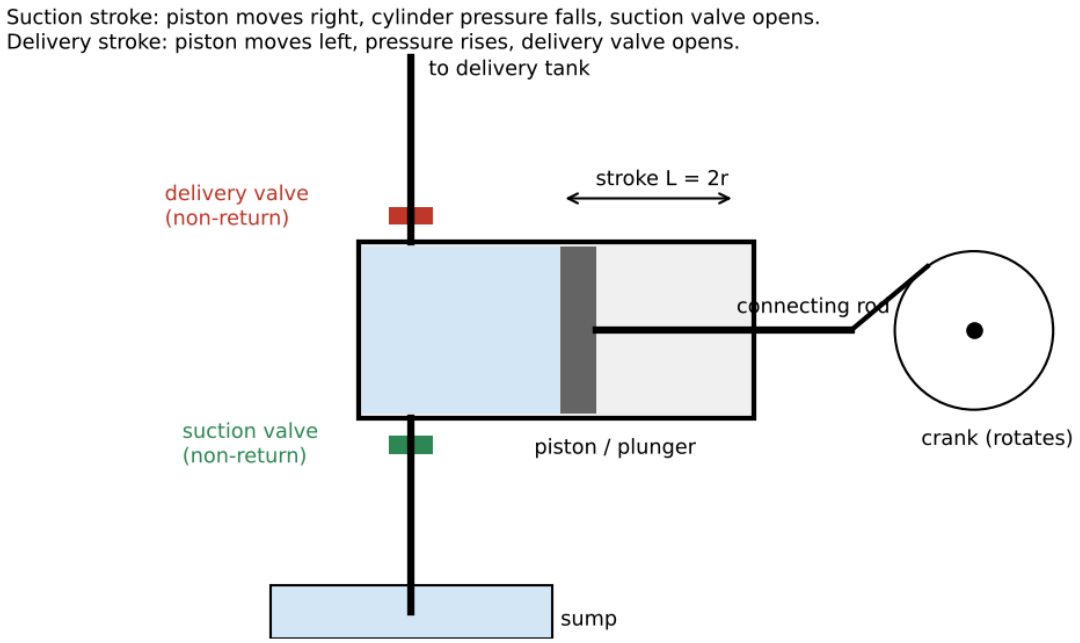
1. Piston area: π × **0.25** × **0.25** / 4 = **0.04909** m^2
2. Theoretical discharge: Q_{th} = kALN/60 = **1** × **0.04909** × **0.4** × **50** / 60 = **0.01636** m^3/s
3. Slip: **0.01636** - **0.015** = **0.00136** m^3/s
4. Percentage slip: 100 × (**0.01636** - **0.015**) / **0.01636** = **8.33** %
5. C_d: **0.015** / **0.01636** = **0.917**
6. Theoretical power: $P = \rho g Q_{th}(H_s + H_d)$ = 1000 × 9.81 × **0.01636** × (**0** + **15**) / 1000 = **2.41** kW

Enter the heads printed in your paper; power with the actual Q is $\rho g Q_{act}(H_s + H_d)$.

Before you start: positive-displacement pumps

A reciprocating pump traps a fixed volume of liquid in a cylinder and pushes it out with a piston, like a syringe. Each stroke delivers a known volume, so the theoretical discharge is just (swept volume) × (strokes per second). It can make very high pressure but gives a small, pulsating flow.

Some liquid leaks back through the valves before they close, so the actual discharge is a little less: the difference is **slip**.



Single-acting reciprocating pump

Working principle of a reciprocating pump (2022 3a, 2023 4c, 2019 8b)

Hydraulic machines: see A5. A **reciprocating pump** is a positive-displacement pump in which a piston or plunger moves to and fro in a close-fitting cylinder. **Main parts:** cylinder with piston, piston rod, connecting rod and crank (driven by a motor), suction pipe with suction (non-return) valve, delivery pipe with delivery (non-return) valve, strainer.

1. **Suction stroke:** the crank turns from inner dead centre (θ = 0) to outer dead centre (θ = 180°); the piston moves away from the valves, the volume increases and a partial vacuum forms in the cylinder. Atmospheric pressure on the sump pushes liquid up the suction pipe, opens the suction valve and fills the cylinder; the delivery valve stays closed.
2. **Delivery stroke:** the crank turns from 180° to 360°; the piston moves back towards the valves and the pressure rises. The suction valve closes and the delivery valve opens; the liquid is forced up the delivery pipe.
3. The cycle repeats, so the delivery is intermittent (pulsating). Air vessels on the suction and delivery pipes smooth the flow and save power.

Slip, percentage slip and negative slip (2023 4c, 2019 8b, 2020 2a)

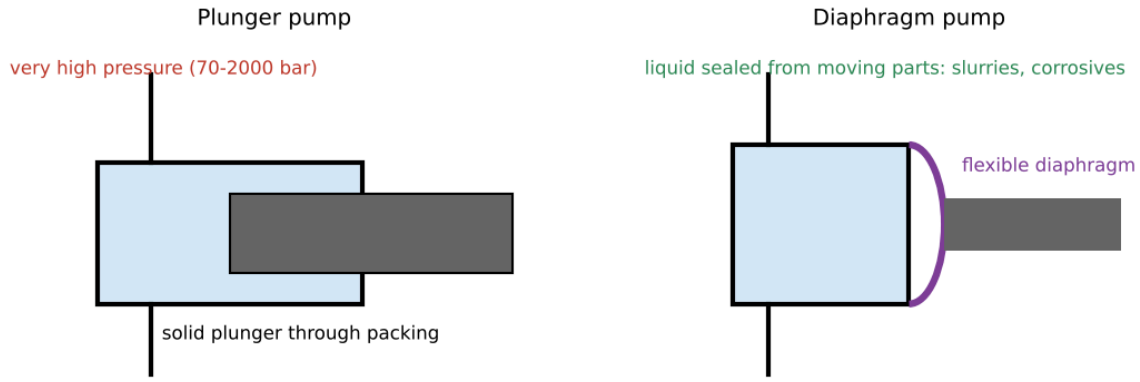
Q_{th} = ALN/60 (single acting) Q_{th} = 2ALN/60 (double acting) Slip = $Q_{th} - Q_{act}$ % slip = $(Q_{th} - Q_{act})/Q_{th} \times 100 = (1 - C_d) \times 100$

Slip is the difference between the theoretical and actual discharge, caused by leakage past the piston and valves and by the delay in closing the valves; $C_d = Q_{act}/Q_{th}$ (volumetric efficiency, 0.95-0.98); slip is about 2% in a good pump. **Negative slip** occurs when $Q_{act} > Q_{th}$ ($C_d > 1$). **Reason:** with a long suction pipe, short delivery pipe and high speed, the inertia (momentum) of the long column of liquid in the suction pipe is large compared with the pressure on the delivery side; near the end of the suction stroke the moving liquid keeps flowing and pushes open the delivery valve before the delivery stroke starts. Some liquid goes straight to the delivery pipe, so more is delivered than the swept volume.

Classification; centrifugal vs reciprocating (2021 4c)

- According to the sides of the piston in contact with water: **single-acting** (one side, one suction and one delivery pipe) and **double-acting** (both sides, two suction and two delivery valves; twice the discharge).
- According to the number of cylinders: single-cylinder, double-cylinder (two-throw) and triple-cylinder (three-throw, cranks 120° apart, smoother flow).
- According to the element: piston pump, plunger pump, diaphragm pump; with or without air vessels.

Centrifugal pump	Reciprocating pump
rotodynamic: energy from a rotating impeller	positive displacement: piston pushes trapped liquid
continuous, steady flow	intermittent, pulsating flow
large discharge, small to medium head	small discharge, high head
handles viscous, dirty liquids, slurries	clean, low-viscosity liquids only (valves clog)
high speed, direct coupled to a motor; compact, low cost	low speed, belt or gear drive; needs 6-7 times the floor area, high cost
needs priming	self-priming
simple, few parts, low maintenance; uniform torque	complex (valves, crank); high maintenance; uneven torque
efficiency high at design point	efficiency can be low at low head (about 40%)



Plunger and diaphragm pumps

Plunger pumps vs diaphragm pumps (2023 4b)

	Plunger pump	Diaphragm pump
Principle	solid plunger moves through a stationary packing/seal in the cylinder	flexible diaphragm (rubber, thermoplastic, PTFE) flexed by a rod, hydraulic oil or air (AODD) changes the chamber volume
Advantages	very high pressures (70-2000 bar); pressure can change without changing the flow; high efficiency; accurate metering	liquid never touches moving parts or seals, so no leakage; handles slurries (up to 75% solids), abrasive, corrosive and viscous liquids; self-priming, can run dry; good suction lift
Disadvantages	pulsating flow; packing wears and leaks; not for abrasive or very acidic liquids; heavy, high maintenance	limited pressure and flow; diaphragm fatigues and must be replaced; pulsating; air-operated types use much compressed air
Applications	high-pressure cleaning, oil-well service, hydraulic testing, boiler feed, chemical injection, sludge	chemical dosing, paints, mine and site dewatering, food and pharma transfer, slurries, fuel pumps

Numerical 1: single-acting pump (2020 2b)

Given: D = 0.25 m, L = 0.35 m, N = 60 rpm, Q_{act} = 0.0165 m^3/s, h_s = 5 m, h_d = 20 m.

Step (write this)	Why / what it means
$A = (\pi/4)(0.25)^2 = 0.04909\ m^2$	
$Q_{th} = ALN/60 = 0.04909 \times 0.35 \times 60/60 =$ **$0.01718\ m^3/s = 17.18\ L/s$**	Theoretical discharge.
$C_d = Q_{act}/Q_{th} =$ **0.960**	
$Slip = 0.01718 - 0.0165 =$ **$0.00068\ m^3/s$ (0.68 L/s)**; % slip = $(1 - 0.960) \times 100 =$ **3.96%**	
$Power = \rho g Q_{th}(h_s + h_d) = 1000 \times 9.81 \times 0.01718 \times 25 =$ **4.21 kW**	Theoretical power (Bansal uses Q_{th}).

Numerical 2: double-acting pump, negative slip (2021 4d)

Step (write this)	Why / what it means
$Q_{th} = 2ALN/60 = 2 \times 0.04909 \times 0.5 \times 40/60 = 0.03272\ m^3/s$	Piston rod neglected.
$Q_{act} = 2/60 = 0.03333\ m^3/s$	
$Slip = 0.03272 - 0.03333 =$ **$-0.00061\ m^3/s$**; % slip = -1.86%; $C_d = 1.019$	Negative slip: actual > theoretical.
$Power = \rho g Q_{th}(h_s + h_d) = 9810 \times 0.03272 \times 23 =$ **7.38 kW**	Bansal Problem 20.3.

Numerical 3: single-acting pump (2023 4d)

Step (write this)	Why / what it means
$Q_{th} = 0.04909 \times 0.40 \times 50/60 =$ **$0.01636\ m^3/s$**	

Slip = 0.01636 - 0.015 = **0.00136 m^3/s**; % slip = 0.00136/0.01636 = **8.33%**	
C_d = 0.015/0.01636 = **0.917**	
Power = $\rho g Q_{th} H$ = 9810 × 0.01636 × H = 160.5H W	The paper gives no lift H.
Assume total lift H = 15 m (as in Shameem sir's Problem 1): Power = **2.41 kW**	Flag: state the assumed head.

Numerical 4: three-throw pump (2019 8c)

Step (write this)	Why / what it means
Slip 2%: $Q_{th} = Q_{act}/0.98 = 0.1/0.98 = 0.10204$ m^3/s	
Three-throw: $Q_{th} = 3ALN/60 \Rightarrow N = 60 \times 0.10204/(3 \times 0.04909 \times 0.5)$ = **83.2 rpm**	"Determine the pump ..." read as the pump speed.
Total head $H = H_{st} + h_{fs} + h_{fd} + V_d^2/2g$ = 100 + 1 + 16 + $1.4^2/19.62$ = **117.1 m**	
Power = $\rho g Q_{th} H/n$ = 9810 × 0.10204 × 117.1/0.9 = **130.2 kW**	Shameem sir's Problem 6.

> **Memory trick:** Slip sign: "Long suction, short delivery, high speed = negative slip" (the suction column overshoots).

A7 Centrifugal pump numericals: pumps in series, multistage pumps, affinity laws Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q3(d)** [8] Pump at 80 gpm, maximum head 90 ftH_2O, 6 hp at 750 rpm; system head rises to 120 ftH_2O: new impeller speed.
- **2022 Q4(d)** [8] Double-stage pump, impeller 30 cm, 2000 rpm, lifts 5 m^3/s to 25 m, η 80%: stages and diameter of a similar multistage pump lifting 7 m^3/s to 200 m at 1500 rpm.
- **2021 Q3(d)** [12] Single-stage pump, impeller 50 cm, 1800 rpm, lifts 4 m^3/s to 25 m, η 70%: stages and impeller diameter of a similar multistage pump lifting 5 m^3/s to 180 m at 1300 rpm.
- **2020 Q3(d)** [8] Water from a deep well, total head 100 m; identical pumps at 900 rpm, specific speed 30, discharge 0.20 m^3/s: number of pumps.
- **2019 Q7(d)** [10] Pump at 100 gpm, head 100 ftH_2O, 8 hp at 875 rpm; a new heat exchanger raises the head to 150 ftH_2O: new impeller speed.

CALCULATOR Affinity laws: new speed for a new head (2019 7(d), 2023 3(d))

Change any number; every step is recalculated and shown with the numbers put in.

1. New speed: $N_2 = N_1\sqrt{(H_2/H_1)}$ = **875** × √(**150** / **100**) = **1072** rpm
2. New flow: $Q \propto N$ = **100** × **1072** / **875** = **122.5** gpm
3. New power: $P \propto N^3$ = **8** × pow(**1072** / **875**, 3) = **14.70** hp

CALCULATOR Multistage pump from specific speed (2021 3(d), 2022 4(d))

Change any number; every step is recalculated and shown with the numbers put in.

1. Specific speed of the given pump: $N_s = N\sqrt{Q}/H^{3/4}$ = **1800** × √(**4**) / pow(**25**, 0.75) = **322.0**
2. Head per stage at the same N_s: $H = (N\sqrt{Q}/N_s)^{4/3}$ = pow(**1300** × √(**5**) / **322**, 4 / 3) = **18.80** m
3. Stages (raw): **180** / **18.8** = **9.58**
4. Stages (round up): ceil(**9.576**) = **10**
5. Actual head per stage: **180** / **10** = **18.00** m
6. New impeller diameter: $D_2 = D_1(N_1/N_2)\sqrt{(H_2/H_1)}$ = **0.5** × (**1800** / **1300**) × √(**18** / **25**) = **0.587** m

2022 4(d): the given pump is double-stage lifting 25 m, so enter H = 12.5 m per stage. Efficiency is not needed. Pumps in series (2020 3(d)): H per pump = $(N\sqrt{Q}/N_s)^{4/3}$, number = total head / that, rounded up.

Before you start: how to solve multistage pump problems

Similar pumps have the same specific speed $N_s = N\sqrt{Q}/H^{3/4}$, where H is the head of **one stage**.

In a multistage pump the stages are in series: the same Q goes through each and the heads add.

So: (1) find N_s from the given pump, (2) find the head per stage of the new pump from N_s, (3) number of stages = total head / head per stage (round up), (4) diameter from the head coefficient $\sqrt{H}/(DN)$ = constant.

The efficiency given in these questions is not needed (it would only be used for power).

Numerical 1: number of identical pumps in series (2020 3d)

Step (write this)	Why / what it means
$N_s = N\sqrt{Q}/H_m^{3/4} \Rightarrow H_m^{3/4} = N\sqrt{Q}/N_s = 900\sqrt{0.20}/30 = 13.42$	Head of one pump.
$H_m = 13.42^{4/3} = 31.9$ m	
Number of pumps = 100/31.9 = 3.14	Pumps in series share the total head.
4 pumps in series (3 would give only 96 m); each then works at 25 m	Round up.

Numerical 2: single-stage to multistage pump (2021 3d)

Step (write this)	Why / what it means
Given pump: $N_s = 1800\sqrt{4}/25^{3/4} = 3600/11.18 = 322.0$	D_1 = 0.50 m, H_1 = 25 m.
New pump, same N_s: $H_{stage}^{3/4} = 1300\sqrt{5}/322.0 = 9.03 \Rightarrow H_{stage} = 18.8$ m	
Stages = 180/18.8 = 9.58 ⇒ **10 stages**, each 180/10 = 18 m	Round up.
Head coefficient: $\sqrt{H_1}/(D_1N_1) = \sqrt{H_2}/(D_2N_2) \Rightarrow D_2 = D_1(N_1/N_2)\sqrt{(H_2/H_1)}$	u = πDN/60 ∝ √H.
$D_2 = 0.50 \times (1800/1300) \times \sqrt{(18/25)} =$ **0.587 m = 58.7 cm**	Efficiency 70% is not needed.

Numerical 3: double-stage to multistage pump (2022 4d)

Reading: the given pump is double-stage and lifts the water 25 m in total, so each stage gives 12.5 m.

Step (write this)	Why / what it means
$N_s = 2000\sqrt{5}/12.5^{3/4} = 4472/6.65 = 672.7$	Per-stage head 12.5 m.
New: $H_{stage}^{3/4} = 1500\sqrt{7}/672.7 = 5.90 \Rightarrow H_{stage} = 10.66$ m	
Stages = 200/10.66 = 18.8 ⇒ **19 stages**, each 200/19 = 10.53 m	
$D_2 = 0.30 \times (2000/1500) \times \sqrt{(10.53/12.5)} =$ **0.367 m = 36.7 cm**	

Side note: If the 25 m is taken as the head of each of the two stages (some solutions do this), N_s = 400, head per stage 21.3 m, 200/21.3 = 9.4 ⇒ 10 stages of 20 m and D = 0.30(2000/1500)√(20/25) = 35.8 cm. State your reading on the script.

Numerical 4: affinity laws, new impeller speed (2023 3d)

Step (write this)	Why / what it means
$H_2/H_1 = (N_2/N_1)^2 \Rightarrow N_2 = N_1\sqrt{(H_2/H_1)}$	Head ∝ N^2.
$N_2 = 750\sqrt{(120/90)} = 750 \times 1.1547 =$ **866 rpm**	An increase of 15.5%.
Then $Q_2 = 80 \times 1.1547 = 92.4$ gpm and $P_2 = 6 \times 1.1547^3 = 9.24$ hp	Extra: check the motor can supply it.

Numerical 5: brewery expansion (2019 7d)

Step (write this)	Why / what it means
$N_2 = N_1\sqrt{(H_2/H_1)} = 875\sqrt{(150/100)} = 875 \times 1.2247 =$ **1072 rpm**	About 23% faster (Shameem sir's Problem 2).
$Q_2 = 100 \times 1.2247 = 122.5$ gpm; $P_2 = 8 \times 1.2247^3 = 14.7$ hp	Power almost doubles.

Memory trick: Multistage: "N_s first, head per stage second, count stages third, diameter last".

A5 Centrifugal pump: definition, parts, casings, working, heads and efficiencies, work done by the impeller Pass set Very likely · 87% Easy

PAST-PAPER QUESTIONS

- **2023 Q3(b)** [9] Define pump. Illustrate and describe the working principle of a single-stage centrifugal pump.
- **2023 Q3(c)** [10] Derive an expression for the work done by the impeller on the liquid per second per unit weight.
- **2022 Q3(c)** [10] Describe the casing with guide blades of a centrifugal pump with a schematic diagram.
- **2022 Q3(b)** [10] Short notes: manometric efficiency, static head, mechanical efficiency, manometric head.
- **2021 Q3(a)** [5] What are hydraulic machines? Basic difference between a turbine and a pump.
- **2021 Q3(b)** [10] Define centrifugal pump. Main parts, with sketches.
- **2021 Q3(c)** [8] Define: mechanical efficiency, manometric head, static head, manometric efficiency.
- **2019 Q7(c)** [7] Define pump. Describe with neat sketches the working principle of a centrifugal pump.
- **2019 Q7(a)** [8] Define: static suction head, static delivery head, water horsepower, pump efficiency.

★ REMEMBER IT

- **Pump vs turbine**: a pump puts energy *into* the liquid (mechanical → pressure head); a turbine takes it out.
- Centrifugal pump parts: **I-C-S-D**: Impeller (curved vanes, spins the liquid), Casing (volute or with guide blades turns velocity into pressure), Suction pipe with foot valve and strainer, Delivery pipe with valve.
- Efficiencies chain: shaft power → (mechanical η) → power given to liquid by impeller → (manometric η = $gH_m/V_{w2}u_2$) → useful power ρgQH_m. Overall $\eta_o = \eta_{mano} \times \eta_{mech}$.
- Work done per unit weight = $V_{w2}u_2/g$ (radial entry, $V_{w1} = 0$).

Before you start: what a pump does

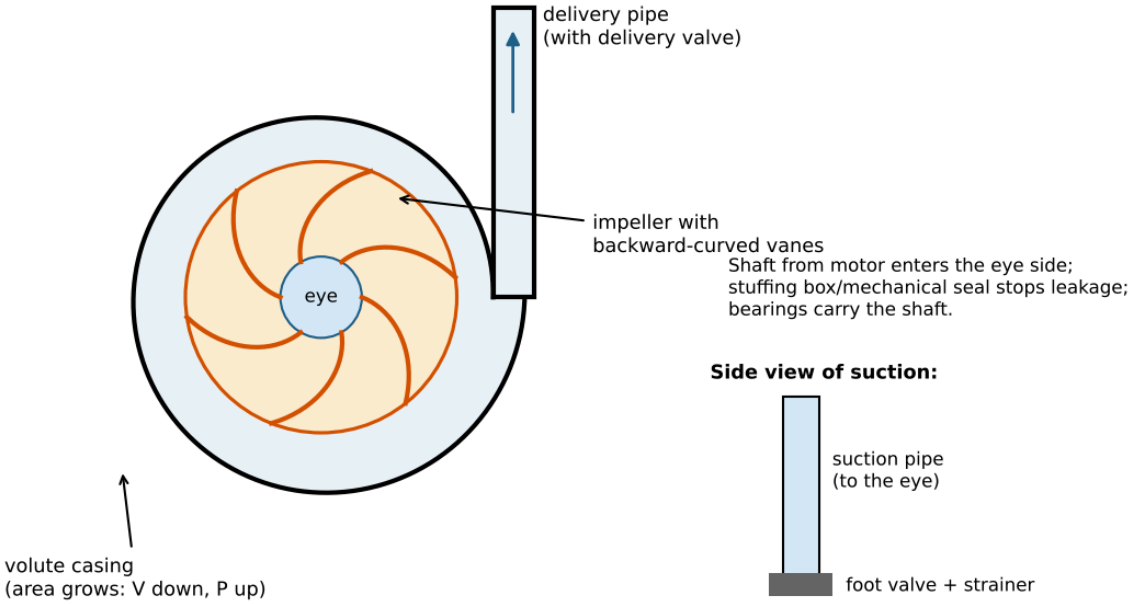
A **hydraulic machine** converts energy between a liquid and a machine. A **turbine** takes hydraulic energy from water and gives mechanical energy (shaft work); a **pump** takes mechanical energy from a motor and gives hydraulic energy (pressure) to a liquid, so that it can be lifted to a height or pushed through pipes.

In a **centrifugal pump** a spinning impeller whirls the liquid (a forced vortex). The centrifugal action throws the liquid outward at high speed; the casing then slows it down and turns the velocity into pressure. Head is measured in metres of liquid: H = pressure/ρg.

Hydraulic machines; turbine vs pump (2021 3a)

Hydraulic machines are machines that convert hydraulic energy (energy of a liquid) into mechanical energy (turbines) or mechanical energy into hydraulic energy (pumps). Examples: Pelton, Francis and Kaplan turbines; centrifugal and reciprocating pumps.

Turbine	Pump
converts hydraulic energy of water into mechanical (shaft) energy	converts mechanical energy into hydraulic (pressure) energy of the liquid
water flows from high head to low head; drives a generator	liquid is raised from low head to high head; driven by a motor
flow (reaction turbine) is usually radially inward / axial	flow in a centrifugal pump is radially outward
output: power; input: head and flow	output: head and flow; input: power



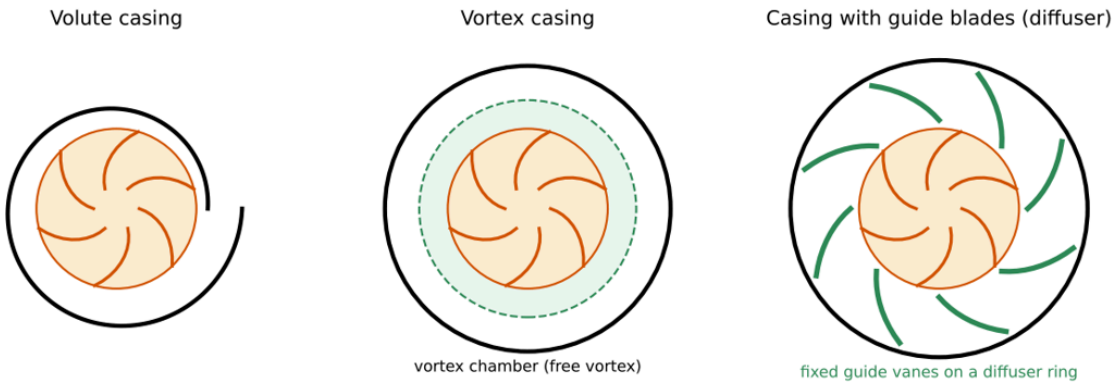
Centrifugal pump: main parts

Centrifugal pump: definition, main parts and working (2021 3b, 2023 3b, 2019 7c)

Pump: a mechanical device that raises the energy (pressure) of a liquid using mechanical energy, to move it from a low level or pressure to a high one. **Centrifugal pump:** a rotodynamic pump in which the energy is added by the centrifugal action of a rotating impeller; it is the reverse of a radially inward-flow reaction turbine.

Part	Function
Impeller	rotating wheel with backward-curved vanes, fixed on the shaft; gives the liquid its velocity and pressure. Types: **closed** (shrouds on both sides, clean water), **semi-open** (one shroud, liquids with some solids), **open** (no shrouds, sewage, paper pulp)
Casing	air-tight passage around the impeller; collects the liquid and converts its kinetic energy into pressure energy (volute, vortex, or with guide blades)
Suction pipe with foot valve and strainer	connects the sump to the eye of the impeller; the strainer keeps out debris; the foot valve (one-way) holds the water in the pipe when the pump stops, which helps priming
Delivery pipe with delivery valve	carries the liquid from the casing outlet to the delivery point; the valve controls the flow
Shaft, bearings, coupling	transmit the motor torque to the impeller and keep it aligned
Stuffing box / mechanical seal, wearing rings	stop leakage where the shaft passes through the casing and between impeller and casing

Working (single-stage): (1) The pump is **primed**: suction pipe, casing and impeller are filled with the liquid so that no air is present. (2) The delivery valve is kept closed and the motor is started; the impeller rotates and creates a forced vortex, which raises the pressure at the outlet and lowers it at the eye. (3) The delivery valve is opened. The low pressure at the eye lets atmospheric pressure push liquid up the suction pipe into the eye. (4) The vanes throw the liquid radially outward at high velocity and pressure. (5) In the casing the area grows, so the velocity falls and the pressure rises further; the liquid leaves through the delivery pipe. The flow is continuous and steady as long as the impeller turns.

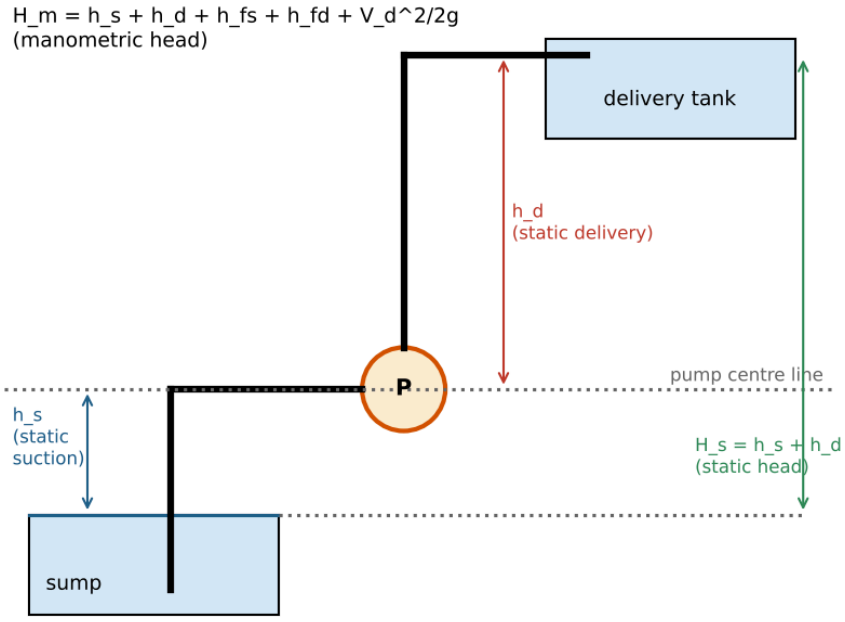


Three types of casing

Casing with guide blades (2022 3c)

In this casing the impeller is surrounded by a ring of fixed **guide blades** (vanes) mounted on a **diffuser** ring. The blade angles match the direction of the liquid leaving the impeller, so the liquid enters the guide passages without shock. The passages between the blades widen gradually, so the velocity falls and the pressure rises; the liquid then passes into the surrounding casing (usually concentric) and to the delivery pipe. Because the conversion of kinetic energy to pressure is gradual and shock-free, efficiency is high; this casing is used in **multistage** pumps (the guide passages lead the liquid to the eye of the next impeller). The other casings: **volute** (spiral casing whose area increases steadily to the outlet: simple, but eddies reduce efficiency) and **vortex** (a

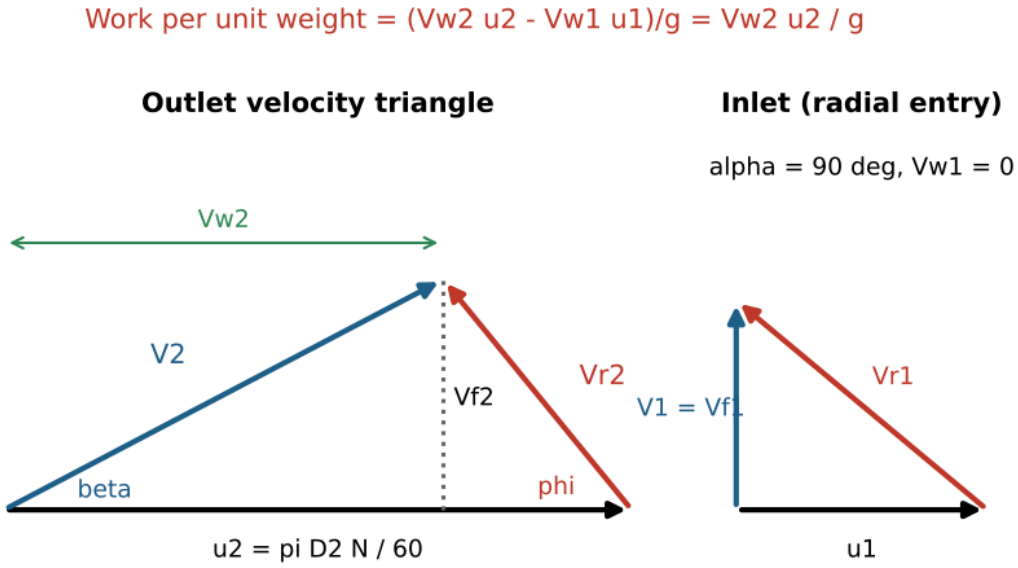
circular whirlpool chamber between impeller and volute where a free vortex forms, improving efficiency).



Heads in a pumping system

Heads and efficiencies (2021 3c, 2022 3b, 2019 7a)

Term	Definition / formula
Static suction head h_s	vertical height of the pump centre line above the liquid surface in the sump (suction lift). If the liquid is above the pump it is a positive static suction head
Static delivery head h_d	vertical height of the liquid surface in the delivery tank (or outlet) above the pump centre line
Static head H_s	$H_s = h_s + h_d$: vertical height between the sump and the delivery liquid surfaces
Manometric head H_m	head against which a centrifugal pump has to work: $H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/2g$ = (total head at outlet) − (total head at inlet) = $V_{w2}u_2/g$ − losses in the impeller and casing
Manometric efficiency η_{man}	manometric head / head imparted by the impeller = $\mathbf{gH_m/(V_{w2}u_2)}$
Mechanical efficiency η_m	power delivered by the impeller to the liquid / power input at the shaft = $(W/g)V_{w2}u_2/(1000$ S.P.$)$
Overall (pump) efficiency η_o	water power / shaft power = ρgQH_m/S.P. = $\eta_{man} \times \eta_m$
Water horsepower (WHP)	power actually given to the liquid: WHP (water horsepower) = $\rho gQH/746$ (SI, hp) = $QH/3960$ (Q in gpm, H in ft)
Brake horsepower (BHP)	power supplied to the pump shaft: BHP (brake horsepower) = WHP/η_o



Velocity triangles at the impeller inlet and outlet

Work done by the impeller per second per unit weight (2023 3c)

Assumptions: liquid enters the impeller eye radially ($\alpha = 90°$, $V_{w1} = 0$, $V_1 = V_{f1}$); no shock at entry; uniform velocity over the vane passage; no losses. Symbols: N rpm, D_1, D_2 inlet/outlet diameters, $u = \pi DN/60$ vane (tangential) velocity, V absolute velocity, V_w whirl (tangential) component, V_f flow (radial) component, V_r relative velocity.

Step (write this)	Why / what it means
Mass flow = ρQ kg/s; angular momentum per second at inlet = $\rho Q V_{w1} r_1$, at outlet = $\rho Q V_{w2} r_2$	Moment of momentum.
Torque on the liquid T = rate of change of angular momentum = $\rho Q(V_{w2} r_2 - V_{w1} r_1)$	Newton's second law for rotation.
Work done per second = $T\omega = \rho Q(V_{w2} r_2 \omega - V_{w1} r_1 \omega) = \rho Q(V_{w2} u_2 - V_{w1} u_1)$	$u = r\omega$.
Weight of liquid per second = $\rho g Q$	Divide to get work per unit weight.
Work done per unit weight = $(1/g)(V_{w2} u_2 - V_{w1} u_1)$	Euler's head (general).
= $V_{w2} u_2 / g$ (radial entry, $V_{w1} = 0$)	Work done per second = $(W/g)V_{w2}u_2$, $W = \rho g Q$.

Also $Q = \pi D_1 B_1 V_{f1} = \pi D_2 B_2 V_{f2}$ (B = vane width). From the velocity triangles, $V_{w2}u_2 = (V_2^2 + u_2^2 - V_{r2}^2)/2$, which gives the **fundamental equation**: work/unit weight = $(V_2^2 - V_1^2)/2g + (u_2^2 - u_1^2)/2g + (V_{r1}^2 - V_{r2}^2)/2g$ = rise in kinetic head + rise in pressure head from centrifugal action + from the slowing of relative velocity.

> **Memory trick:** Pump work = "whirl times wheel over g": $V_{w2}u_2/g$. Efficiency chain: manometric (hydraulic) × mechanical = overall.

A1 Pipe flow theory: laminar and turbulent flow, Reynolds number, entrance region, losses, Darcy-Weisbach, laminar velocity profile Pass set

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q6(c)** [4] Differentiate between major and minor losses in pipe flow.
- **2023 Q6(a)** [6] Short notes: irrotational flow region, hydrodynamic entrance region, hydrodynamically fully developed region.
- **2023 Q6(b)** [15] With an appropriate force balance show that for fully developed laminar flow in a circular pipe $u(r) = 2V_{avg}(1 - r^2/R^2)$.
- **2022 Q1(b)** [15] What are the different types of losses when a fluid flows through a pipe? Derive the expression for the loss of head due to friction in pipes.
- **2022 Q1(a)** [10] Define hydraulic diameter. Show the different flow regions of a fluid flowing through a pipe and how they differ with Reynolds number.
- **2021 Q1(a)** [10] What are the different kinds of losses in pipe flow? Briefly discuss.
- **2021 Q1(b)** [10] State and discuss Bernoulli's equation. How can it be derived from Euler's equation?
- **2020 Q1(a)** [12] What do you understand by laminar flow and turbulent flow? Derive an expression for the loss of head due to friction in pipes.
- **2019 Q2(a)** [12] Briefly discuss: (i) hydrodynamic entrance region and fully developed flow, (ii) minor losses in pipe flow, (iii) working principle of turbine flow meters.
- **2019 Q1(c)** [15] With an appropriate force balance show that for fully developed laminar flow in a circular pipe $u(r) = 2V_{avg}(1 - r^2/R^2)$.

★ REMEMBER IT

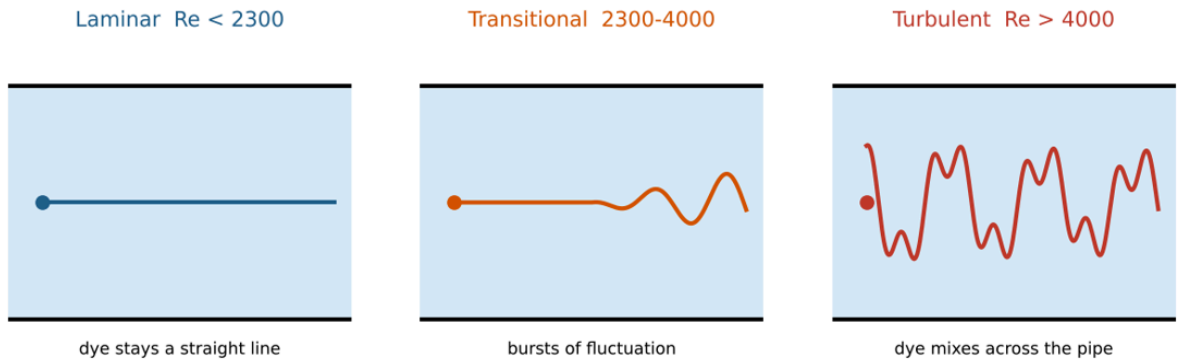
- **Re = ρVD/μ**: below 2300 laminar (smooth layers), 2300–4000 transitional, above 4000 turbulent (eddies).
- **Major loss** = wall friction along the pipe, $h_f = f(L/D)(V^2/2g)$ (Darcy-Weisbach). **Minor losses** = fittings: entry, exit, bends, valves, sudden expansion/contraction, $h = KV^2/2g$.
- Laminar: parabolic profile $u = 2V_{avg}(1 - r^2/R^2)$, so $u_{max} = 2V_{avg}$ and $f = 64/Re$. Entrance length: laminar ≈ 0.05Re D; turbulent ≈ 10D.

Before you start: what pipe flow is about

A pipe carries a fluid from one place to another. Because the fluid sticks to the wall (the **no-slip condition**), the velocity is zero at the wall and largest at the centre. The friction this causes uses up energy, which shows up as a fall in pressure along the pipe. We measure that lost energy as a **head loss** h (metres of the fluid): energy lost per unit weight of fluid.

Three quantities run the whole topic: the **Reynolds number** Re = ρVD/μ (decides laminar or turbulent), the **friction factor** f (how rough the flow is), and the **head loss** $h_f = f(L/D)(V^2/2g)$.

Laminar and turbulent flow (2020 1a), and how the regions depend on Re (2022 1a)

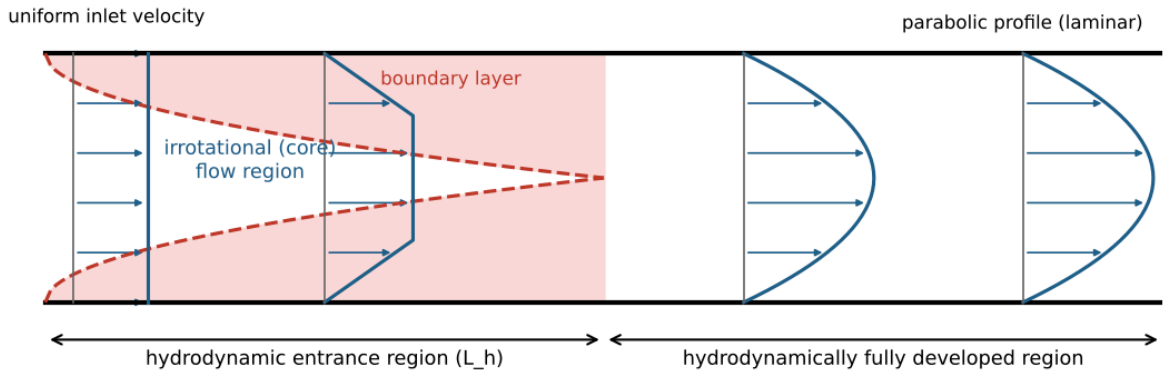


Reynolds' dye experiment: laminar, transitional and turbulent flow

	Laminar flow	**Turbulent flow**
Motion	fluid moves in smooth parallel layers (laminae); no mixing between layers	fluid particles move in irregular paths with eddies; strong mixing across the pipe
Reynolds number (pipe)	Re < 2300	Re > 4000 (2300-4000 is transitional: flow switches between the two)
Velocity profile	parabolic, $u_{max} = 2V_{avg}$	flat (full) in the core, very steep near the wall, $u_{max} \approx 1.2V_{avg}$
Shear stress	only viscous: $\tau = \mu\, du/dy$	viscous + turbulent (eddy) stress, much larger
Friction factor	$f = 64/Re$, independent of roughness	$f = f(Re, \varepsilon/D)$ from Colebrook equation / Moody chart
Head loss	$h_f \propto V$	$h_f \propto V^{1.75}$ to V^2
Example	oil or honey in a thin tube, blood in capillaries	water in city mains, smoke from a chimney, air in ducts

Reynolds number $Re = \rho V_{avg} D/\mu = V_{avg} D/\nu$ = inertia force / viscous force. When inertia wins (high Re) small disturbances grow into eddies: turbulent. When viscosity wins (low Re) disturbances are damped: laminar. The value at which the flow becomes turbulent is the **critical Reynolds number**, taken as 2300 for a circular pipe.

Hydraulic diameter $D_h = 4A_c/p$, where A_c is the flow area and p the **wetted perimeter** (length of wall touched by the fluid). It lets circular-pipe formulas (Re, f, h_f) be used for non-circular ducts. Circle: $D_h = 4(\pi D^2/4)/(\pi D) = D$. Square of side a: $D_h = a$. Rectangle a × b: $2ab/(a + b)$. Annulus: $D_o - D_i$.



Development of the velocity boundary layer in a pipe

Entrance region, fully developed flow and the irrotational core (2019 2a(i), 2023 6a, 2022 1a)

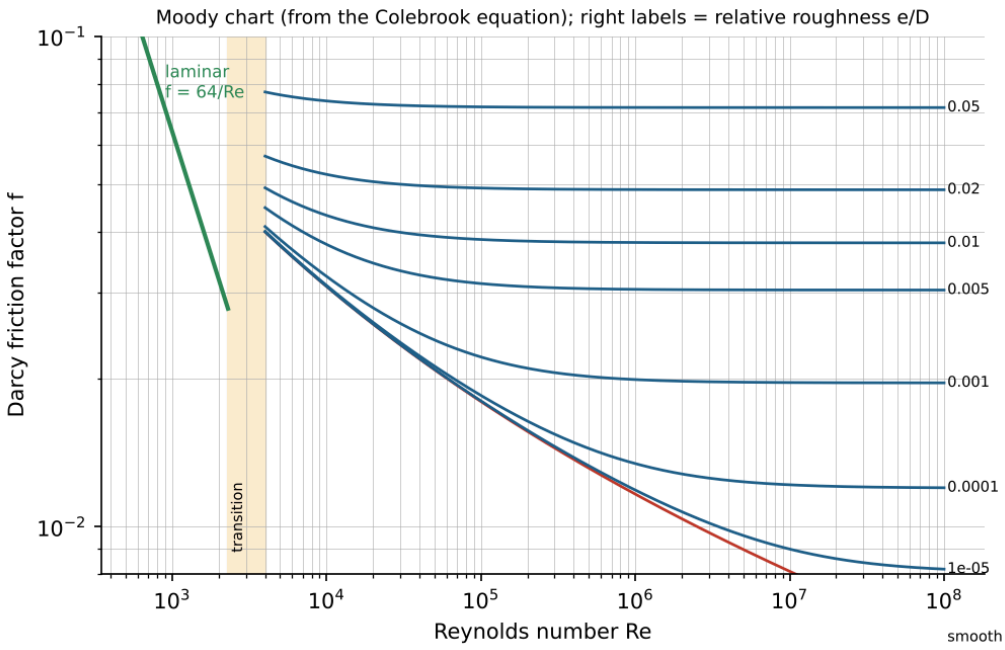
- **Irrotational (core) flow region:** near the inlet, the central part of the flow outside the boundary layer. Viscous effects are negligible there, the velocity is uniform across the radius and the fluid particles do not rotate.
- **Hydrodynamic entrance region:** the length from the pipe inlet to where the velocity profile becomes fully developed. The boundary layer grows from the wall until it meets at the centre; the core accelerates (to keep the mass flow) and the profile keeps changing. Its length $L_h \approx 0.05$ Re D for laminar flow and ≈ 10D for turbulent flow. Wall shear (and f) is highest at the inlet and falls along it.
- **Hydrodynamically fully developed region:** beyond the entrance length, the velocity profile no longer changes in the flow direction (du/dx = 0). The profile is parabolic in laminar flow and flatter in turbulent flow; the wall shear stress and f are constant, so the pressure falls linearly.
- **How the regions differ with Re:** at higher Re the boundary layer grows more slowly relative to D, so the laminar entrance region is longer ($L_h \propto$ Re); in turbulent flow the mixing makes the entrance short and nearly independent of Re. The fully developed profile is parabolic for Re < 2300 and flat for Re > 4000.

Losses in pipe flow (2021 1a, 2022 1b, 2023 6c, 2019 2a(ii))

Major loss is the loss of head due to friction along a straight pipe of constant area: $h_f = f(L/D)(V^2/2g)$ (Darcy-Weisbach). **Minor losses** are losses due to a change in the magnitude or direction of the velocity at fittings; they are written $h_m = K_L V^2/2g$ with a loss coefficient K_L found by experiment.

Minor loss	Head loss (Bansal)	Note
Sudden enlargement	$(V_1 - V_2)^2/2g$	eddies in the corner after the step
Sudden contraction	$0.5V_2^2/2g$ ($C_c = 0.62$)	loss is mostly from the expansion after the vena contracta
Entrance of a pipe	$0.5V^2/2g$ (sharp edge)	rounded entrance: about 0.03
Exit of a pipe	$V^2/2g$	all kinetic energy is lost in the tank
Obstruction	$[A/(C_c(A - a)) - 1]^2V^2/2g$	partly closed valve, orifice
Bend, elbow, fittings	$kV^2/2g$	k from tables (90° elbow about 0.3-0.9)

Major loss	Minor loss
due to wall friction along the pipe length	due to sudden change of velocity (size or direction) at fittings
$h_f = fLV^2/(2gD)$, grows with length	$h_m = KV^2/2g$, does not depend on length
dominant in long pipes	dominant in short pipes with many fittings; may be neglected in long pipes (L > 1000D)
found from the Moody chart / Colebrook	found from experimental K values



Moody chart drawn from the Colebrook equation

Colebrook equation (turbulent): $1/\sqrt{f} = -2.0 \log[(\varepsilon/D)/3.7 + 2.51/(Re\sqrt{f})]$. It is implicit, so either iterate or read f from the Moody chart, which plots it against Re for each relative roughness ε/D.

Derivation: loss of head due to friction (Darcy-Weisbach) (2020 1a, 2022 1b)

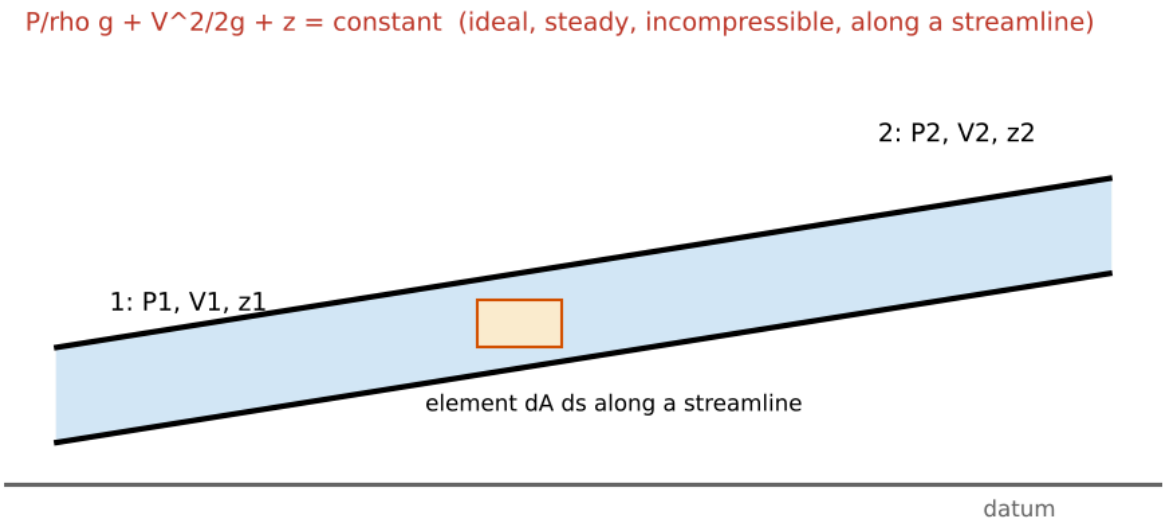
Take a horizontal pipe of diameter D with steady uniform flow V between sections 1 and 2, length L. Let the frictional resistance per unit wetted area per unit velocity squared be f′ (Froude's experiments: resistance $\propto V^2$).

Step (write this)	Why / what it means
Bernoulli 1 → 2: $p_1/\rho g + V^2/2g + z = p_2/\rho g + V^2/2g + z + h_f \Rightarrow h_f = (p_1 - p_2)/\rho g$	Same V (same area) and same z: the pressure drop is the head loss.
Frictional resistance $F = f' \times$ wetted area $\times V^2 = f'(\pi DL)V^2$	Resistance acts on the wall area πDL.
Force balance on the fluid (no acceleration): $(p_1 - p_2)(\pi D^2/4) = f'\pi DLV^2$	Net pressure force = friction force.
$p_1 - p_2 = f'(4L/D)V^2$	
$h_f = (f'/\rho g)(4L/D)V^2$	Divide by ρg.

Put $f'/\rho = f/8$ (Darcy f) or $f'/\rho = f_F/2$ ($f_F = f/4$)	Definition of the friction factor.
$h_f = f(L/D)(V^2/2g) = 4f_F LV^2/(2gD)$	Darcy-Weisbach equation.

Side note: Bansal writes $h_f = 4fLV^2/(2gD)$ where his f is the Fanning (coefficient of friction) = $f_{Darcy}/4$. Cengel and the lecture use the Darcy f with f = 64/Re for laminar flow. Write one, and say which f it is.

Bernoulli's equation from Euler's equation (2021 1b)

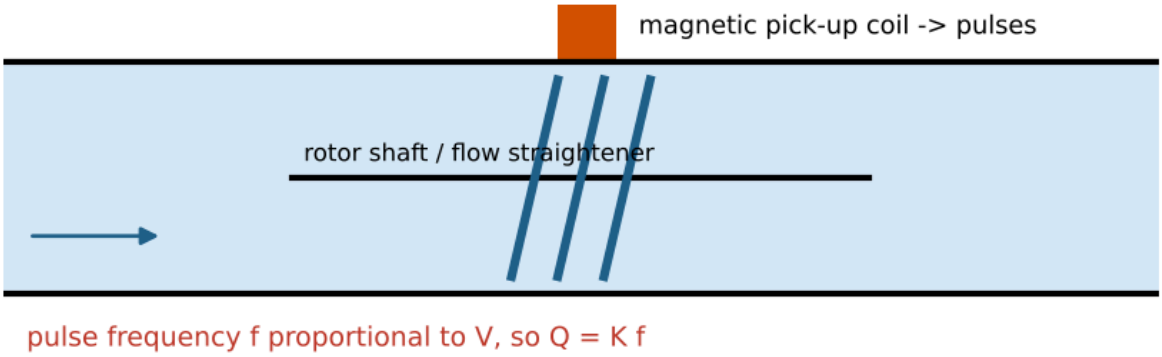


Fluid element on a streamline

Statement: in steady, ideal (non-viscous), incompressible, irrotational flow, the total energy per unit weight is the same at every point on a streamline: $p/\rho g + V^2/2g + z$ = constant (pressure head + velocity head + datum head). **Assumptions:** ideal fluid, steady, incompressible, irrotational, along one streamline. For real fluids add the loss: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2 + h_L$.

Step (write this)	Why / what it means
Element of area dA, length ds along a streamline; forces: pressure $pdA - (p + dp)dA = -dp\,dA$; weight component $-\rho g\,dA\,ds\cos\theta$	θ = angle between the streamline and the vertical, so $ds\cos\theta = dz$.
Mass × acceleration = $\rho dA\,ds \times V\,dV/ds$ (steady flow: $a_s = V\,dV/ds$)	Newton's second law along the streamline.
$-dp\,dA - \rho g\,dA\,dz = \rho\,dA\,V\,dV$	Substitute $ds\cos\theta = dz$.
$dp/\rho + g\,dz + V\,dV = 0$	Euler's equation of motion.
Integrate (ρ constant): $p/\rho + gz + V^2/2 =$ constant	Incompressible.
$p/\rho g + V^2/2g + z =$ constant	Bernoulli's equation, divide by g.

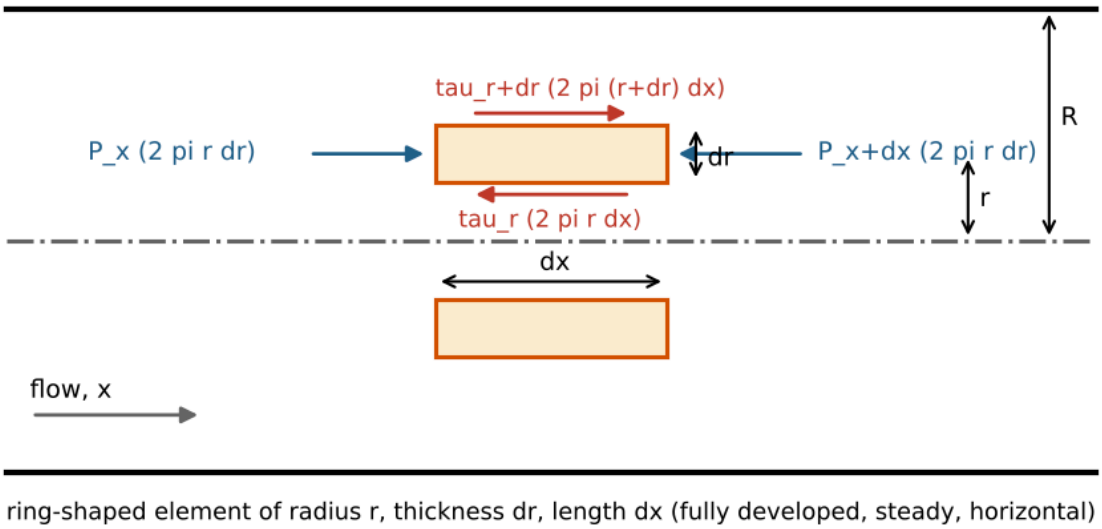
Turbine flow meter (2019 2a(iii))



Turbine flow meter

A multi-bladed rotor is mounted on bearings along the pipe axis. The fluid strikes the angled blades and spins the rotor at a speed proportional to the fluid velocity. A magnetic (or Hall-effect) pick-up outside the pipe produces one pulse each time a blade passes; the pulse frequency f is proportional to the volumetric flow rate, Q = Kf (K = meter factor from calibration). Counting pulses gives the total volume. Accurate (±0.5%), used for fuels, LNG, water meters; needs clean fluid and straight pipe upstream.

Velocity profile of fully developed laminar flow (2019 1c, 2023 6b)

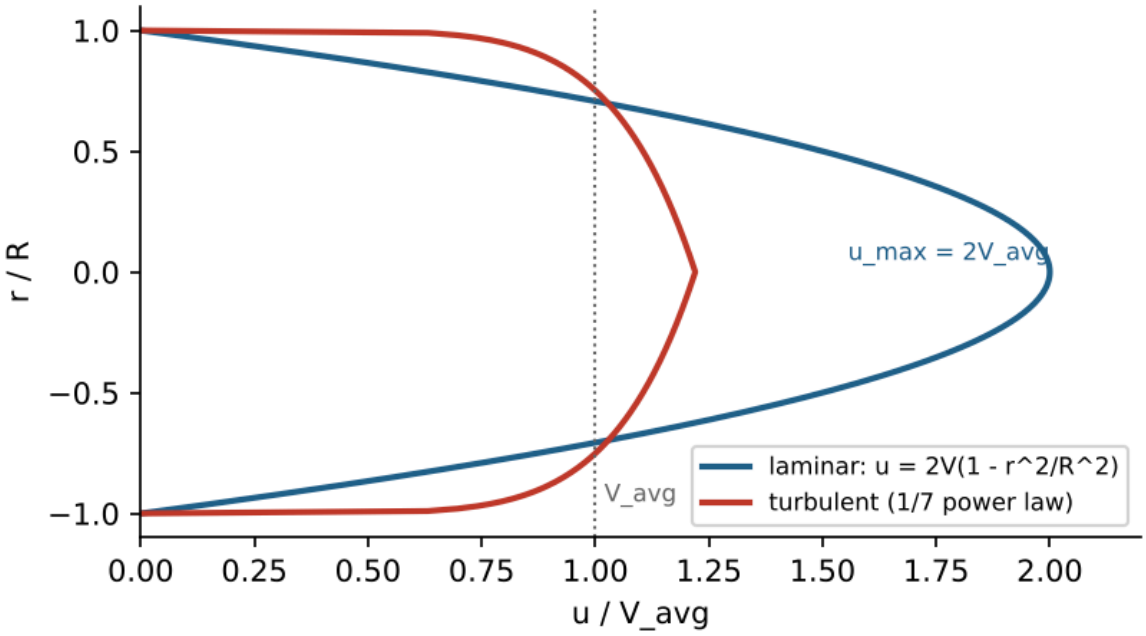


Force balance on a ring-shaped element

Step (write this)	Why / what it means
Ring element: radius r, thickness dr, length dx. Fully developed, steady: no acceleration, so the net force is zero	Only pressure and viscous shear act (horizontal pipe).
$(2\pi r\ dr\ P)_x - (2\pi r\ dr\ P)_{x+dx} + (2\pi r\ dx\ \tau)_{r+dr} - (2\pi r\ dx\ \tau)_r = 0$	Pressure on the two ends, shear on the outer and inner faces.
Divide by $2\pi dr\ dx$: $r(P_{x+dx} - P_x)/dx = [(r\tau)_{r+dr} - (r\tau)_r]/dr$	Let dx, dr → 0.

$r\,dP/dx = d(r\tau)/dr$, with $\tau = -\mu\,du/dr$	u falls as r grows, so du/dr < 0 and τ is positive.
$(\mu/r)\,d/dr(r\,du/dr) = dP/dx$	Left side depends only on r, right side only on x: both must be the same constant.
Integrate once: $r\,du/dr = (1/2\mu)(dP/dx)r^2 + C_1$	dP/dx is a constant.
$du/dr = (r/2\mu)(dP/dx) + C_1/r$; integrate again: $u = (r^2/4\mu)(dP/dx) + C_1 \ln r + C_2$	
Boundary conditions: $du/dr = 0$ at $r = 0$ (symmetry) ⇒ $C_1 = 0$; $u = 0$ at $r = R$ (no slip) ⇒ $C_2 = -(R^2/4\mu)(dP/dx)$	In r would be infinite at the centre, so C_1 must be zero.
$u(r) = -(R^2/4\mu)(dP/dx)(1 - r^2/R^2)$	Parabola; dP/dx < 0 so u > 0.
$V_{avg} = (2/R^2)\int_0^R u\,r\,dr = -(R^2/8\mu)(dP/dx)$	Average over the area: $\int(1 - r^2/R^2)r\,dr = R^2/4$.
$u(r) = 2V_{avg}(1 - r^2/R^2)$, $u_{max} = 2V_{avg}$	Replace $-(R^2/4\mu)dP/dx$ by $2V_{avg}$.

Useful results from the same analysis: $\Delta P = 8\mu L V_{avg}/R^2 = 32\mu L V_{avg}/D^2$; writing $\Delta P = f(L/D)\rho V^2/2$ gives **f = 64/Re**; flow rate $\dot{V} = \pi D^4 \Delta P/(128\mu L)$ (Hagen-Poiseuille).



Laminar and turbulent velocity profiles

Memory trick: Laminar profile: “two-V, one-minus-r-squared”. The four steps of the proof: balance, separate (both sides constant), integrate twice, two boundary conditions (centre flat, wall zero).

A3 Dimensional analysis: dimensions, homogeneity, Rayleigh's method, Buckingham π theorem

Next to add | Very likely · 87% | Hard

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [4] State Buckingham's π theorem.
- **2022 Q2(a)** [6] What is meant by dimensional homogeneity? Discuss with an example.
- **2021 Q2(a)** [8] State and discuss Buckingham-π theorem. How does it differ from Rayleigh's method?
- **2021 Q2(d)** [13] ΔP in a pipe of diameter D and length l in turbulent flow depends on V, ν, ρ and roughness k. Using Buckingham's π theorem obtain an expression for ΔP.
- **2019 Q1(a)** [5] What is meant by qualitative description of a physical quantity? Discuss in brief.
- **2019 Q1(b)** [15] Incompressible fluid (ρ, μ) in a long horizontal round pipe of diameter D, average velocity V, roughness ε: develop a non-dimensional relationship between wall shear stress τ_w and the other parameters.

UNDERSTAND **Read this to understand; you do not need to write it**

Dimensional analysis lets you find the *form* of a law without solving it. Write every quantity in M, L, T. **Buckingham π**: n variables and m basic dimensions give (n − m) dimensionless groups. Pick m **repeating variables** (usually ρ, V, D: one fluid property, one flow, one geometry) and combine each remaining variable with them to make a π group. Pipe friction always ends as $\Delta P/(\rho V^2) = \varphi(Re, L/D, \varepsilon/D)$.

Before you start: why dimensional analysis

Every physical quantity has dimensions built from a few fundamental ones: mass M, length L, time T (and temperature θ). Velocity = LT^{-1}, force = MLT^{-2}, pressure = $ML^{-1}T^{-2}$, density = ML^{-3}, dynamic viscosity μ = $ML^{-1}T^{-1}$, kinematic viscosity ν = L^2T^{-1}.

Dimensional analysis groups many variables into a few dimensionless numbers. That cuts the number of experiments and lets a small model predict a full-size machine.

Qualitative description of a physical quantity (2019 1a)

A physical quantity is described in two ways. The **qualitative** description gives its **dimension**, i.e. what kind of quantity it is, without a number: e.g. length L, mass M, time T, velocity LT^{-1}. The **quantitative** description gives its size as a number with a **unit**, e.g. 5 m, 3 m/s. Primary (fundamental) dimensions are M, L, T, θ, current I, amount of light C, amount of matter N; all others are secondary dimensions formed from them.

Dimensional homogeneity (2022 2a)

An equation is **dimensionally homogeneous** when every additive term has the same dimensions (law of dimensional homogeneity). Such an equation is valid in any system of units, and it can be checked or made non-dimensional by dividing by any one term.

Example: Bernoulli $p/\rho g + V^2/2g + z = H$. $[p/\rho g] = (ML^{-1}T^{-2})/(ML^{-3} \cdot LT^{-2}) = L$; $[V^2/2g] = L^2T^{-2}/LT^{-2} = L$; $[z] = L$. Every term is a length (head), so it is homogeneous. Another: $V = \sqrt{(2gH)}$:

$[V] = LT^{-1}$ and $[\sqrt{(gH)}] = \sqrt{(LT^{-2}L)} = LT^{-1}$.

Buckingham π theorem and Rayleigh's method (2021 2a, 2023 5a)

Statement: if a physical phenomenon depends on n variables and these involve m fundamental dimensions, the variables can be arranged into (n − m) independent dimensionless groups called π terms: $f(X_1, \ldots, X_n) = 0$ becomes $\varphi(\pi_1, \ldots, \pi_{n-m}) = 0$.

1. List all n variables and write their dimensions; find m.
2. Pick m **repeating variables**: together they must contain all m dimensions, must not form a dimensionless group, and should be one geometric (D), one flow (V) and one fluid property (ρ). Never pick the dependent variable.
3. Each π = (repeating variables with unknown powers) × one remaining variable; set the powers of M, L, T to zero and solve.
4. Write $\pi_1 = \varphi(\pi_2, \pi_3, \ldots)$ and rearrange.

Rayleigh's method	Buckingham π method
assumes $X = K \cdot X_1^a X_2^b X_3^c \ldots$ and matches powers of M, L, T	groups variables into (n − m) π terms using repeating variables
gives one equation in all the variables	gives a set of independent dimensionless groups
becomes awkward when there are more than 3-4 independent variables (more unknown powers than equations)	systematic for any number of variables
the form of the function is assumed (a product of powers)	no form is assumed; the function φ must come from experiment

Numerical 1: wall shear stress in a pipe (2019 1b)

$\tau_w = f(V, \varepsilon, \rho, \mu, D)$: n = 6 variables; dimensions M, L, T so m = 3; n − m = 3 π terms. Repeating variables: V, ρ, D.

Variable	τ_w	V	ε	ρ	μ	D
Dimension	$ML^{-1}T^{-2}$	LT^{-1}	L	ML^{-3}	$ML^{-1}T^{-1}$	L

Step (write this)	Why / what it means
$\pi_1 = \tau_w V^a \rho^b D^c$: M: 1 + b = 0; T: -2 - a = 0; L: -1 + a - 3b + c = 0 ⇒ a = -2, b = -1, c = 0	$\pi_1 = \tau_w/(\rho V^2)$.
$\pi_2 = \mu V^a \rho^b D^c$: M: 1 + b = 0; T: -1 - a = 0; L: -1 + a - 3b + c = 0 ⇒ a = -1, b = -1, c = -1	$\pi_2 = \mu/(\rho VD) = 1/Re$.
$\pi_3 = \varepsilon V^a \rho^b D^c$ ⇒ a = b = 0, c = -1	$\pi_3 = \varepsilon/D$, relative roughness.
$\tau_w/(\rho V^2) = \varphi(Re, \varepsilon/D)$	Invert π_2 to the familiar Re. Since $f = 8\tau_w/\rho V^2$, this is why f = f(Re, ε/D) on the Moody chart.

Numerical 2: pressure difference in turbulent pipe flow (2021 2d)

$\Delta P = f(D, l, V, \nu, \rho, k)$: n = 7, m = 3, so 4 π terms. Repeating: D, V, ρ.

Step (write this)	Why / what it means

$\pi_1 = D^a V^b \rho^c \Delta P$: M: c + 1 = 0; T: -b - 2 = 0; L: a + b - 3c - 1 = 0 ⇒ c = -1, b = -2, a = 0	$\pi_1 = \Delta P/(\rho V^2)$.
$\pi_2 = D^a V^b \rho^c l$ ⇒ a = -1, b = c = 0	$\pi_2 = l/D$.
$\pi_3 = D^a V^b \rho^c \nu$: M: c = 0; T: -b - 1 = 0; L: a + b + 2 = 0 ⇒ b = -1, a = -1	$\pi_3 = \nu/(VD) = 1/Re$.
$\pi_4 = D^a V^b \rho^c k$ ⇒ a = -1	$\pi_4 = k/D$.
$\Delta P = \rho V^2\, \varphi(l/D,\ VD/\nu,\ k/D)$	Experiments show $\Delta P \propto l/D$, so $\Delta P = \rho V^2 (l/D)\varphi_1(Re, k/D)$, the Darcy form with $f = 2\varphi_1$.

Memory trick: Repeating variables are “D-V-ρ” (Diameter, Velocity, rho): one for size, one for motion, one for fluid. Each π then reads like a famous number.

A9 Compressors, fans and blowers Next to add Likely · 67% Easy

PAST-PAPER QUESTIONS

- **2023 Q2(c)** [15] Purpose of a compressor; working principles of centrifugal and reciprocating compressors.
- **2022 Q3(d)** [5] Basic differences between fan, compressor and blower.
- **2019 Q6(b)** [8] Purpose of a compressor; working principle of centrifugal and reciprocating compressors.
- **2019 Q6(a)** [5] Differentiate between a compressor and a blower.
- **2019 Q8(d)** [5] Differentiate between forced and induced draft fans.

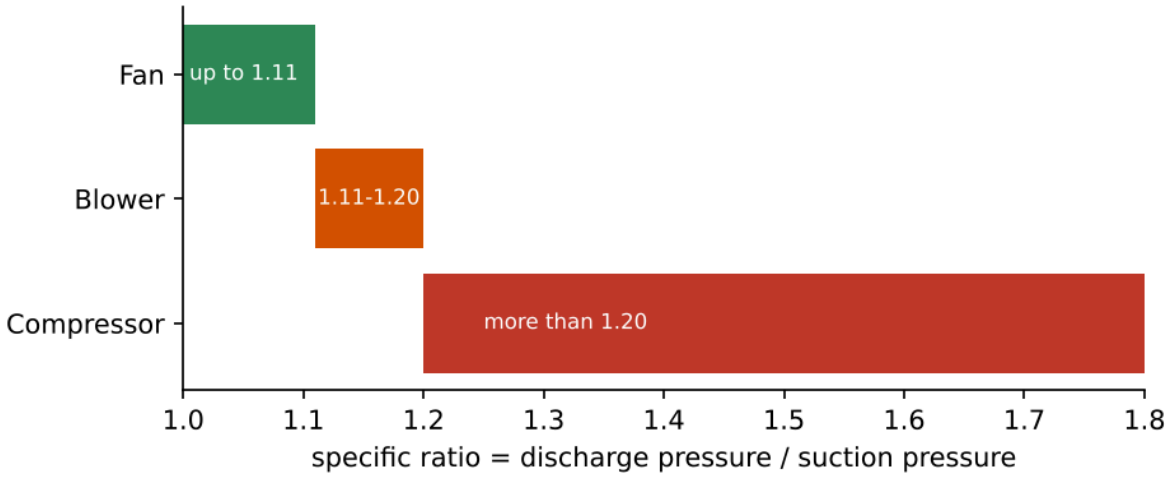
★ REMEMBER IT

- **Fan < blower < compressor** by pressure ratio: fan up to 1.1, blower 1.1–1.2, compressor above 1.2. Forced draft fan pushes air *into* the furnace (positive pressure); induced draft fan pulls gases *out* at the chimney (negative pressure).

Before you start: machines that move gases

Fans, blowers and compressors all raise the pressure of a gas; they differ in how much. A small rise moves air (ventilation); a large rise stores energy in compressed air.

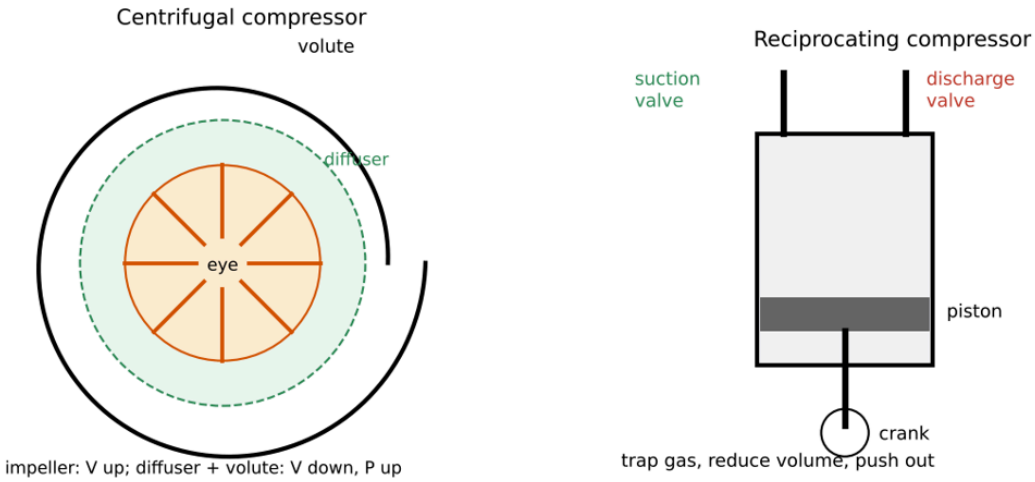
Two ways to raise pressure: squeeze a trapped volume (positive displacement) or give the gas speed and then slow it down (dynamic).



Fan, blower and compressor by specific ratio

Fan vs blower vs compressor (2020 3a, 2022 3d, 2019 6a)

Item	Fan	Blower	Compressor
Specific ratio ($p_{discharge}/p_{suction}$, ASME)	up to 1.11	1.11 to 1.20	above 1.20
Pressure rise	up to about 1136 mm WC	1136-2066 mm WC	above 2066 mm WC (several bar to hundreds of bar)
Gas density change	negligible (incompressible)	small	large (compressible); needs cooling/intercooling
Flow	very large volume, low pressure	medium	smaller volume, high pressure
Examples	ventilation, cooling towers, FD (forced draught)/ID fans, HVAC	furnace air, pneumatic conveying, aeration, cleaning	compressed air plants, refrigeration, gas pipelines, gas turbines



Centrifugal and reciprocating compressors

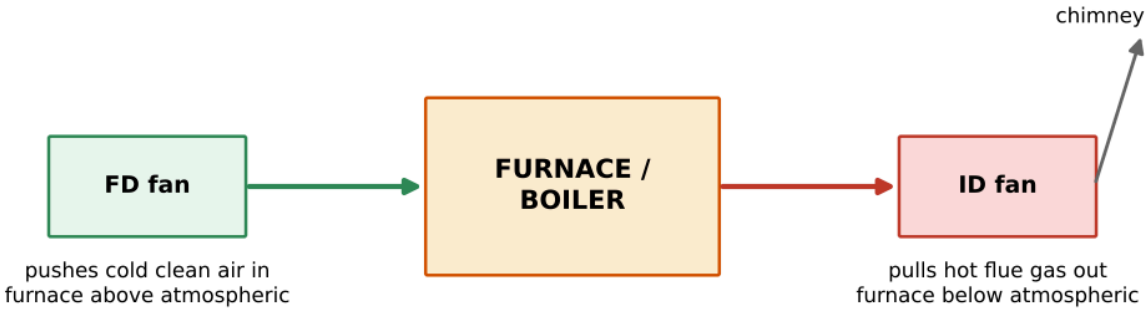
Purpose of a compressor; centrifugal and reciprocating compressors (2023 2c, 2019 6b)

A **compressor** increases the pressure of a gas by reducing its volume. **Purposes:** compressed air for pneumatic tools, drills, paint spraying and instrument/control systems; refrigeration and air-conditioning (raises refrigerant vapour pressure); gas turbines and jet engines (combustion air); transport of natural gas in pipelines; supercharging engines; liquefaction and storage of gases; chemical processes that need high pressure and temperature.

Centrifugal compressor (dynamic): gas enters the eye of a high-speed impeller axially; the radial vanes throw it outward, increasing its velocity (and some pressure). It then enters the **diffuser**, a passage of increasing area where the velocity falls and the kinetic energy changes into pressure; the **volute** collects the gas and continues the conversion to the outlet. Pressure ratio per stage about 2-4; several stages in series for higher ratios. Smooth, pulsation-free, high flow, compact, little maintenance; less efficient at small flows and subject to surge.

Reciprocating compressor (positive displacement): a piston moves in a cylinder driven by crank and connecting rod. **Suction stroke:** piston moves down, cylinder pressure falls below suction pressure, the spring-loaded suction valve opens and gas fills the cylinder. **Compression stroke:**

both valves closed, the piston moves up and the gas is compressed (polytropically, $pV^n = C$). **Delivery:** when the pressure exceeds the delivery pressure the discharge valve opens and gas is pushed out; the gas left in the clearance volume re-expands on the next suction stroke. Single- or double-acting, single- or multi-stage with intercoolers between stages. High pressure ratios, any flow size, but pulsating flow, heavy foundation and more maintenance.



Forced draft and induced draft fans

Forced draft vs induced draft fans (2019 8d)

Forced draft (FD) fan	Induced draft (ID) fan
placed at the bottom/inlet of the furnace, before the air preheater	placed at the outlet, between the boiler (dust collector) and the chimney
pushes (forces) fresh air into the furnace	sucks (induces) flue gases out of the furnace
furnace pressure above atmospheric (risk of gas leakage out)	furnace pressure below atmospheric (air leaks in)
handles cold, clean air: smaller, cheaper, less power	handles hot, dusty, larger volume of gas: bigger fan, more power, erosion
better flame distribution and air control	draft does not depend on the chimney height

> **Memory trick:** “1.11, 1.20”: fan below 1.11, blower up to 1.20, compressor beyond. FD = “Forces air in”, ID (induced draught (or inside diameter)) = “Inhales gas out”.

A2 Pipe flow numericals: laminar head loss and pumping power, pipes in series and parallel, pipe schedules Pass set Likely · 60% Medium

PAST-PAPER QUESTIONS

- **2023 Q6(d)** [10] Water at 10 °C (ρ = 999.7 kg/m^3, μ = 1.307×10^{-3} kg/m s) flows in a 0.2 cm diameter, 15 m long pipe at 1.2 m/s: pressure drop, head loss, pumping power.
- **2021 Q1(c)** [15] Two pipes each 300 m long, diameters 0.30 m and 0.15 m, carry 0.085 m^3/s from a reservoir. Ratio of head lost in series to parallel connection. Neglect minor losses.
- **2019 Q2(b)** [14] Water at 5 °C (ρ = 1000 kg/m^3, μ = 1.519×10^{-3} kg/m s) flows in a 0.3 cm diameter, 9 m long horizontal pipe at 0.9 m/s: head loss, pressure drop, pumping power.
- **2019 Q2(c)** [9] Flow area and wall thickness of: (i) 3-nom sch 10S, (ii) 3-nom sch 40, (iii) 3-nom XXS (double extra strong (pipe schedule)).

CALCULATOR Laminar pipe: head loss, pressure drop, pumping power (2019 2(b), 2023 6(d))

Change any number; every step is recalculated and shown with the numbers put in.

1. Reynolds number: Re = ρVD/μ = **999.7** × **1.2** × **0.002** / **0.001307** = **1836**
2. Friction factor (laminar): f = 64/Re = 64 / **1836** = **0.03486**
3. Head loss: h_f = f(L/D)(V²/2g) = **0.03486** × (**15** / **0.002**) × **1.2** × **1.2** / (2 × 9.81) = **19.19** m
4. Pressure drop: ΔP = ρgh_f = **999.7** × 9.81 × **19.19** / 1000 = **188.2** kPa
5. Flow rate: π × **0.002** × **0.002** / 4 × **1.2** = **3.770 × 10^{-6}** m³/s
6. Pumping power: W = QΔP = **3.770 × 10^{-6}** × **188.2** × 1000 = **0.710** W

Check Re < 2300 first; f = 64/Re is valid only for laminar flow.

CALCULATOR Two pipes in series vs in parallel (2021 1(c))

Change any number; every step is recalculated and shown with the numbers put in.

1. Series: h ∝ Q²(1/D_1^5 + 1/D_2^5): **0.085** × **0.085** × (1 / pow(**0.3**,5) + 1 / pow(**0.15**,5)) = **98** × k
2. Parallel split Q_1/Q_2 = $(D_1/D_2)^{2.5}$: pow(**0.3** / **0.15**, 2.5) = **5.657**
3. Q_2 in parallel: **0.085** / (**5.657** + 1) = **0.01277** m³/s
4. Parallel: h ∝ Q_2^2/D_2^5: **0.01277** × **0.01277** / pow(**0.15**,5) = **2.15** × k
5. Head loss ratio series/parallel: **98.12** / **2.147** = **45.7**

Same f and L for both pipes, so k = 8fL/(π²g) cancels in the ratio.

Before you start: the recipe for every pipe numerical

1. Find Re = $\rho VD/\mu$. 2. Choose f: f = 64/Re if Re < 2300, else Moody chart/Colebrook. 3. Head loss $h_L = f(L/D)V^2/2g$. 4. Pressure drop $\Delta P = \rho g h_L$. 5. Pumping power $W = \dot{V}\Delta P = \rho g \dot{V} h_L$, where $\dot{V} = (\pi D^2/4)V$.

Always convert cm and mm to metres first. The diameters in these questions are tiny (2-3 mm), which is why the flow is laminar even for water.

Numerical 1: water at 5 °C in a 0.3 cm, 9 m pipe at 0.9 m/s (2019 2b)

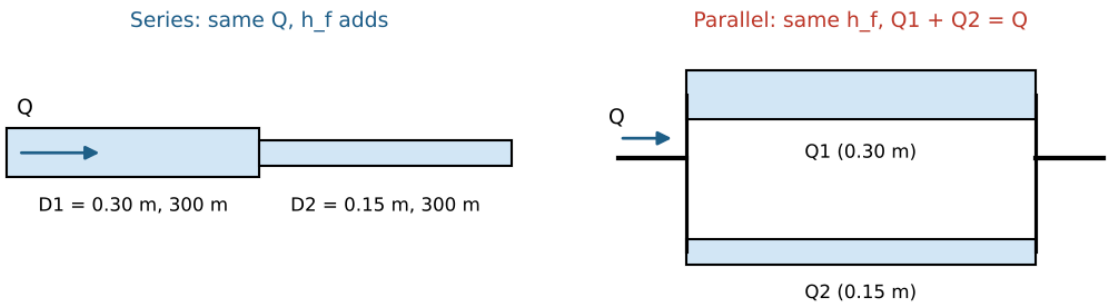
Given: $\rho = 1000$ kg/m^3, $\mu = 1.519\times10^{-3}$ kg/m s, D = 0.003 m, L = 9 m, V = 0.9 m/s. **Find:** h_L, ΔP, $\dot{W}$.

Step (write this)	Why / what it means
$Re = \rho VD/\mu = 1000(0.9)(0.003)/1.519\times10^{-3} = 1777 < 2300$	Laminar, so f = 64/Re can be used.
$f = 64/1777 = 0.0360$	
$h_L = f(L/D)V^2/2g = 0.0360 \times (9/0.003) \times 0.9^2/(2 \times 9.81) =$ **4.46 m**	Head loss.
$\Delta P = \rho g h_L = 1000(9.81)(4.46) =$ **43.7 kPa**	Check: $32\mu LV/D^2 = 32(1.519\times10^{-3})(9)(0.9)/0.003^2$ = 43,747 Pa.
$\dot{V} = (\pi/4)(0.003)^2(0.9) = 6.36\times10^{-6}$ m^3/s	
$\dot{W} = \dot{V}\Delta P = 6.36\times10^{-6} \times 43{,}747 =$ **0.28 W**	Cengel Example 8-2.

Numerical 2: water at 10 °C in a 0.2 cm, 15 m pipe at 1.2 m/s (2023 6d)

Step (write this)	Why / what it means
$Re = 999.7(1.2)(0.002)/1.307\times10^{-3} = 1836 < 2300$	Laminar.
$\Delta P = 32\mu LV/D^2 = 32(1.307\times10^{-3})(15)(1.2)/0.002^2 =$ **188.2 kPa**	Same as f = 64/Re = 0.03486 in Darcy-Weisbach.
$h_L = \Delta P/\rho g = 188{,}208/(999.7 \times 9.81) =$ **19.2 m**	
$\dot{V} = (\pi/4)(0.002)^2(1.2) = 3.77\times10^{-6}$ m^3/s; $\dot{W} = \dot{V}\Delta P =$ **0.71 W**	Pumping power.

Numerical 3: two pipes in series and in parallel (2021 1c)



The two pipes in series and in parallel

Given: L = 300 m each, D_1 = 0.30 m, D_2 = 0.15 m, Q = 0.085 m^3/s, same f, minor losses neglected. Use $h_f = fLV^2/(2gD)$ with $V = 4Q/\pi D^2$, so $\mathbf{h_f = 8fLQ^2/(\pi^2 gD^5) = kQ^2/D^5}$ with $k = 8fL/\pi^2 g$ (same for both pipes).

Step (write this)	Why / what it means
Series: same Q in both; $H_s = kQ^2(1/D_1^5 + 1/D_2^5)$	Losses add.
Parallel: same head H_p in each: $kQ_1^2/D_1^5 = kQ_2^2/D_2^5 \Rightarrow Q_1/Q_2 = (D_1/D_2)^{2.5} = 2^{2.5} = 5.657$	Equal loss in the two branches.
$Q_1 + Q_2 = 0.085 \Rightarrow Q_2 = 0.085/6.657 = 0.01277$, $Q_1 = 0.07223\ m^3/s$	Flows add.
$H_p = kQ_1^2/D_1^5 = kQ^2/(D_1^{2.5} + D_2^{2.5})^2$	Same thing in one line.
$H_s/H_p = (1/D_1^5 + 1/D_2^5)(D_1^{2.5} + D_2^{2.5})^2$	k and Q cancel.
$= (411.5 + 13{,}169)(0.04930 + 0.008714)^2 = 13{,}580 \times 0.003365 =$ **45.7**	Head lost in series is about 46 times that in parallel.

Numerical 4: flow area and wall thickness of 3-inch nominal pipes (2019 2c)

For nominal sizes 1/8 to 12 inch, the outside diameter is fixed by the nominal size (3-inch nominal: **OD = 3.500 in = 88.9 mm**) and the schedule fixes the wall thickness t (ASME B36.10M / B36.19M tables). ID (induced draught (or inside diameter)) = OD − 2t, flow area = $\pi ID^2/4$.

Pipe	Wall thickness t	Inside diameter	Flow area
3-nom sch 10S	0.120 in = 3.05 mm	3.260 in = 82.80 mm	8.347 in^2 = 5.385×10^{-3} m^2
3-nom sch 40	0.216 in = 5.49 mm	3.068 in = 77.93 mm	7.393 in^2 = 4.769×10^{-3} m^2
3-nom XXS	0.600 in = 15.24 mm	2.300 in = 58.42 mm	4.155 in^2 = 2.680×10^{-3} m^2

Side note: Wall thicknesses are from the standard pipe tables (the exam says charts are supplied on request). Higher schedule = thicker wall = smaller bore for the same OD.

Memory trick: Pipes in parallel share the head, pipes in series share the flow. Q splits as $D^{2.5}$.

Section B: Compressible flow and heat exchangers

B5 Heat exchanger LMTD numericals: area, length, outlet temperature, fouling

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [20] Water 20 → 80 °C by oil 160 → 140 °C, D_i 20 mm, U = 500, q = 3000 W: length. After 3 years water leaves at 65 °C: new q, oil outlet, U and water-side fouling factor.
- **2023 Q8(c)** [12] Milk 250 L/h chilled 38.6 → 13 °C by ground water at 10 °C, 0.72 m^3/h; ρ_{milk} 1030, c_p 3860: UA, length for a 50 mm pipe with U = 1000; water outlet temperature.
- **2022 Q7(c)** [13] Water 68 kg/min heated 35 → 75 °C by oil (c_p 1.9 kJ/kg°C) 110 → 75 °C in a counter-flow double pipe; U = 320 W/m^2°C: area.
- **2021 Q7(c)** [15] Engine oil 80 → 50 °C, water in at 20 °C; D_i 2.5 cm; $\dot{m}_w$ 0.08, $\dot{m}_{oil}$ 0.16 kg/s; h_w 100, h_{oil} 80 W/m^2°C; fouling 0.00018 each; c_w 4180, c_{oil} 2020: tube length.
- **2020 Q5(c)** [12] Counter-flow double tube: water 20 → 80 °C at 1.2 kg/s heated by geothermal water at 160 °C, 2 kg/s; tube 1.5 cm; U = 640 W/m^2K: length.
- **2020 Q6(b)** [10] Counter-flow shell-and-tube: water 0.8 kg/s from 30 to 80 °C; oil 120 → 85 °C; U = 125 W/m^2°C: area.
- **2019 Q4(c)** [15] Gas-turbine lube oil cooler: water 0.2 kg/s in D_i 25 mm, oil 0.1 kg/s in annulus D_o 45 mm; oil 100 → 60 °C, water in at 30 °C: tube length (properties given).

CALCULATOR LMTD method: heat, area and tube length (2020 6(b), 2022 7(c), 2023 7(c))

Change any number; every step is recalculated and shown with the numbers put in.

1. Heat duty: $Q = \dot{m}_c c_c (T_{c,o} - T_{c,i})$ = **1.133** × **4180** × (**75** - **35**) = **189488** W
2. ΔT_1: **1** == 1 ? **110** - **75** : **110** - **35** = **35.00** °C
3. ΔT_2: **1** == 1 ? **75** - **35** : **75** - **75** = **40.00** °C
4. LMTD: $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$ = abs(**35** - **40**) < 10^{-6} ? **35** : (**35** - **40**) / ln(**35** / **40**) = **37.44** °C
5. Area: $A = Q/(U\Delta T_{lm})$ = **189500** / (**320** × **37.44**) = **15.81** m²
6. Tube length: $L = A/(\pi D)$ = **15.81** / (π × **0.02**) = **251.7** m

68 kg/min = 1.1333 kg/s. If the hot outlet is unknown, find it from $Q = \dot{m}_h c_h (T_{h,i} - T_{h,o})$ first.

Before you start: the LMTD (log mean temperature difference) recipe

1. Energy balance for the unknown flow or outlet temperature. 2. ΔT_1, ΔT_2 at the two ends. 3. ΔT_{lm}. 4. $A = Q/(U\Delta T_{lm})$. 5. $L = A/(\pi D)$.

Where c_p of water is not given, use 4.18 kJ/kg K (Cengel).

Numerical 1: geothermal water heater (2020 5c)

Step (write this)	Why / what it means
$Q = \dot{m}_c c_{pc} \Delta T_c = 1.2 \times 4.18 \times (80 - 20) = 301.0$ kW	$c_{p,water}$ = 4.18 kJ/kg K.
$T_{h,o} = 160 - 301.0/(2 \times 4.31) = 125.1$ °C	c_p of geothermal water 4.31 kJ/kg K (Cengel Example 11-4; not given in the paper).
$\Delta T_1 = 160 - 80 = 80$ °C; $\Delta T_2 = 125.1 - 20 = 105.1$ °C; $\Delta T_{lm} = (80 - 105.1)/\ln(80/105.1) = 92.0$ °C	Counter flow.
$A_s = Q/(U \Delta T_{lm}) = 301{,}000/(640 \times 92.0) = 5.11$ m^2	
$L = A_s/(\pi D) = 5.11/(\pi \times 0.015) =$ **108.5 m**	Cengel: 109 m. (With c_p = 4.18 for both, L = 107 m.)

Numerical 2: shell-and-tube water heater (2020 6b)

Step (write this)	Why / what it means
$Q = 0.8 \times 4180 \times (80 - 30) = 167.2$ kW	
$\Delta T_1 = 120 - 80 = 40$ °C; $\Delta T_2 = 85 - 30 = 55$ °C; $\Delta T_{lm} = (40 - 55)/\ln(40/55) = 47.1$ °C	Counter flow.
$A = Q/(U \Delta T_{lm}) = 167{,}200/(125 \times 47.1) =$ **28.4 m^2**	Treated as pure counter flow, as the question says (no F correction).

Numerical 3: water heated by oil (2022 7c)

Step (write this)	Why / what it means
$\dot{m}_w = 68/60 = 1.133$ kg/s; $Q = 1.133 \times 4180 \times (75 - 35) = 189.5$ kW	
$\Delta T_1 = 110 - 75 = 35$; $\Delta T_2 = 75 - 35 = 40$; $\Delta T_{lm} = (40 - 35)/\ln(40/35) = 37.44$ °C	
$A = 189{,}500/(320 \times 37.44) =$ **15.8 m^2**	Holman Example 10-4 (15.84 m^2). Oil flow = Q/(1900 × 35) = 2.85 kg/s (not asked).

Numerical 4: engine oil cooler with fouling (2021 7c)

Step (write this)	Why / what it means
$Q = \dot{m}_{oil} c_{oil} \Delta T = 0.16 \times 2020 \times (80 - 50) = 9696$ W	
$T_{w,o} = 20 + 9696/(0.08 \times 4180) = 49.0$ °C	Water outlet.
$1/U = 1/h_w + R_{f,w} + R_{f,oil} + 1/h_{oil} = 0.01 + 0.00018 + 0.00018 + 0.0125 = 0.02286 \Rightarrow U = 43.7$ W/m^2°C	Thin wall, $A_i \approx A_o$.
$\Delta T_1 = 80 - 49.0 = 31.0$; $\Delta T_2 = 50 - 20 = 30$; $\Delta T_{lm} = 30.5$ °C	Counter flow.
$A = 9696/(43.7 \times 30.5) = 7.27$ m^2; $L = A/(\pi D_i) = 7.27/(\pi \times 0.025) =$ **92.5 m**	The small h values make U low and the tube long.

Numerical 5: oil-heated water exchanger and fouling after 3 years (2023 7c)

Step (write this)	Why / what it means
(i) $\Delta T_1 = 160 - 80 = 80$; $\Delta T_2 = 140 - 20 = 120$; $\Delta T_{lm} = (80 - 120)/\ln(80/120) = 98.65$ °C	Counter flow.
$A = q/(U\Delta T_{lm}) = 3000/(500 \times 98.65) = 0.0608$ m^2; $L = A/(\pi D_i) = 0.0608/(\pi \times 0.020) =$ **0.968 m**	
Heat capacity rates (unchanged): $C_c = 3000/60 = 50$ W/K; $C_h = 3000/20 = 150$ W/K	From the design.
(ii) $q = C_c(65 - 20) =$ **2250 W**; $T_{h,o} = 160 - 2250/150 =$ **145 °C**	
$\Delta T_1 = 160 - 65 = 95$; $\Delta T_2 = 145 - 20 = 125$; $\Delta T_{lm} = 109.3$ °C	
$U = q/(A\Delta T_{lm}) = 2250/(0.0608 \times 109.3) =$ **338 W/m^{2}K**	Same area.
$R_f = 1/U_{dirty} - 1/U_{clean} = 1/338.4 - 1/500 =$ **9.5×10^{-4} m^{2}K/W**	Incropera Problem 11.?; water-side fouling.

Numerical 6: gas-turbine lubricating oil cooler (2019 4c)

Step (write this)	Why / what it means
$q = \dot{m}_h c_h(T_{h,i} - T_{h,o}) = 0.1 \times 2131 \times 40 = 8524$ W; $T_{c,o} = 30 + 8524/(0.2 \times 4178) = 40.2$ °C	Energy balance.
Water (inner tube): $Re = 4\dot{m}/(\pi D_i \mu) = 4(0.2)/(\pi \times 0.025 \times 725\times10^{-6}) = 14{,}050$: turbulent	Water μ given as 725×10^{-6} N s/m^2.
$Nu = 0.023Re^{0.8}Pr^{0.4} = 0.023(14{,}050)^{0.8}(4.85)^{0.4} = 90.0$; $h_i = 90.0 \times 0.625/0.025 = 2250$ W/m^{2}K	Dittus-Boelter, fluid heated.
Oil (annulus): $D_h = D_o - D_i = 0.020$ m; $Re = 4\dot{m}/[\pi(D_o + D_i)\mu] = 4(0.1)/(\pi \times 0.070 \times 0.0325) = 56$: laminar	Annulus hydraulic diameter.
$D_i/D_o = 0.56 \Rightarrow Nu_i \approx 5.56$ (inner surface, outer insulated, Incropera Table 8.2); $h_o = 5.56 \times 0.138/0.020 = 38.4$ W/m^{2}K	
$U = 1/(1/2250 + 1/38.4) = 37.7$ W/m^{2}K	Thin wall.
$\Delta T_{lm} = [(100 - 40.2) - (60 - 30)]/\ln(59.8/30) = 43.2$ °C	
$L = q/(U\pi D_i \Delta T_{lm}) = 8524/(37.7 \times \pi \times 0.025 \times 43.2) =$ **66.6 m**	Incropera Example 11.1 (about 66 m; differences are rounding).

Numerical 7: milk chiller (2023 8c)

Part (ii), the water outlet temperature, is needed first.

Step (write this)	Why / what it means
$\dot{m}_{milk} = (250/3600) \times 10^{-3} \times 1030 = 0.07153$ kg/s; $C_m = 0.07153 \times 3860 = 276.1$ W/K	

$Q = 276.1 \times (38.6 - 13) = 7068$ W	
$\dot{m}_w = 0.72/3600 \times 1000 = 0.200$ kg/s; $C_w = 0.2 \times 4180 = 836$ W/K	
(ii) $T_{w,o} = 10 + 7068/836 =$ **18.5 °C**	
$\Delta T_1 = 38.6 - 18.5 = 20.15$; $\Delta T_2 = 13 - 10 = 3$; $\Delta T_{lm} = (20.15 - 3)/\ln(20.15/3) = 9.00$ °C	Counter flow.
(i) **$UA = Q/\Delta T_{lm} = 7068/9.00 = 785$ W/K**	
$A = 785/1000 = 0.785$ m^2; $L = A/(\pi D) = 0.785/(\pi \times 0.05) =$ **5.0 m**	

> **Memory trick:** Always do the energy balance first: you cannot get ΔT_{lm} without all four temperatures.

B4 Heat exchanger fundamentals: classification, temperature profiles, overall U, fouling factor, LMTD derivations

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [5] What is heat exchange? Its applications.
- **2023 Q8(a)** [8] Temperature profiles for counter and parallel flow; why counter flow is more effective.
- **2023 Q7(b)** [10] Show that $Q = UA_s\Delta T_{lm}$ for a counter-flow heat exchanger.
- **2022 Q7(a)** [8] Classification of heat exchangers; short note on the fouling factor.
- **2022 Q8(c)** [11] Derive the overall heat transfer coefficient of a heat exchanger including fouling factors.
- **2022 Q7(b)** [14] Derive LMTD (log mean temperature difference) for a counter-flow heat exchanger.
- **2021 Q8(a)** [6] Define heat exchanger. Classification of heat exchangers.
- **2021 Q8(b)** [14] What is effectiveness of a heat exchanger? Show that $U = 1/(1/h_i + 1/h_o)$ for a double-pipe heat exchanger.
- **2021 Q7(a)** [15] Show that $Q = UA_s\Delta T_{lm}$ for a parallel-flow heat exchanger.
- **2020 Q5(a)** [6] What is the fouling factor? Discuss.
- **2019 Q4(a)** [10] Define heat exchanger. Temperature profiles for counter and parallel flow. Why is counter flow more effective?
- **2019 Q4(b)** [10] Develop an expression for LMTD of a counter-flow heat exchanger.

★ REMEMBER IT

- Counter flow beats parallel flow: the cold fluid can leave **hotter** than the hot fluid's outlet, and ΔT stays more even along the length, so the same Q needs less area.
- Fouling factor $R_f = 1/U_{dirty} - 1/U_{clean}$: scale, rust, deposits add resistance in series. Overall: $1/U = 1/h_i + R_f + \text{wall} + 1/h_o$ (thin wall).
- LMTD $= (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$. Parallel: $\Delta T_1 = T_{h,in} - T_{c,in}$, $\Delta T_2 = T_{h,out} - T_{c,out}$. Counter: $\Delta T_1 = T_{h,in} - T_{c,out}$, $\Delta T_2 = T_{h,out} - T_{c,in}$.

Before you start: the job of a heat exchanger

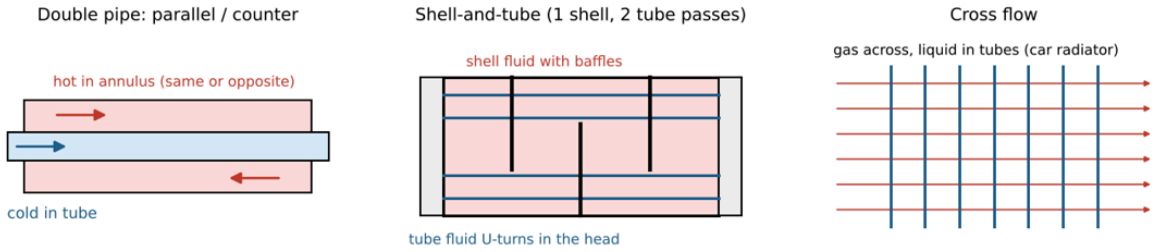
A **heat exchanger** moves heat from a hot fluid to a cold fluid, usually through a metal wall so the fluids do not mix: car radiators, condensers, evaporators, oil coolers, boiler economisers. The heat flow is $Q = UA\Delta T$, but ΔT changes along the exchanger, so we use the proper mean: the **log mean temperature difference**.

$C = \dot{m}c_p$ (W/K) is the **heat capacity rate**: how many watts a stream absorbs per degree of temperature change.

Definition, heat exchange and applications, classification (2021 8a, 2022 7a, 2023 7a)

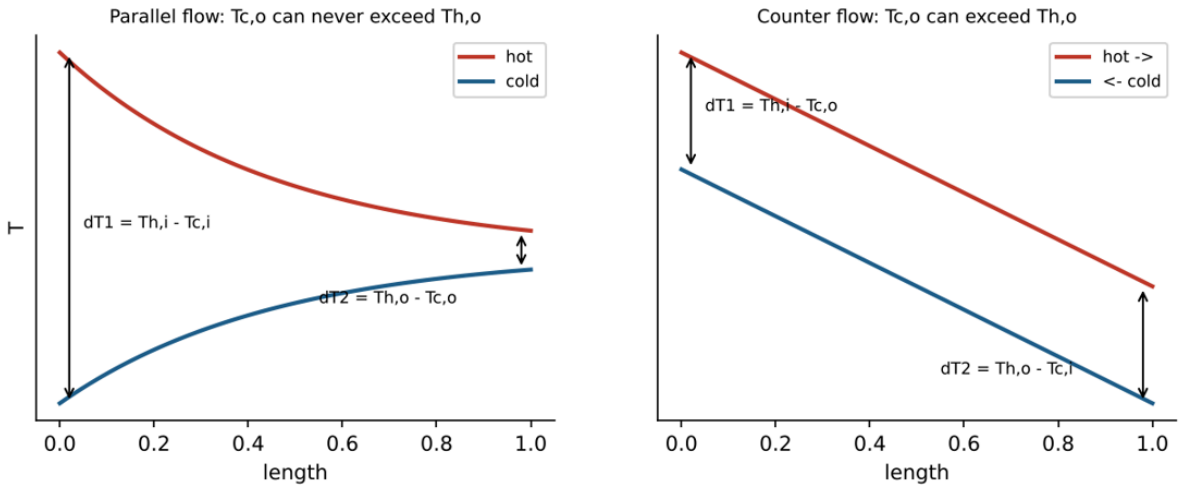
Heat exchanger: a device that transfers heat between two (or more) fluids at different temperatures, usually separated by a solid wall so that they do not mix. **Heat exchange** is this transfer of thermal energy from the hotter to the colder fluid, by convection on each side and conduction through the wall (radiation in furnaces). **Applications:** radiators and air-conditioning coils, refrigerator condensers and evaporators, boiler economisers, superheaters and air

preheaters, steam power plant condensers, oil coolers in engines and turbines, milk pasteurisers and chillers, chemical and petroleum process heaters, solar water heaters, waste-heat recovery.



Common types of heat exchanger

Basis	Types
Nature of heat exchange	**recuperator** (transfer type: fluids separated by a wall, e.g. shell-and-tube); **regenerator** (storage type: hot and cold fluids flow alternately over a matrix that stores heat, e.g. Ljungström air preheater); **direct contact** (fluids mix, e.g. cooling tower, jet condenser, open feed-water heater)
Flow arrangement	**parallel flow** (both enter at the same end and flow the same way); **counter flow** (opposite directions); **cross flow** (at right angles, mixed or unmixed); **multipass** (shell-and-tube with several tube passes)
Construction	double pipe (concentric tube); shell-and-tube; plate (gasketed plates); plate-fin and tube-fin; regenerative
Compactness	compact (area density > 700 m^2/m^3, e.g. car radiator) and non-compact
Phase change / service	condensers, evaporators, boilers, coolers, heaters



Temperature profiles in parallel-flow and counter-flow exchangers

Temperature profiles; why counter flow is more effective (2019 4a, 2023 8a)

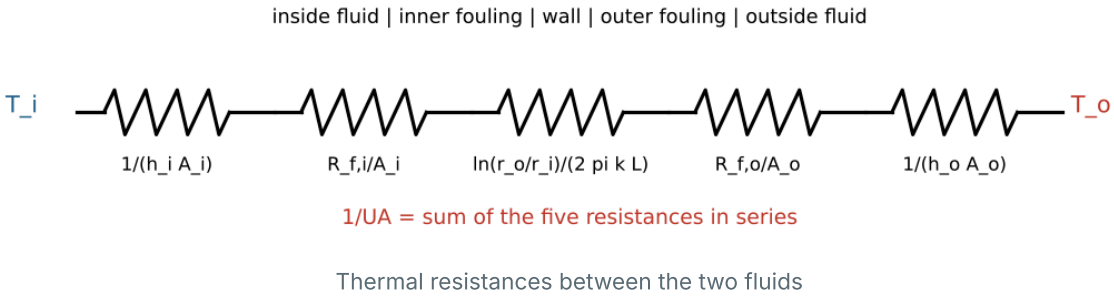
In **parallel flow** the temperature difference is very large at the inlet and becomes small at the outlet; the cold outlet can never be hotter than the hot outlet (both approach the same temperature). In **counter flow** the hot inlet meets the cold outlet, so ΔT is more nearly uniform along the length, and the cold fluid can leave hotter than the hot fluid leaves.

1. For the same inlet temperatures, the counter-flow LMTD is larger, so for the same U and A more heat is transferred (or less area is needed for the same duty).

2. The cold fluid can be heated to nearly the hot inlet temperature ($T_{c,o} > T_{h,o}$ possible), which is impossible in parallel flow.
3. The more uniform ΔT means less thermal stress and less entropy generation (less irreversibility).
4. For the same NTU (number of transfer units) and C_r, $\varepsilon_{counter} > \varepsilon_{parallel}$; parallel flow has $\varepsilon \le 1/(1 + C_r)$ (only 0.5 for equal C) while counter flow tends to 1 (see B6).

Fouling factor (2020 5a, 2022 7a)

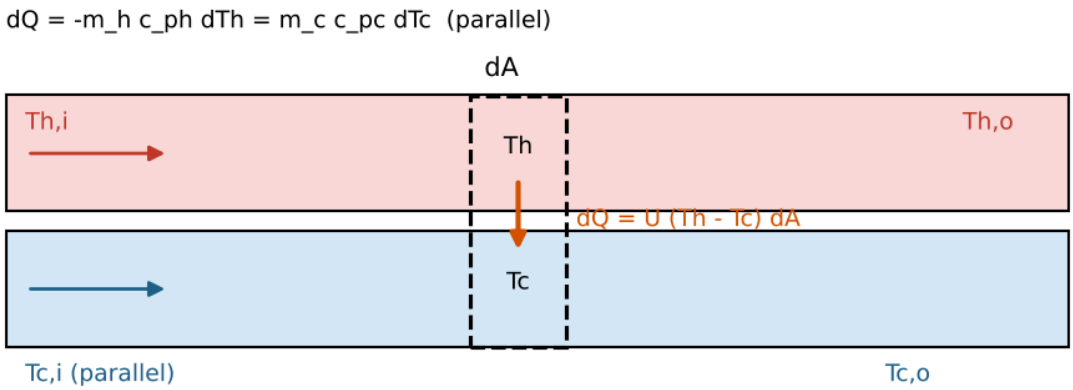
Fouling is the deposit of unwanted material (scale from hard water, rust, sludge, oil films, biological growth, soot) on heat transfer surfaces during operation. The deposit has low thermal conductivity, so it adds a thermal resistance and reduces the heat transfer (and raises the pressure drop). The **fouling factor** R_f (m^2K/W) is the measure of this extra resistance: $\mathbf{R_f = 1/U_{dirty} - 1/U_{clean}}$. Typical values: 0.0001 (clean water below 50 °C) to 0.0009 m^2K/W (fuel oil, river water). It depends on fluid, temperature, velocity and time; designers add it to size an exchanger that still works after fouling, and cleaning is scheduled when U falls too far.



Overall heat transfer coefficient with fouling (2022 8c) and for a double-pipe exchanger (2021 8b)

Step (write this)	Why / what it means
Heat passes in series: inside convection → inside fouling → wall conduction → outside fouling → outside convection	Same Q through each.
$R_{conv,i} = 1/h_iA_i$; $R_{f,i}/A_i$; $R_{wall} = \ln(D_o/D_i)/2\pi kL$; $R_{f,o}/A_o$; $R_{conv,o} = 1/h_oA_o$	$A_i = \pi D_iL$, $A_o = \pi D_oL$.
$Q = (T_h - T_c)/R_{total} = UA\Delta T \Rightarrow 1/UA = R_{total}$	Definition of U.
$\mathbf{1/UA_s = 1/U_iA_i = 1/U_oA_o = 1/h_iA_i + R_{f,i}/A_i + \ln(D_o/D_i)/2\pi kL + R_{f,o}/A_o + 1/h_oA_o}$	With fouling (2022 8c).
Clean, thin, highly conducting wall: $R_f = 0$, ln term ≈ 0, $A_i \approx A_o = A$	Double-pipe tube.
$\mathbf{1/U = 1/h_i + 1/h_o \Rightarrow U = 1/(1/h_i + 1/h_o)}$	2021 8b. U is always less than the smaller h.

Effectiveness (2021 8b): ε = actual heat transfer/maximum possible heat transfer = $Q/[C_{min}(T_{h,i} - T_{c,i})]$ (see B6).



Element of a heat exchanger

Q = UAΔT$_{lm}$ for parallel flow (2021 7a)

Assumptions: steady flow, no heat loss to the surroundings, constant U and specific heats, negligible kinetic/potential energy changes and axial conduction.

Step (write this)	Why / what it means
Element dA: $dQ = -\dot{m}_h c_{ph} dT_h = \dot{m}_c c_{pc} dT_c$	Hot cools, cold warms, both along x.
$dT_h = -dQ/C_h$, $dT_c = dQ/C_c \Rightarrow d(T_h - T_c) = -dQ(1/C_h + 1/C_c)$	
$dQ = U(T_h - T_c)dA \Rightarrow d(T_h - T_c)/(T_h - T_c) = -U\,dA(1/C_h + 1/C_c)$	Rate equation.
Integrate inlet (1) to outlet (2): $\ln(\Delta T_2/\Delta T_1) = -UA_s(1/C_h + 1/C_c)$	$\Delta T_1 = T_{h,i} - T_{c,i}$, $\Delta T_2 = T_{h,o} - T_{c,o}$.
Overall balance: $1/C_h = (T_{h,i} - T_{h,o})/Q$, $1/C_c = (T_{c,o} - T_{c,i})/Q$	$Q = C_h\Delta T_h = C_c\Delta T_c$.
$\ln(\Delta T_2/\Delta T_1) = -(UA_s/Q)[(T_{h,i} - T_{c,i}) - (T_{h,o} - T_{c,o})] = -(UA_s/Q)(\Delta T_1 - \Delta T_2)$	Add the two.
$Q = UA_s\Delta T_{lm}$, $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$	Log mean temperature difference.

LMTD for counter flow; Q = UAΔT$_{lm}$ (2022 7b, 2019 4b, 2023 7b)

Step (write this)	Why / what it means
Take x from the hot inlet. Both fluids' temperatures fall along x (the cold fluid flows the other way): $dQ = -C_h dT_h = -C_c dT_c$	Sign of dT_c changes compared with parallel flow.
$d(T_h - T_c) = -dQ(1/C_h - 1/C_c)$	
With $dQ = U(T_h - T_c)dA$: $d(\Delta T)/\Delta T = -U\,dA(1/C_h - 1/C_c)$	
Integrate: $\ln(\Delta T_2/\Delta T_1) = -UA_s(1/C_h - 1/C_c)$	$\Delta T_1 = T_{h,i} - T_{c,o}$ (x = 0), $\Delta T_2 = T_{h,o} - T_{c,i}$ (x = L).
$1/C_h - 1/C_c = [(T_{h,i} - T_{h,o}) - (T_{c,o} - T_{c,i})]/Q = (\Delta T_1 - \Delta T_2)/Q$	Rearranging the four temperatures.
$Q = UA_s(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$	Same form; only the end ΔTs differ.

If $C_h = C_c$: ΔT is constant and $\Delta T_{lm} = \Delta T_1 = \Delta T_2$	Special case.

Memory trick: “Ends, not middles”: ΔT_1 and ΔT_2 are the hot-minus-cold differences at the two physical ends. Parallel: in-in and out-out. Counter: hot-in minus cold-out, hot-out minus cold-in.

B1 Compressible flow basics: definitions, Mach number classes, speed of sound, stagnation properties, adiabatic Bernoulli

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [6] Define: compressible flow, sonic flow, shock waves.
- **2023 Q1(c)** [9] Define sound. Prove that c = √(γP/ρ) for one-dimensional isentropic flow of an ideal gas.
- **2023 Q1(b)** [9] What is stagnation pressure? Derive an expression for it in compressible flow.
- **2022 Q5(a)** [11] Differences between compressible and incompressible flow; classification of compressible flow by Mach number.
- **2021 Q5(a)** [6] What is compressible flow? Differentiate between compressible and incompressible flow.
- **2021 Q5(c)** [4] Critical properties and critical ratios; their significance.
- **2021 Q5(d)** [14] Air in a duct: stagnation 100 kPa abs and 35 °C, static 52 kPa abs. Velocity considering and ignoring compressibility. k = 1.4, R = 287.
- **2020 Q4(b)** [6] Differentiate between compressible and incompressible flow.
- **2019 Q5(a)** [4] Define: subsonic flow, Mach number.
- **2019 Q5(c)** [10] Prove that the velocity of sound in a compressible fluid is c = √(K/ρ), K = bulk modulus.
- **2019 Q5(b)** [9] Derive Bernoulli's equation for compressible flow when the process is adiabatic.

LEARN IT FROM ZERO

Compressible flow, from zero

Below about Mach 0.3 air behaves like water: density stays nearly constant. Faster than that the gas squeezes noticeably, so density, temperature and pressure all change together and the ordinary Bernoulli equation fails.

The key speed is the **speed of sound c = √(kRT)**, the speed at which a small pressure wave travels. **Mach number Ma = V/c**: < 1 subsonic, ≈ 1 sonic (transonic 0.8–1.2), > 1 supersonic, > 5 hypersonic.

Stagnation (total) properties are what you would get if you stopped the gas without losses: $T_0/T = 1 + (k - 1)Ma^2/2$ and $P_0/P = (T_0/T)^{k/(k-1)}$. A Pitot tube in a fast gas reads P_0.

CALCULATOR **Velocity from stagnation and static pressure (2021 5(d))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Static temperature: $T = T_0(P/P_0)^{(k-1)/k}$ = **308.1** × pow(**52** / **100**, (**1.4** - 1) / **1.4**) = **255.63** K
2. c_p: **1.4** × **287** / (**1.4** - 1) = **1004.5** J/kg K
3. Velocity (compressible): $V = \sqrt{[2c_p(T_0 - T)]}$ = √(2 × **1005** × (**308.1** - **255.6**)) = **324.8** m/s
4. Mach number: **324.8** / √(**1.4** × **287** × **255.6**) = **1.013**
5. Density at stagnation: **100** × 1000 / (**287** × **308.1**) = **1.1307** kg/m³

6. Velocity ignoring compressibility: $V = \sqrt{[2(P_0 - P)/\rho]} = \sqrt{(2 \times (\mathbf{100} - \mathbf{52}) \times 1000 / \mathbf{1.131})} = \mathbf{291.4}$ m/s

Before you start: when density is no longer constant

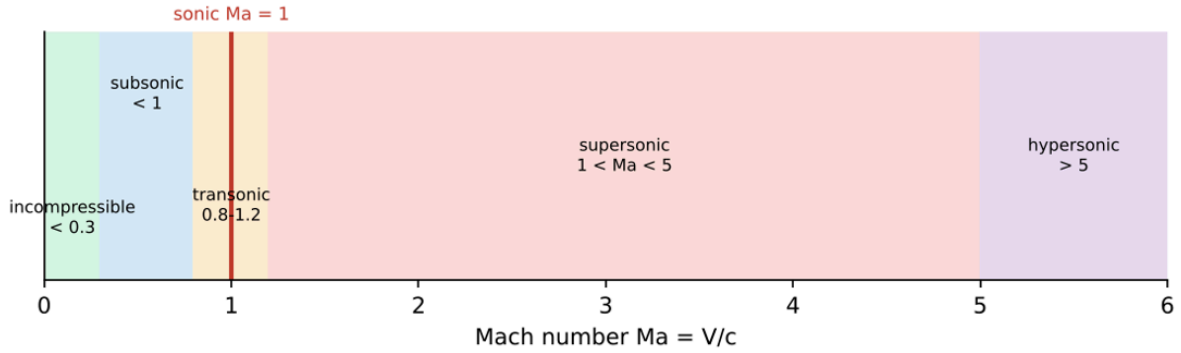
For liquids and slow gases the density hardly changes, so Bernoulli works. When a gas moves fast (above about 100 m/s, Ma > 0.3) its density changes noticeably with pressure and temperature: the flow is **compressible**, and kinetic energy and enthalpy trade places (fast gas is cold gas).

The key speed is the **speed of sound** c, the speed at which a small pressure disturbance travels. The **Mach number** Ma = V/c compares the flow speed with it. Answer source: the teachers' notes do not cover this unit, so these answers follow Cengel & Cimbala (Fluid Mechanics, ch. 12) and Bansal ch. 15, the books in the syllabus.

Compressible vs incompressible flow (2020 4b, 2021 5a, 2022 5a)

Compressible flow is flow in which the density of the fluid changes appreciably from point to point ($\Delta\rho/\rho$ > about 5%), e.g. gas flow at high speed in nozzles, turbines, jet engines. **Incompressible flow** has (practically) constant density.

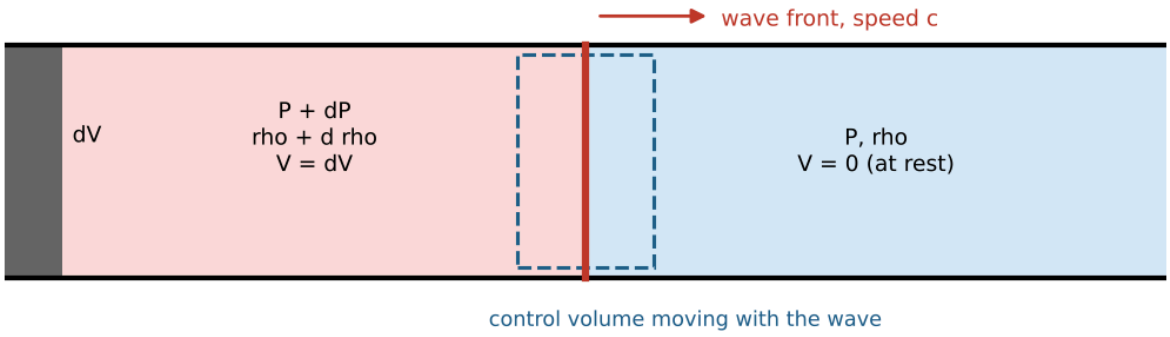
Compressible flow	Incompressible flow
ρ varies: ρ = f(p, T)	ρ = constant
gases at Ma > 0.3; high-speed flows, nozzles, shocks	liquids; gases at Ma < 0.3
energy equation includes enthalpy: $h + V^2/2$ = constant; thermodynamics is coupled to the flow	mechanical energy (Bernoulli) is enough; temperature decoupled
continuity ρAV = constant	AV = constant
pressure waves travel at finite speed c; shocks and choking can occur	pressure signals travel instantly (c → ∞)
Mach number is the key parameter	Reynolds number is the key parameter



Classification of flow by Mach number

Classification by Mach number and definitions (2022 5a, 2023 1a, 2019 5a)

Term	Meaning
Mach number Ma	ratio of the local flow velocity to the local speed of sound: Ma = V/c (square root of inertia force/elastic force)
Incompressible	Ma < 0.3: density change below about 5%
Subsonic flow	Ma < 1: flow slower than sound; disturbances travel upstream
Sonic flow	Ma = 1: flow velocity equals the speed of sound (e.g. at the throat of a choked nozzle)
Transonic	0.8 < Ma < 1.2: both subsonic and supersonic regions present
Supersonic	1 < Ma < 5: disturbances cannot travel upstream; Mach cone, shocks
Hypersonic	Ma > 5: strong heating, dissociation of gas
Compressible flow	flow with significant density variation (above)
Shock wave	a very thin region (a few μm) in a supersonic flow across which pressure, temperature and density rise suddenly and velocity falls; irreversible (see B3)



Pressure wave in a duct: control volume moving with the wave

Speed of sound: c = √(K/ρ) and c = √(γP/ρ) (2023 1c, 2019 5c)

Sound is a weak (infinitesimal) pressure wave travelling through a medium; its speed is the **speed of sound** c. A piston in a duct gives the gas a small velocity dV; a wave moves into the still gas at speed c. Take a control volume moving with the wave: gas enters at c with P, ρ and leaves at c − dV with P + dP, ρ + dρ.

Step (write this)	Why / what it means
Mass: $\rho A c = (\rho + d\rho)A(c - dV) \Rightarrow c\,d\rho - \rho\,dV = 0$ (drop $d\rho dV$)	$dV = c\,d\rho/\rho$.
Momentum: $PA - (P + dP)A = \rho Ac[(c - dV) - c] \Rightarrow dP = \rho c\,dV$	Net force = mass flow × change of velocity.
Combine: $dP = \rho c\,(c\,d\rho/\rho) = c^2 d\rho \Rightarrow \mathbf{c^2 = (\partial P/\partial \rho)_s}$	Process is isentropic (small, fast, reversible, adiabatic).
Bulk modulus $K = \rho\,dP/d\rho \Rightarrow \mathbf{c = \sqrt{(K/\rho)}}$	2019 5c (any fluid; water: $\sqrt{(2.2\times10^9/1000)} = 1483$ m/s).

Ideal gas, isentropic: $P/\rho^{\gamma} = C \Rightarrow dP/d\rho = \gamma C\rho^{\gamma-1} = \gamma P/\rho$	Differentiate.
$c = \sqrt{(\gamma P/\rho)} = \sqrt{(\gamma RT)}$	2023 1c. Air at 300 K: $\sqrt{(1.4 \times 287 \times 300)} = 347$ m/s.

Stagnation pressure (2023 1b)

Stagnation (total) pressure P_0 is the pressure a moving fluid would reach if it were brought to rest isentropically (adiabatically and without friction), e.g. at the nose of a Pitot tube. Similarly T_0 and ρ_0.

Step (write this)	Why / what it means
Steady adiabatic energy equation: $h + V^2/2 = h_0 \Rightarrow c_pT + V^2/2 = c_pT_0$	No work, no heat.
$T_0/T = 1 + V^2/(2c_pT)$; with $c_p = kR/(k - 1)$ and $c^2 = kRT$: **$T_0/T = 1 + (k - 1)Ma^2/2$**	
Isentropic: $P_0/P = (T_0/T)^{k/(k-1)}$	Stagnation is reached isentropically.
$P_0 = P[1 + (k - 1)Ma^2/2]^{k/(k-1)}$	Stagnation pressure.
Binomial expansion: $P_0 = P + (\rho V^2/2)[1 + Ma^2/4 + (2 - k)Ma^4/24 + \ldots]$	Bansal form; the bracket is the compressibility correction.
For $Ma \to 0$: $P_0 = P + \rho V^2/2$	Incompressible (Bernoulli) result.

Bernoulli's equation for adiabatic compressible flow (2019 5b)

Step (write this)	Why / what it means
Euler's equation: $dp/\rho + V\,dV + g\,dz = 0$	From A1.
Adiabatic (isentropic): $p/\rho^k = C \Rightarrow \rho = (p/C)^{1/k}$	
$\int dp/\rho = C^{1/k}\int p^{-1/k}dp = C^{1/k}p^{(k-1)/k}\,k/(k - 1) = [k/(k - 1)]p/\rho$	Use $C^{1/k} = p^{1/k}/\rho$.
$[k/(k - 1)](p/\rho g) + V^2/2g + z = \text{constant}$	Bernoulli for adiabatic flow; isothermal flow would give $(p/\rho g)\ln p$ instead.
Between two sections: $[k/(k - 1)](p_1/\rho_1 g) + V_1^2/2g + z_1 = [k/(k - 1)](p_2/\rho_2 g) + V_2^2/2g + z_2$	Used in A4 (air venturi).

Critical properties and ratios (2021 5c)

Critical properties (marked *) are the values of P, T, ρ, V at the section where Ma = 1. The **critical ratios** compare them with stagnation values: $T^*/T_0 = 2/(k + 1)$, $P^*/P_0 = [2/(k + 1)]^{k/(k-1)}$, $\rho^*/\rho_0 = [2/(k + 1)]^{1/(k-1)}$. For air (k = 1.4): **0.8333, 0.5283, 0.6339**. **Significance:** they fix the throat conditions of a choked nozzle; if the back pressure is below $P^* = 0.528P_0$ the flow chokes and the mass flow is maximum; A* is the reference area for A/A* tables; they are used to design nozzles and to check whether flow can become supersonic.

Numerical: velocity with and without compressibility (2021 5d)

Given: $P_0 = 100$ kPa, $T_0 = 308.15$ K, $P = 52$ kPa, $k = 1.4$, $R = 287$ ($c_p = kR/(k - 1) = 1004.5$ J/kg K).

Step (write this)	Why / what it means
Compressible: $T = T_0(P/P_0)^{(k-1)/k} = 308.15(0.52)^{0.2857} = 255.6$ K	Isentropic from stagnation.
$V = \sqrt{[2c_p(T_0 - T)]} = \sqrt{[2 \times 1004.5 \times 52.5]} =$ **324.8 m/s**	Ma = 324.8/√(1.4 × 287 × 255.6) = 1.01.
Ignoring compressibility: $\rho = \rho_0 = P_0/RT_0 = 100{,}000/(287 \times 308.15) = 1.131$ kg/m^3	Density fixed at its stagnation value.
$P_0 - P = \rho V^2/2 \Rightarrow V = \sqrt{[2 \times 48{,}000/1.131]} =$ **291.4 m/s**	Bernoulli.
Error = (324.8 - 291.4)/324.8 = 10%	Compressibility matters at this speed.

Side note: If the static density P/RT = 0.709 kg/m^3 is used instead, the incompressible answer is 368 m/s; state which density you use.

Memory trick: “c = root kRT”; stagnation: “one plus k-minus-one-over-two Ma squared” for T, raise to k/(k − 1) for P. Air critical: 0.833, 0.528, 0.634.

B3 Shock waves: normal and oblique shocks, Rayleigh line, normal-shock numericals

Next to add | Very likely · 93% | Medium

PAST-PAPER QUESTIONS

- **2023 Q2(a)** [8] Show that the Rayleigh line is $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$.
- **2023 Q2(b)** [12] Same as 2022 6(c) but helium (k = 1.667, R = 2.0769 kJ/kg K).
- **2022 Q6(a)** [6] Short notes: shock wave, normal shock wave, oblique shock wave.
- **2022 Q6(b)** [17] Show that $Ma_2^2 = (Ma_1^2 + 2/(k - 1))/(2kMa_1^2/(k - 1) - 1)$ across a normal shock.
- **2022 Q6(c)** [12] Air, normal shock at Ma = 2.5, upstream 61.64 kPa and 262.15 K: downstream pressure, temperature, velocity, Mach number, stagnation pressure.
- **2021 Q6(b)** [5] Differentiate between normal and oblique shock waves.
- **2021 Q6(c)** [14] Air enters a C-D (converging-diverging) nozzle at 1 MPa, 300 K, low velocity; normal shock at the exit plane at Ma = 2: pressure, temperature, Mach number, velocity after the shock.
- **2020 Q5(b)** [12] Normal shock in air: upstream Ma 2.2, 106 bar, 16 °C: downstream Mach number, pressure, temperature (k = 1.4).

UNDERSTAND **Read this to understand; you do not need to write it**

A **normal shock** is a very thin jump, square to the flow, where supersonic flow suddenly becomes subsonic. Pressure, temperature and density jump up; velocity and Mach drop; stagnation temperature stays the same (adiabatic) but **stagnation pressure drops** (it is irreversible, entropy rises). An **oblique shock** is tilted, turns the flow, and the flow behind it can stay supersonic.

CALCULATOR **Normal shock calculator (2020 5(b), 2021 6(c), 2022 6(c), 2023 2(b))**

Change any number; every step is recalculated and shown with the numbers put in.

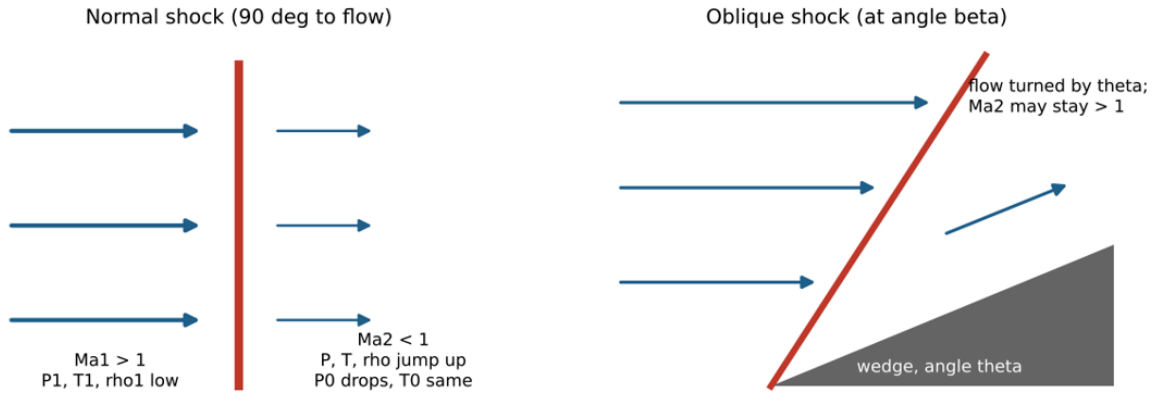
1. Downstream Mach Ma_2: $Ma_2^2 = [Ma_1^2 + 2/(k - 1)]/[2kMa_1^2/(k - 1) - 1] = \sqrt{((\mathbf{2.5} \times \mathbf{2.5} + 2 / (\mathbf{1.4} - 1)) / (2 \times \mathbf{1.4} \times \mathbf{2.5} \times \mathbf{2.5} / (\mathbf{1.4} - 1) - 1))} = \mathbf{0.513}$
2. Downstream pressure: $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2) = \mathbf{61.64} \times (1 + \mathbf{1.4} \times \mathbf{2.5} \times \mathbf{2.5}) / (1 + \mathbf{1.4} \times \mathbf{0.513} \times \mathbf{0.513}) = \mathbf{439.2}$ kPa
3. Downstream temperature: $T_2/T_1 = [1 + (k - 1)Ma_1^2/2]/[1 + (k - 1)Ma_2^2/2] = \mathbf{262.1} \times (1 + (\mathbf{1.4} - 1) / 2 \times \mathbf{2.5} \times \mathbf{2.5}) / (1 + (\mathbf{1.4} - 1) / 2 \times \mathbf{0.513} \times \mathbf{0.513}) = \mathbf{560.3}$ K
4. Downstream velocity: $\mathbf{0.513} \times \sqrt{(\mathbf{1.4} \times \mathbf{287} \times \mathbf{560.3})} = \mathbf{243.4}$ m/s
5. Downstream stagnation pressure: $\mathbf{439.2} \times \text{pow}(1 + (\mathbf{1.4} - 1) / 2 \times \mathbf{0.513} \times \mathbf{0.513}, \mathbf{1.4} / (\mathbf{1.4} - 1)) = \mathbf{525.6}$ kPa
6. Upstream stagnation pressure: $\mathbf{61.64} \times \text{pow}(1 + (\mathbf{1.4} - 1) / 2 \times \mathbf{2.5} \times \mathbf{2.5}, \mathbf{1.4} / (\mathbf{1.4} - 1)) = \mathbf{1053.2}$ kPa
7. Stagnation pressure lost: $100 \times (1 - \mathbf{525.6} / \mathbf{1053}) = \mathbf{50.1}$ %

Helium (2023 2(b)): k = 1.667, R = 2076.9 J/kg K.

Before you start: what a shock is

Supersonic flow cannot 'feel' an obstacle or a high back pressure ahead of it, because pressure signals travel only at the speed of sound. So the flow adjusts suddenly, across a very thin wave: a **shock wave**. Across it the velocity drops and P, T, ρ jump up. It is irreversible: entropy rises and stagnation pressure falls, but stagnation temperature stays the same (adiabatic).

All normal-shock numericals: (1) Ma_2 from Ma_1, (2) P_2/P_1 and T_2/T_1, (3) $V_2 = Ma_2\sqrt{(kRT_2)}$, (4) $P_{02} = P_2[1 + (k - 1)Ma_2^2/2]^{k/(k-1)}$. Use formulas or the normal-shock table.



Normal and oblique shocks

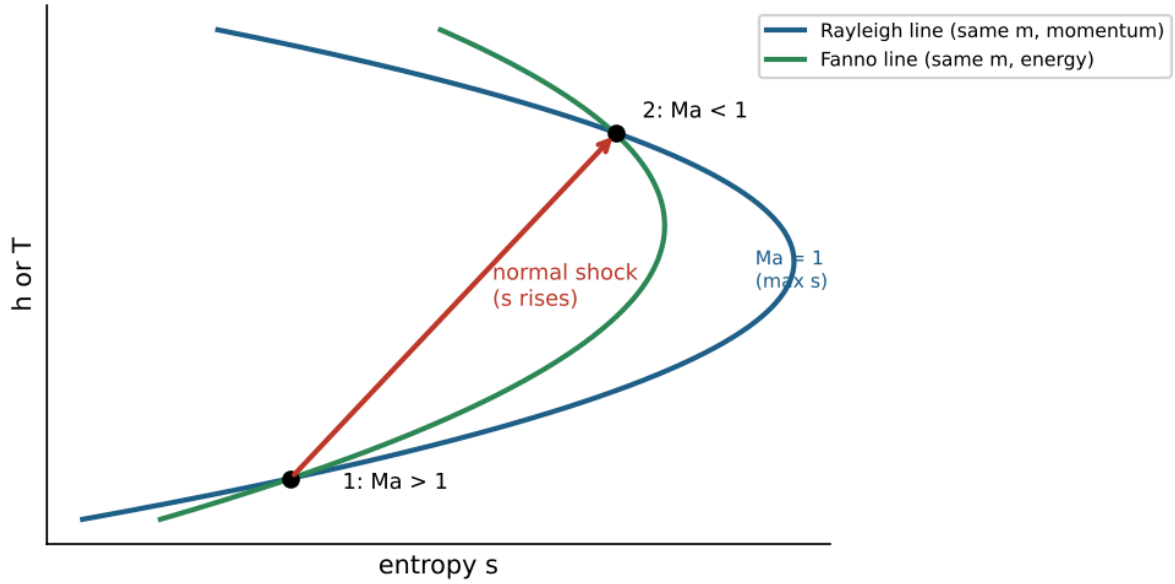
Shock wave, normal shock and oblique shock (2022 6a, 2021 6b)

- **Shock wave:** an extremely thin region (about 0.1-1 μm) in a supersonic flow across which the flow properties change abruptly: velocity and Mach number fall, pressure, temperature, density and entropy rise; the process is adiabatic and irreversible.
- **Normal shock wave:** a shock perpendicular to the flow direction; the flow is always supersonic upstream and subsonic downstream; flow direction does not change. Occurs in C-D nozzles at off-design back pressure, in supersonic inlets, ahead of blunt bodies.
- **Oblique shock wave:** a shock inclined at an angle β to the flow, formed when supersonic flow is turned into itself (wedge, compression corner). The flow is deflected by the angle θ; the downstream flow is usually still supersonic (weaker shock).

Normal shock	Oblique shock
perpendicular to the flow (β = 90°)	inclined at β < 90°
no change of flow direction	flow turned through angle θ
downstream always subsonic (Ma_2 < 1)	downstream usually supersonic (only the normal component becomes subsonic)
strongest: largest loss of P_0	weaker: smaller entropy rise and loss
1-D analysis	2-D analysis (θ-β-Ma relation)

Rayleigh line (2023 2a)

The **Rayleigh line** is the locus of states (on an h-s or T-s diagram) of a frictionless flow in a constant-area duct with heat transfer: all states have the same mass flux and the same momentum flux ($P + \rho V^2$). The **Fanno line** has the same mass flux and the same stagnation enthalpy (adiabatic flow with friction). A normal shock satisfies all three (mass, momentum, energy), so its upstream and downstream states are the two **intersections** of the Rayleigh and Fanno lines.



Rayleigh and Fanno lines; the normal shock joins their intersections

Step (write this)	Why / what it means
Momentum, constant area, no friction: $P_1 + \rho_1 V_1^2 = P_2 + \rho_2 V_2^2$	Rayleigh line condition.
$\rho V^2 = \rho Ma^2 c^2 = \rho Ma^2 (kP/\rho) = kPMa^2$	Ideal gas, $c^2 = kP/\rho$.
$P_1(1 + kMa_1^2) = P_2(1 + kMa_2^2)$	
$P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$	Equation of the Rayleigh line (also the normal-shock pressure ratio).

Mach number relation across a normal shock (2022 6b)

Step (write this)	Why / what it means
Energy (T_0 constant): $T_2/T_1 = [1 + (k - 1)Ma_1^2/2]/[1 + (k - 1)Ma_2^2/2]$	Adiabatic, no work.
Momentum (above): $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$	
Continuity $\rho_1 V_1 = \rho_2 V_2$ with $\rho = P/RT$, $V = Ma\sqrt{(kRT)}$: $P_1 Ma_1/\sqrt{T_1} = P_2 Ma_2/\sqrt{T_2} \Rightarrow P_2/P_1 = (Ma_1/Ma_2)\sqrt{(T_2/T_1)}$	
Equate the two P_2/P_1: $(1 + kMa_1^2)/(1 + kMa_2^2) = (Ma_1/Ma_2)\sqrt{\{[1 + (k - 1)Ma_1^2/2]/[1 + (k - 1)Ma_2^2/2]\}}$	One equation in Ma_1 and Ma_2.
Square and rearrange: $Ma_1^2[1 + (k - 1)Ma_1^2/2]/(1 + kMa_1^2)^2 = Ma_2^2[1 + (k - 1)Ma_2^2/2]/(1 + kMa_2^2)^2$	Same function of Ma on each side.
This is a quadratic in Ma_2^2; one root is $Ma_2 = Ma_1$ (no shock); the other root is	Divide out the trivial root.
$Ma_2^2 = [Ma_1^2 + 2/(k - 1)] / [2kMa_1^2/(k - 1) - 1]$	Check: $Ma_1 = 2$, $k = 1.4$: $(4 + 5)/(28 - 1) = 0.333$, $Ma_2 = 0.577$.

Numerical 1: normal shock at Ma 2.2 (2020 5b)

Step (write this)	Why / what it means
$Ma_2^2 = (2.2^2 + 5)/(7 \times 2.2^2 - 1) = 9.84/32.88 = 0.2993 \Rightarrow \mathbf{Ma_2 = 0.547}$	k = 1.4: 2/(k − 1) = 5, 2k/(k − 1) = 7.
$P_2/P_1 = (1 + 1.4 \times 4.84)/(1 + 1.4 \times 0.2993) = 7.776/1.419 = 5.48 \Rightarrow \mathbf{P_2 = 5.48 \times 106 = 581\ bar}$	
$T_2/T_1 = (1 + 0.2 \times 4.84)/(1 + 0.2 \times 0.2993) = 1.968/1.060 = 1.857 \Rightarrow \mathbf{T_2 = 1.857 \times 289.15 = 536.9\ K\ (263.8\ °C)}$	

Side note: 106 bar upstream is unusually high (probably meant 1.06 bar, giving 5.81 bar). The ratios do not depend on it, so the answer scales directly.

Numerical 2: shock at the exit of a C-D nozzle (2021 6c)

Step (write this)	Why / what it means
Before the shock (isentropic from P_0 = 1 MPa, T_0 = 300 K to Ma = 2): $P_1 = 1.0/1.8^{3.5} = 0.1278$ MPa, $T_1 = 300/1.8 = 166.7$ K	
$Ma_2^2 = (4 + 5)/(28 - 1) = 0.3333 \Rightarrow \mathbf{Ma_2 = 0.577}$	
$P_2/P_1 = (1 + 5.6)/(1 + 0.4667) = 4.5 \Rightarrow \mathbf{P_2 = 0.575\ MPa}$	
$T_2/T_1 = 1.8/1.0667 = 1.6875 \Rightarrow \mathbf{T_2 = 281.3\ K}$	
$V_2 = Ma_2\sqrt{(kRT_2)} = 0.577\sqrt{(1.4 \times 287 \times 281.3)} = \mathbf{194\ m/s}$	Cengel Example 12-8 style. (P_{02} = 0.721 MPa.)

Numerical 3: air, normal shock at Ma 2.5 (2022 6c)

Step (write this)	Why / what it means
$Ma_2^2 = (6.25 + 5)/(7 \times 6.25 - 1) = 11.25/42.75 = 0.2632 \Rightarrow \mathbf{Ma_2 = 0.513}$	
$P_2/P_1 = (1 + 8.75)/(1 + 0.3684) = 7.125 \Rightarrow \mathbf{P_2 = 439.2\ kPa}$	
$T_2/T_1 = (1 + 1.25)/(1 + 0.05263) = 2.1375 \Rightarrow \mathbf{T_2 = 560.3\ K}$	
$V_2 = 0.513\sqrt{(1.4 \times 287 \times 560.3)} = \mathbf{243.4\ m/s}$ (V_1 = 811.4 m/s)	
$P_{01} = 61.64(1 + 0.2 \times 6.25)^{3.5} = 1053$ kPa; $\mathbf{P_{02} = P_2(1 + 0.2 \times 0.2632)^{3.5} = 525.6\ kPa}$	Stagnation pressure halves (P_{02}/P_{01} = 0.499); T_0 = 589.8 K both sides. Cengel Example 12-7.

Numerical 4: the same shock in helium (2023 2b)

k = 1.667, R = 2076.9 J/kg K: 2/(k − 1) = 3.0, 2k/(k − 1) = 5.0, (k − 1)/2 = 0.3335, k/(k − 1) = 2.5.

Step (write this)	Why / what it means
$Ma_2^2 = (6.25 + 3.0)/(5.0 \times 6.25 - 1) = 9.25/30.24 = 0.3059 \Rightarrow \mathbf{Ma_2 = 0.553}$	

$P_2/P_1 = (1 + 1.667 \times 6.25)/(1 + 1.667 \times 0.3059) = 11.42/1.510 = 7.563 \Rightarrow \mathbf{P_2 = 466.2\ kPa}$	
$T_2/T_1 = (1 + 0.3335 \times 6.25)/(1 + 0.3335 \times 0.3059) = 3.084/1.102 = 2.799 \Rightarrow \mathbf{T_2 = 733.7\ K}$	
$V_2 = 0.553\sqrt{(1.667 \times 2076.9 \times 733.7)} = \mathbf{881\ m/s}$ ($V_1 = 2382$ m/s)	
$P_{01} = 61.64(3.084)^{2.5} = 1029$ kPa; $\mathbf{P_{02} = 466.2(1.102)^{2.5} = 594.3\ kPa}$	Helium (higher k) heats more, loses less P_0 than air.

Memory trick: Normal shock: “Mach down, everything else up, except P_0 (down) and T_0 (same)”. For air: $2/(k-1) = 5$ and $2k/(k-1) = 7$, so $Ma_2^2 = (Ma_1^2 + 5)/(7Ma_1^2 - 1)$.

B6 Effectiveness-NTU method Pass set Very likely · 93% Medium

PAST-PAPER QUESTIONS

- **2023 Q8(b)** [15] Same derivation as 2020 6(a).
- **2022 Q8(a)** [6] What is effectiveness of a heat exchanger? Its significance.
- **2022 Q8(b)** [6] Justify: a counter-flow heat exchanger has higher effectiveness than a parallel-flow one under the same conditions.
- **2022 Q8(d)** [12] Counter flow, A = 12.5 m^2: oil (2000 J/kg°C) at 100 °C, 2 kg/s; water (4170) at 20 °C, 0.48 kg/s; U = 400: water outlet temperature and Q.
- **2021 Q7(b)** [5] Physical significance of the NTU (number of transfer units) method.
- **2021 Q8(c)** [15] Counter flow: oil 0.55 kg/s (c_p 2.45 kJ/kg°C) 115 → 40 °C, water 15 → 75 °C, U = 1450: by NTU, water flow, effectiveness, area.
- **2020 Q6(a)** [20] Show that $\varepsilon = [1 - e^{-NTU(1 + C_{min}/C_{max})}]/(1 + C_{min}/C_{max})$ for parallel flow.

LEARN IT FROM ZERO

Effectiveness-NTU, from zero

LMTD (log mean temperature difference) needs all four temperatures. When you know only the **inlets** and the exchanger size, use ε-NTU instead.

- **Heat capacity rates** $C = \dot{m}c_p$ for each fluid; C_{min} is the smaller, $c = C_{min}/C_{max}$.
- **$Q_{max} = C_{min}(T_{h,in} - T_{c,in})$**: the most heat any exchanger could pass (infinitely long counter flow).
- **Effectiveness $\varepsilon = Q/Q_{max}$**.
- **NTU = UA/C_{min}**: the "size" of the exchanger. Bigger NTU, higher ε, with diminishing returns.

Parallel flow: $\varepsilon = [1 - e^{-NTU(1+c)}]/(1 + c)$. Counter flow: $\varepsilon = [1 - e^{-NTU(1-c)}]/[1 - c\, e^{-NTU(1-c)}]$. Then $Q = \varepsilon Q_{max}$ and the outlets follow.

CALCULATOR **ε-NTU: outlet temperatures from inlets only (2022 8(d))**

Change any number; every step is recalculated and shown with the numbers put in.

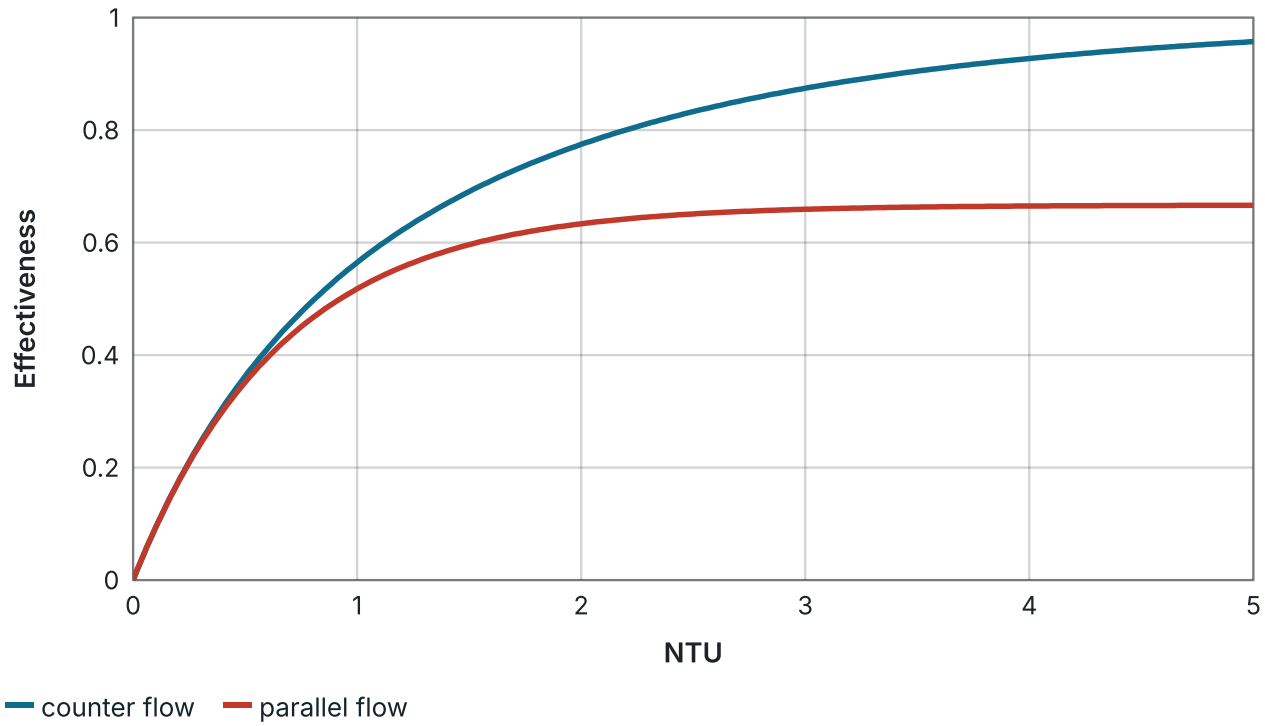
1. C_h: **2** × **2000** = **4000.0** W/K
2. C_c: **0.48** × **4170** = **2001.6** W/K
3. C_{min}: min(**4000**, **2002**) = **2001.6** W/K
4. $c = C_{min}/C_{max}$: **2002** / max(**4000**, **2002**) = **0.5004**
5. NTU: NTU = UA/C_{min} = **400** × **12.5** / **2002** = **2.498**
6. Effectiveness: **1** == 1 ? (1 - exp(-**2.498** × (1 - **0.5004**))) / (1 - **0.5004** × exp(-**2.498** × (1 - **0.5004**))) : (1 - exp(-**2.498** × (1 + **0.5004**))) / (1 + **0.5004**) = **0.8325**
7. Heat transferred: $Q = \varepsilon C_{min}(T_{h,i} - T_{c,i})$ = **0.8325** × **2002** × (**100** - **20**) / 1000 = **133.3** kW
8. Cold out: **20** + **133.3** × 1000 / **2002** = **86.6** °C

9. Hot out: **100** - **133.3** × 1000 / **4000** = **66.7** °C

For c = 1 in counter flow use ε = NTU/(1 + NTU) (the formula above divides by zero).

INTERACTIVE **Effectiveness vs NTU: counter flow beats parallel flow**

Move the sliders and watch the curve change.



Parallel-flow limit 1/(1 + c): **0.667**

2022 8(b): at the same NTU and c, the counter-flow curve is always higher. Parallel flow can never pass 1/(1 + c).

Before you start: why a second method

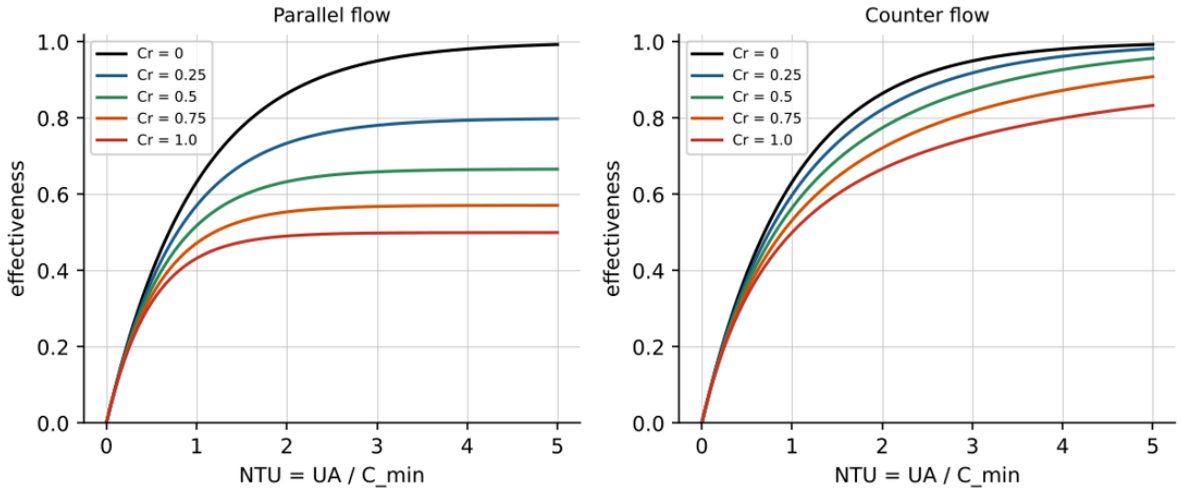
LMTD is easy when all four temperatures are known (design/sizing problem). When an exchanger already exists (area known) and the outlet temperatures are unknown (rating problem), LMTD needs trial and error. The **effectiveness-NTU method** solves it directly. Three numbers: $C_r = C_{min}/C_{max}$, NTU = UA/C_{min} (the 'size' of the exchanger), and $\varepsilon = Q/Q_{max}$ (how good it is).

Effectiveness and its significance (2022 8a, 2021 8b)

Effectiveness ε = actual heat transfer rate / maximum possible heat transfer rate = Q/Q_{max}, with $Q_{max} = C_{min}(T_{h,i} - T_{c,i})$: the heat that would be transferred in an infinitely long counter-flow exchanger, where the fluid with the smaller C would undergo the full temperature difference. **Significance:** $0 \le \varepsilon \le 1$ measures the thermal performance; once ε is known, $Q = \varepsilon C_{min}(T_{h,i} - T_{c,i})$ and both outlet temperatures follow without iteration; it allows comparison of different exchanger types and flow arrangements.

Physical significance of NTU (2021 7b)

NTU = UA_s/C_{min} (number of transfer units) is a dimensionless measure of the heat transfer size of the exchanger: the ratio of the exchanger's heat transfer conductance UA to the heat capacity rate of the weaker fluid. Large NTU means a large area (or high U) compared with the flow: the fluid gets close to the limiting temperature and ε is high. Beyond NTU ≈ 3 ε rises very little, so a bigger exchanger is not economical. The method lets outlet temperatures be found directly when the exchanger size is known.



Effectiveness vs NTU for parallel and counter flow

Counter flow has higher effectiveness (2022 8b)

For the same NTU and C_r, $\varepsilon_{counter} = [1 - e^{-NTU(1-C_r)}]/[1 - C_r e^{-NTU(1-C_r)}]$ is always larger than $\varepsilon_{parallel} = [1 - e^{-NTU(1+C_r)}]/(1 + C_r)$. Example NTU = 2, C_r = 1: parallel 0.49, counter 0.67. As NTU → ∞ parallel flow is limited to $1/(1 + C_r)$ because both outlets approach one common temperature, while counter flow approaches 1 because the cold fluid can be heated up to the hot inlet temperature. Physically, the temperature difference stays large over the whole length in counter flow (larger LMTD). (Only when C_r = 0, e.g. a condensing or boiling fluid, are the two equal: $\varepsilon = 1 - e^{-NTU}$.)

Effectiveness of a parallel-flow exchanger (2020 6a, 2023 8b)

Step (write this)	Why / what it means
From the LMTD derivation (B4): $\ln[(T_{h,o} - T_{c,o})/(T_{h,i} - T_{c,i})] = -UA_s(1/C_h + 1/C_c)$	Parallel flow.
Let the cold fluid be C_{min}: $-(UA_s/C_c)(1 + C_c/C_h) = -NTU(1 + C_r)$	NTU = UA_s/C_{min}, $C_r = C_{min}/C_{max}$. (Same result if hot is C_{min}.)
$\varepsilon = C_c(T_{c,o} - T_{c,i})/[C_{min}(T_{h,i} - T_{c,i})] \Rightarrow T_{c,o} = T_{c,i} + \varepsilon(T_{h,i} - T_{c,i})$	Definition.
Energy balance: $T_{h,o} = T_{h,i} - (C_c/C_h)(T_{c,o} - T_{c,i}) = T_{h,i} - C_r\varepsilon(T_{h,i} - T_{c,i})$	
$T_{h,o} - T_{c,o} = (T_{h,i} - T_{c,i}) - \varepsilon(1 + C_r)(T_{h,i} - T_{c,i})$	Subtract.
$(T_{h,o} - T_{c,o})/(T_{h,i} - T_{c,i}) = 1 - \varepsilon(1 + C_r)$	

$\ln[1 - \varepsilon(1 + C_r)] = -NTU(1 + C_r) \Rightarrow 1 - \varepsilon(1 + C_r) = e^{-NTU(1+C_r)}$	Combine with the first step.
$\boldsymbol{\varepsilon = [1 - e^{-NTU(1 + C_{min}/C_{max})}]/(1 + C_{min}/C_{max})}$, $NTU = U_oA_o/C_{min}$	The required result.

Numerical 1: oil cooler by NTU (2021 8c)

Step (write this)	Why / what it means
$Q = \dot{m}_{oil}c_{oil}\Delta T = 0.55 \times 2450 \times (115 - 40) = 101.06$ kW	
(i) $\dot{m}_w = Q/(c_w\Delta T_w) = 101{,}060/(4180 \times 60) =$ **0.403 kg/s**	c_w = 4.18 kJ/kg K.
$C_{oil} = 1347.5$ W/K $= C_{min}$; $C_w = 1684.4$ W/K; $C_r = 0.80$	
(ii) $\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})] = (115 - 40)/(115 - 15) =$ **0.75**	Oil (C_{min}) has the larger temperature change.
Counter flow: $NTU = [1/(C_r - 1)] \ln[(\varepsilon - 1)/(\varepsilon C_r - 1)] = (1/-0.2) \ln(-0.25/-0.40) = 2.35$	
(iii) $A = NTU \times C_{min}/U = 2.35 \times 1347.5/1450 =$ **2.18 m²**	Check by LMTD: $\Delta T_{lm} = (40 - 25)/\ln(40/25) = 31.9$, $A = 101{,}060/(1450 \times 31.9) = 2.18$ m².

Numerical 2: rating an existing exchanger (2022 8d)

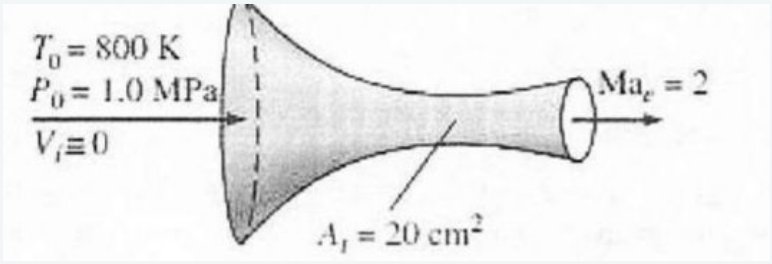
Step (write this)	Why / what it means
$C_h = 2 \times 2000 = 4000$ W/K; $C_c = 0.48 \times 4170 = 2001.6$ W/K $= C_{min}$; $C_r = 0.5004$	
$NTU = UA/C_{min} = 400 \times 12.5/2001.6 = 2.498$	
$\varepsilon = [1 - e^{-2.498(0.4996)}]/[1 - 0.5004e^{-2.498(0.4996)}] = (1 - 0.2871)/(1 - 0.1437) = 0.8325$	Counter-flow formula.
$Q = \varepsilon C_{min}(T_{h,i} - T_{c,i}) = 0.8325 \times 2001.6 \times 80 =$ **133.3 kW**	
$T_{c,o} = 20 + 133{,}300/2001.6 =$ **86.6 °C** (and $T_{h,o} = 100 - 133{,}300/4000 = 66.7$ °C)	Holman Example 10-?; LMTD would need iteration here.

Memory trick: "C-min decides everything": Q_{max}, NTU and ε all use C_{min}. Parallel ε formula: "one minus e to the minus NTU(1 + C_r), over (1 + C_r)".

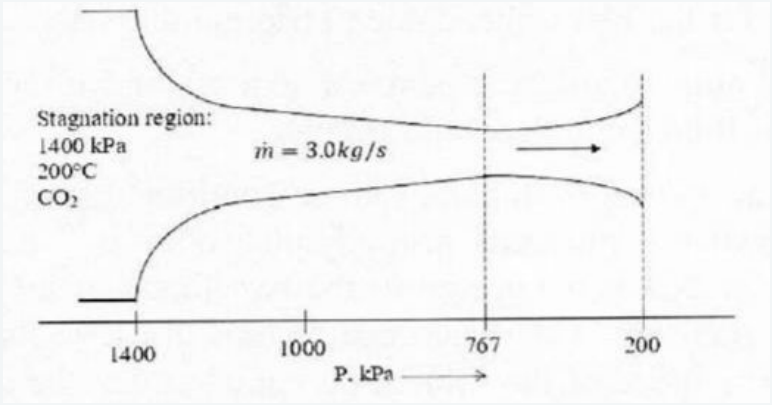
B2 Isentropic flow in nozzles and diffusers: area-velocity relation, choking, back pressure, nozzle numericals Pass set Very likely · 100% Hard

PAST-PAPER QUESTIONS

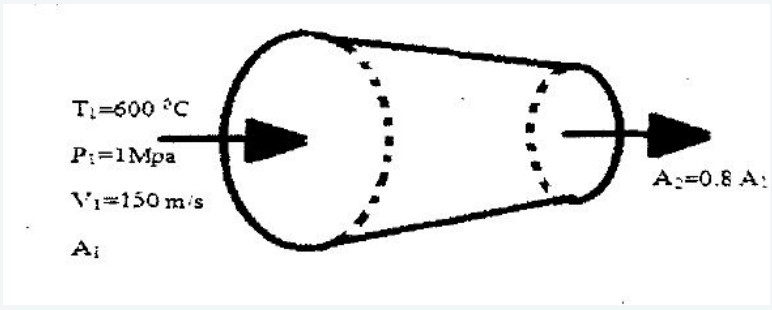
- **2023 Q1(d)** [11] Air enters a C-D (converging-diverging) nozzle at 1.0 MPa, 800 K, negligible velocity, k = 1.4; exit Ma = 2, throat 20 cm^2: throat conditions, exit conditions and exit area, mass flow rate.



- **2022 Q5(b)** [12] Effect of back pressure on mass flow rate and pressure distribution along a converging nozzle, with sketch.
- **2022 Q5(c)** [12] Air enters a nozzle at 0.2 MPa, 350 K, 150 m/s, isentropic: pressure and temperature where V = speed of sound; area ratio to the entrance.
- **2021 Q5(b)** [11] Show mathematically that a converging-diverging duct can be used as a nozzle or a diffuser.
- **2021 Q6(a)** [16] Prove $\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}(2/(k+1))^{(k+1)/[2(k-1)]}$.
- **2020 Q4(a)** [12] Prove that velocity and flow area change in the same direction for supersonic isentropic flow.
- **2020 Q4(c)** [12] CO_2, 3.00 kg/s, enters at 1400 kPa, 200 °C, low velocity, expands to 200 kPa isentropically: density, velocity, area and Mach number where P = 1200 kPa. c_p = 0.846, k = 1.289, R = 0.1889 kJ/kg K.



- **2019 Q5(d)** [12] Air at 1 MPa, 600 °C (150 m/s in the figure) enters a converging nozzle, throat 50 cm^2: mass flow rate for back pressure (i) 0.7 MPa, (ii) 0.4 MPa.

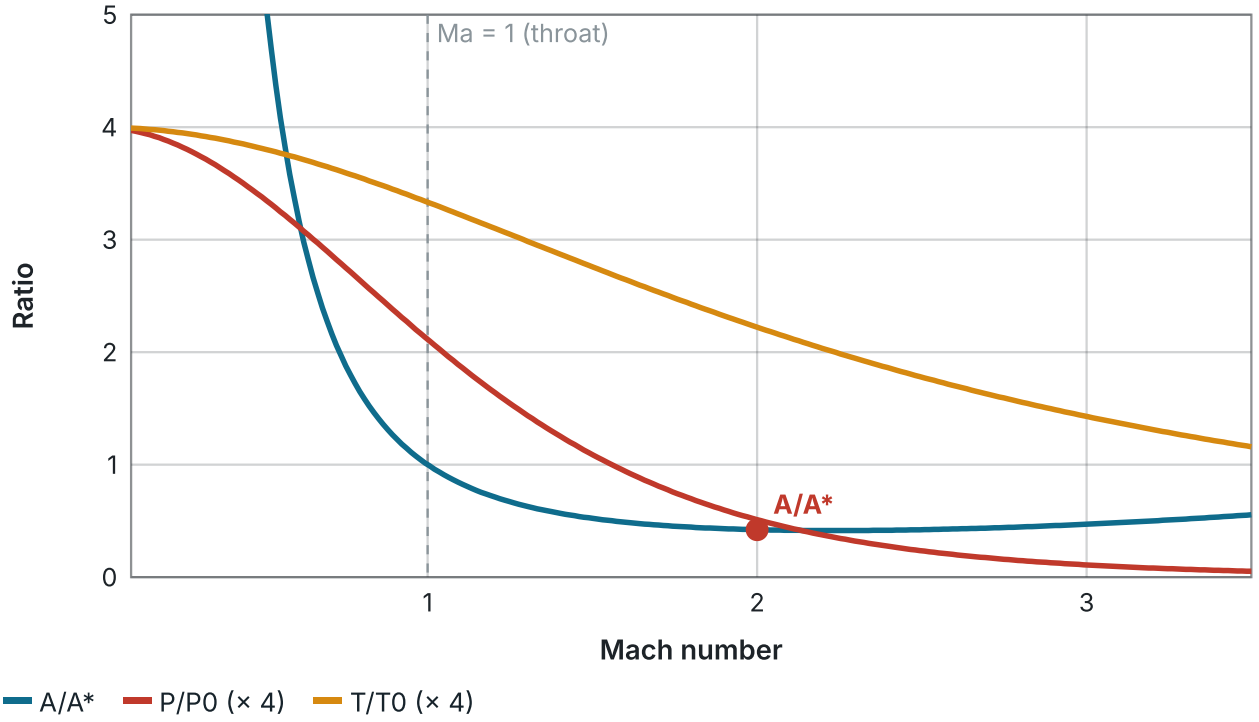


UNDERSTAND **Read this to understand; you do not need to write it**

$dA/A = (Ma^2 - 1)dV/V$ explains every nozzle. Subsonic (Ma < 1): to speed up, the area must *shrink*. Supersonic (Ma > 1): to speed up, the area must *grow*. So reaching supersonic needs a converging-diverging nozzle with Ma = 1 exactly at the throat. Once the throat is sonic, lowering the back pressure further does not raise the mass flow: the nozzle is **choked** ($P^* = 0.528P_0$ for air).

INTERACTIVE **Isentropic flow: area ratio and property ratios vs Mach number**

Move the sliders and watch the curve change.



T/T_0: **0.5556** · P/P_0: **0.1278** · A/A*: **0.4219** · Critical P^*/P_0: **0.5283**

A/A* has its minimum, 1, at Ma = 1: the throat. Each A/A* above 1 has one subsonic and one supersonic Mach number.

CALCULATOR **Converging nozzle mass flow, choked or not (2019 5(d))**

Change any number; every step is recalculated and shown with the numbers put in.

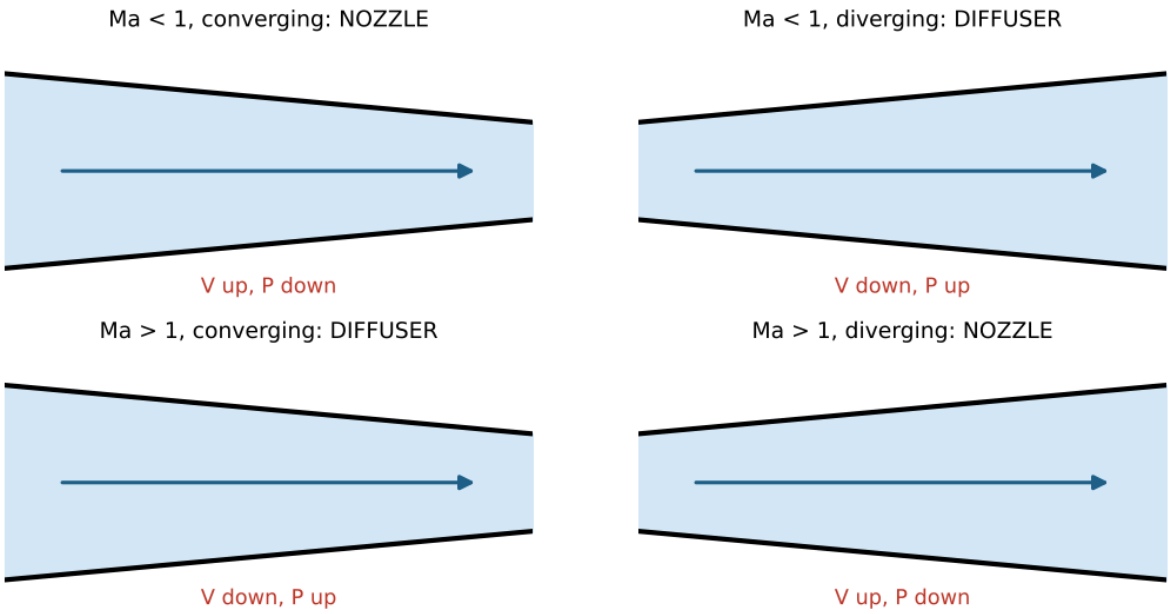
1. Critical pressure P*: $P^* = P_0[2/(k + 1)]^{k/(k-1)}$ = **1046** × pow(2 / (**1.4** + 1), **1.4** / (**1.4** - 1)) = **552.4** kPa
2. Exit pressure: P_e = larger of P_b and P* = max(**700**, **552.4**) = **700.0** kPa
3. Exit Mach: √(2 / (**1.4** - 1) × (pow(**1046** / **700**, (**1.4** - 1) / **1.4**) - 1)) = **0.779**
4. Exit temperature: **884.3** / (1 + (**1.4** - 1) / 2 × **0.7794** × **0.7794**) = **788.5** K
5. Mass flow: $\dot{m} = \rho_e A V_e$ = **700** × 1000 / (**287** × **788.5**) × **0.7794** × √(**1.4** × **287** × **788.5**) × **50** / 10^4 = **6.785** kg/s

Default P_0 and T_0 include the 150 m/s inlet velocity (T_0 = 873.15 + 150²/2010 = 884.3 K, P_0 = 1.0(884.3/873.15)$^{3.5}$ = 1.0456 MPa). Change P_b to 400 kPa: the flow is choked and stops rising.

Before you start: why a supersonic nozzle must diverge

For a liquid, a narrower pipe always means faster flow. For a gas this is true only while the flow is subsonic. Above Ma = 1 the gas expands (its density falls) faster than the velocity rises, so it needs a **bigger** area to go faster. A rocket nozzle therefore converges to a throat (Ma = 1) and then diverges.

Use stagnation values (P_0, T_0) as the reference: they stay constant through an isentropic nozzle.



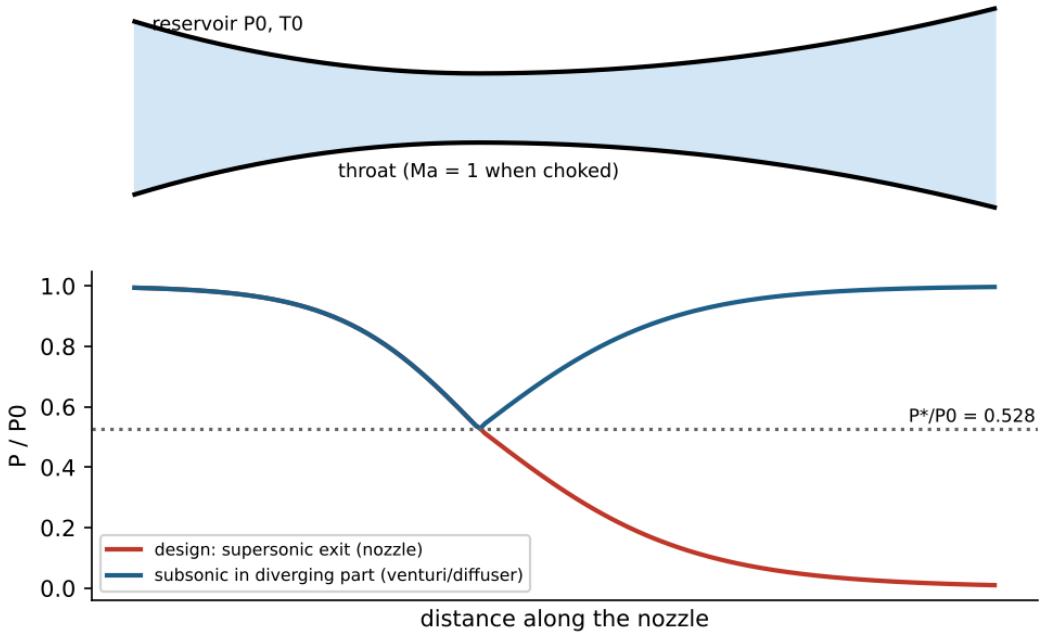
Nozzles and diffusers in subsonic and supersonic flow

Area-velocity relation: velocity and area in supersonic flow, and C-D duct as nozzle or diffuser (2020 4a, 2021 5b)

Step (write this)	Why / what it means
Continuity: ρAV = constant ⇒ $d\rho/\rho + dA/A + dV/V = 0$	Log-differentiate.
Euler (isentropic, horizontal): $dP = -\rho V\, dV$	Momentum.
$dP = c^2 d\rho \Rightarrow d\rho/\rho = dP/(\rho c^2) = -V\, dV/c^2 = -Ma^2 dV/V$	Speed of sound.
$dA/A = -dV/V - d\rho/\rho = -dV/V + Ma^2 dV/V$	Substitute.
$dA/A = (Ma^2 - 1)\, dV/V$ and $dA/A = [(1 - Ma^2)/\rho V^2]\, dP$	Area-velocity relation.

Case	Sign	Result
Ma < 1	$Ma^2 - 1 < 0$: dA and dV opposite	converging (dA < 0) speeds the flow up (nozzle); diverging slows it (diffuser)
Ma > 1	$Ma^2 - 1 > 0$: dA and dV **same sign**	diverging (dA > 0) speeds the flow up (nozzle); converging slows it (diffuser). This proves 2020 4a
Ma = 1	dA = 0	sonic speed can occur only where the area is a minimum: the throat

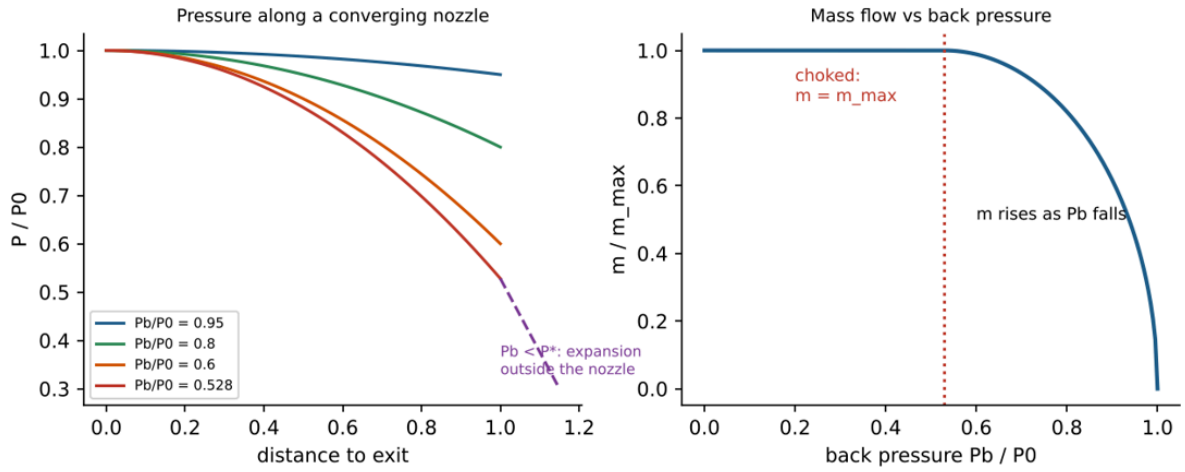
C-D duct as nozzle or diffuser (2021 5b): subsonic flow entering a converging-diverging duct speeds up in the converging part; if Ma reaches 1 at the throat, it keeps speeding up in the diverging part (supersonic) and the duct is a **nozzle** (dV > 0 throughout). If instead supersonic flow enters, the converging part slows it to Ma = 1 at the throat and the diverging part slows it further as subsonic flow: the duct is a **diffuser** (dV < 0 throughout, pressure rises). Which one depends on the inlet Mach number and back pressure; this follows directly from the sign of $(Ma^2 - 1)$.



Pressure along a converging-diverging nozzle

Maximum mass flow rate (2021 6a)

Step (write this)	Why / what it means
$\dot{m} = \rho AV = (P/RT)\ A\ Ma\sqrt{(kRT)} = PAMa\sqrt{(k/RT)}$	Ideal gas and V = Ma c.
$P = P_0[1 + (k - 1)Ma^2/2]^{-k/(k-1)}$, $T = T_0[1 + (k - 1)Ma^2/2]^{-1}$	Isentropic relations.
$\dot{m} = AP_0Ma\sqrt{(k/RT_0)} / [1 + (k - 1)Ma^2/2]^{(k+1)/[2(k-1)]}$	Powers: $k/(k-1) - 1/2 = (k+1)/[2(k-1)]$.
For fixed P_0, T_0, $\dot{m}/A$ is largest where A is smallest; setting $d(\dot{m}/A)/dMa = 0$ gives Ma = 1	Maximum flux at sonic throat.
Put Ma = 1, A = A*: $1 + (k - 1)/2 = (k + 1)/2$	
$\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}\ [2/(k + 1)]^{(k+1)/[2(k-1)]}$	Air: $\dot{m}_{max} = 0.6847A^*P_0/\sqrt{(RT_0)}$.



Converging nozzle: pressure distribution and mass flow vs back pressure

Effect of back pressure on a converging nozzle (2022 5b)

1. **$P_b = P_0$:** no flow; pressure is P_0 everywhere.
2. **$P_0 > P_b > P^*$:** subsonic flow throughout; exit pressure $P_e = P_b$; lowering P_b increases the velocity and the mass flow.
3. **$P_b = P^*$ (= $0.528P_0$ for air):** Ma = 1 at the exit; the mass flow reaches its maximum: the nozzle is **choked**.
4. **$P_b < P^*$:** the exit stays at $P_e = P^*$, $Ma_e = 1$ and $\dot{m} = \dot{m}_{max}$: lowering P_b further has no effect inside the nozzle because pressure signals cannot travel upstream through the sonic exit. The gas expands from P^* to P_b outside the nozzle through expansion waves.

Mass flow can be raised only by increasing P_0, lowering T_0 or enlarging the throat.

Numerical 1: where the air velocity equals the speed of sound (2022 5c)

Step (write this)	Why / what it means
$T_0 = T_1 + V_1^2/2c_p = 350 + 150^2/(2 \times 1005) = 361.2$ K	Stagnation from inlet.
$P_0 = P_1(T_0/T_1)^{k/(k-1)} = 0.2(361.2/350)^{3.5} = 0.2233$ MPa	
$T^* = 0.8333T_0 =$ **301.0 K**; $P^* = 0.5283P_0 =$ **0.118 MPa (118.0 kPa)**	Critical ratios.
$Ma_1 = 150/\sqrt{(1.4 \times 287 \times 350)} = 150/375.0 = 0.400$	
$A_1/A^* = (1/0.4)[(2/2.4)(1 + 0.2 \times 0.16)]^3 = 1.590 \Rightarrow$ **$A^*/A_1 = 0.629$**	Cengel Example 12-? (0.1176 MPa, 300.9 K, 0.629).

Numerical 2: CO_2 nozzle at P = 1200 kPa (2020 4c)

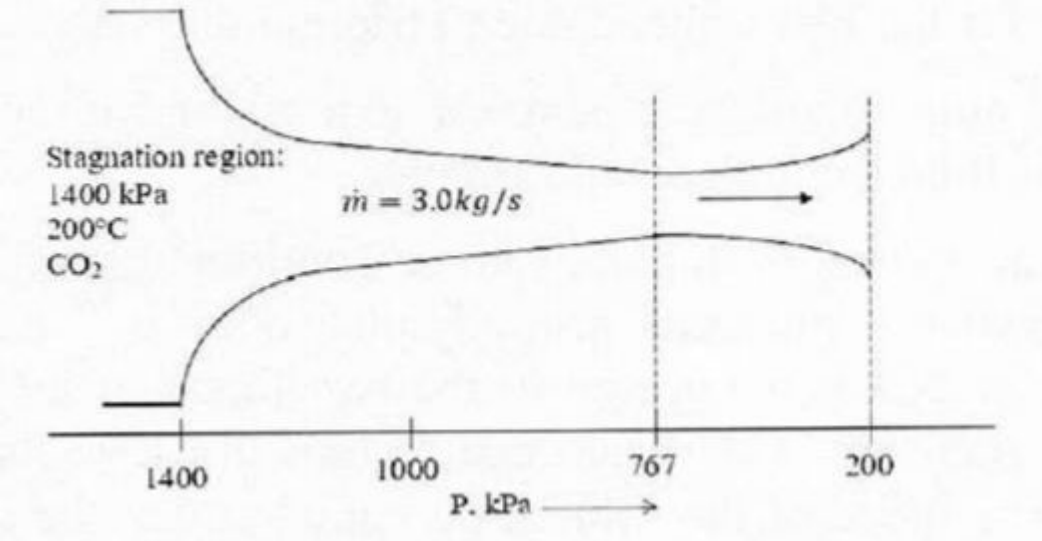


Figure of 2020 4(c)

Step (write this)	Why / what it means
$T = T_0(P/P_0)^{(k-1)/k} = 473.15(1200/1400)^{0.2242} = 457.1$ K	(k − 1)/k = 0.289/1.289 = 0.2242; low inlet velocity so inlet = stagnation.
$V = \sqrt{[2c_p(T_0 - T)]} = \sqrt{[2 \times 846 \times 16.07]} =$ **164.9 m/s**	
$\rho = P/RT = 1200/(0.1889 \times 457.1) =$ **13.9 kg/m^3**	
$A = \dot{m}/\rho V = 3.00/(13.9 \times 164.9) = 1.309\times10^{-3}$ m^2 = **13.1 cm^2**	
$c = \sqrt{(kRT)} = \sqrt{(1.289 \times 188.9 \times 457.1)} = 333.6$ m/s; **Ma = 164.9/333.6 = 0.494**	Cengel Example 12-3. (Throat at 767 kPa where Ma = 1; area 9.6 cm^2.)

Numerical 3: converging-diverging nozzle, exit Ma = 2 (2023 1d)

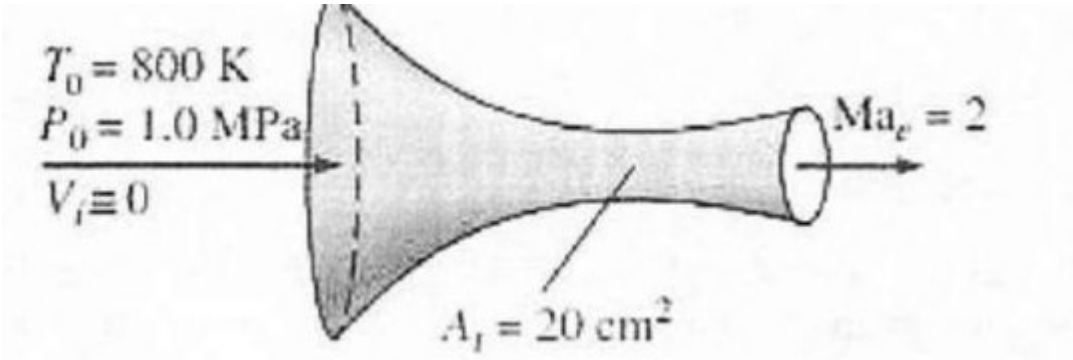


Figure of 2023 1(d)

Step (write this)	Why / what it means
Throat (Ma = 1): $T^* = 0.8333 \times 800 =$ **666.7 K**; $P^* = 0.5283 \times 1.0 =$ **0.5283 MPa**	Inlet velocity negligible: inlet = stagnation.
$\rho^* = P^*/RT^* = 528{,}300/(287 \times 666.7) =$ **2.761 kg/m^3**; $V^* = \sqrt{(kRT^*)} =$ **517.6 m/s**	
Exit Ma = 2: $T_e = 800/(1 + 0.2 \times 4) =$ **444.4 K**; $P_e = 1.0/(1.8)^{3.5} =$ **0.1278 MPa**	Isentropic relations.

$\rho_e = 127{,}800/(287 \times 444.4) =$ **$1.002\ kg/m^3$**; $V_e = 2\sqrt{(1.4 \times 287 \times 444.4)} =$ **$845.2\ m/s$**	
$A_e/A^* = (1/2)[(2/2.4)(1.8)]^3 = 1.6875 \Rightarrow A_e =$ **$33.75\ cm^2$**	
$\dot{m} = \rho^* A^* V^* = 2.761 \times 0.0020 \times 517.6 =$ **$2.86\ kg/s$**	Same at the exit: 1.002 × 0.003375 × 845.2 = 2.86.

Numerical 4: converging nozzle at two back pressures (2019 5d)

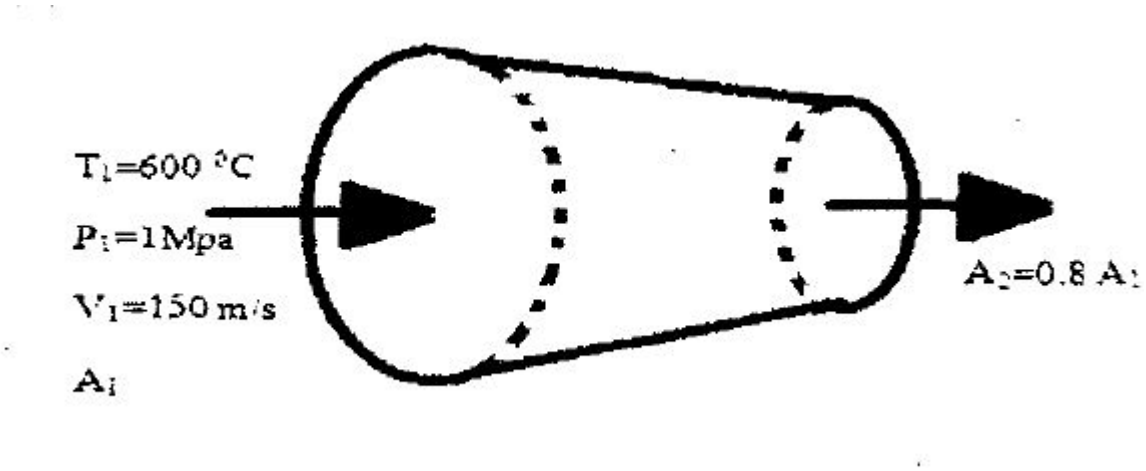


Figure of 2019 5(d): inlet velocity 150 m/s

Step (write this)	Why / what it means
Inlet $T_1 = 873.15\ K$, $P_1 = 1\ MPa$, $V_1 = 150\ m/s$ (figure): $T_0 = 873.15 + 150^2/2010 = 884.3\ K$	Stagnation from the inlet state.
$P_0 = 1.0(884.3/873.15)^{3.5} = 1.0456\ MPa$; $P^* = 0.5283 \times 1.0456 = 0.5524\ MPa$	Critical pressure.
(i) $P_b = 0.7\ MPa > P^*$: not choked, $P_t = 0.7\ MPa$; $T_t = 884.3(0.7/1.0456)^{0.2857} = 788.6\ K$	Throat pressure = back pressure.
$Ma_t = \sqrt{[5(884.3/788.6 - 1)]} = 0.779$; $V_t = 0.779\sqrt{(1.4 \times 287 \times 788.6)} = 438.7\ m/s$; $\rho_t = 3.093\ kg/m^3$	
$\dot{m} = 3.093 \times 0.0050 \times 438.7 =$ **$6.78\ kg/s$**	
(ii) $P_b = 0.4\ MPa < P^*$: choked; $P_t = 0.5524\ MPa$, $T_t = 0.8333 \times 884.3 = 737.0\ K$, $\rho_t = 2.612$, $V_t = \sqrt{(1.4 \times 287 \times 737.0)} = 544.2\ m/s$	Ma = 1 at the throat.
$\dot{m} = 2.612 \times 0.0050 \times 544.2 =$ **$7.11\ kg/s$** $(= \dot{m}_{max})$	Cengel Example 12-5: 6.77 and 7.10 kg/s.

Side note: If the 150 m/s inlet velocity is ignored (stagnation = 1 MPa, 873 K) the choked flow would be 6.84 kg/s; the figure gives the velocity, so use it.

Memory trick: $dA/A = (Ma^2 - 1)dV/V$: “below one, opposite; above one, same”.